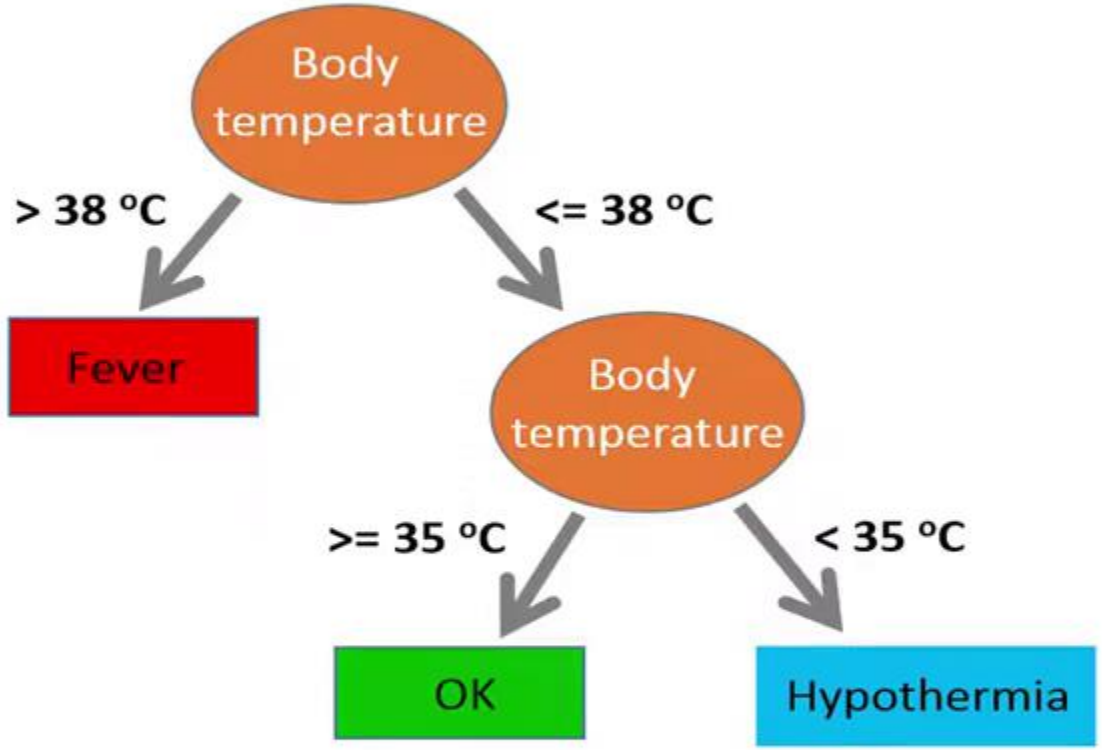
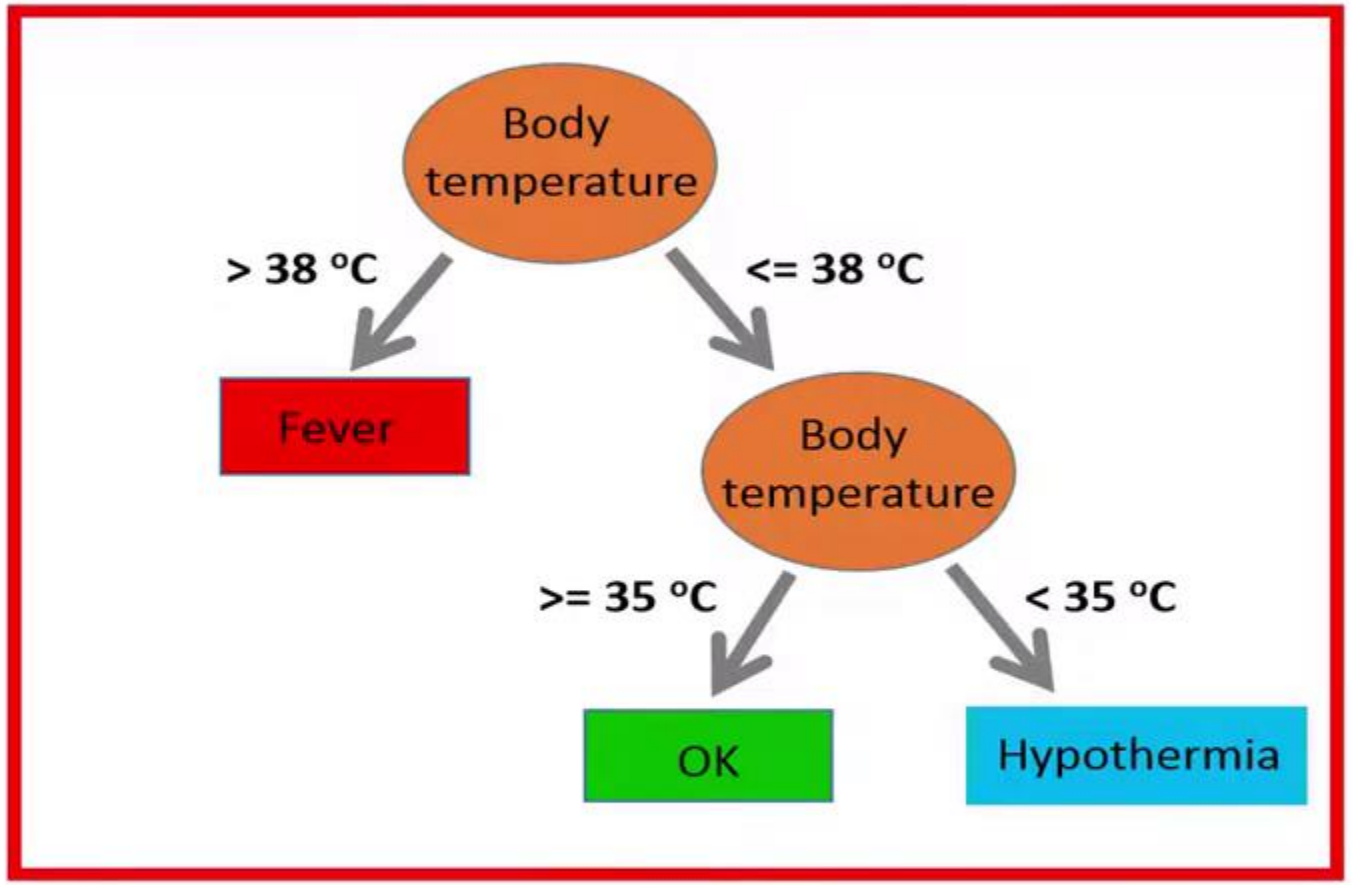
Decision Trees

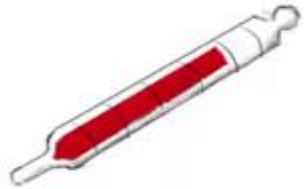
Dr.Hafeez Anwar

Decision tree



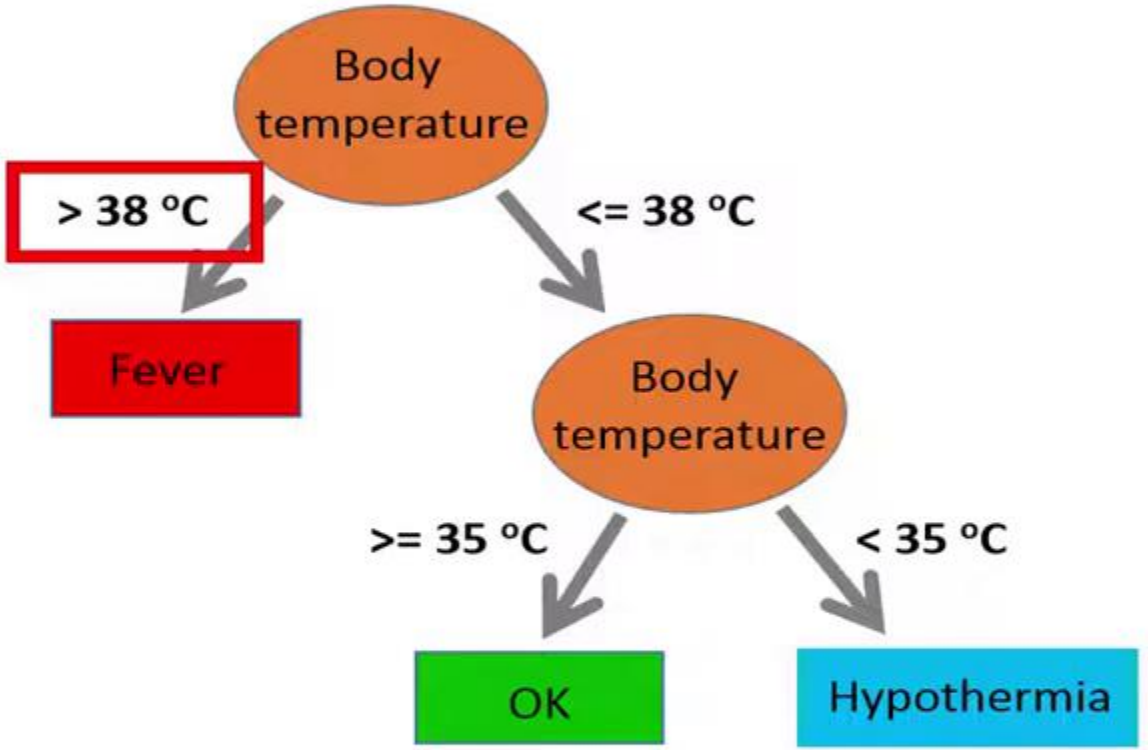
Decision tree

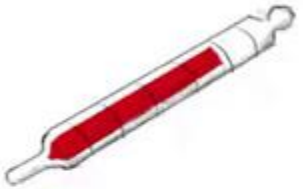




Suppose we have the following decision tree to determine if someone has fever or hypothermia based on a person's body temperature in Celsius.

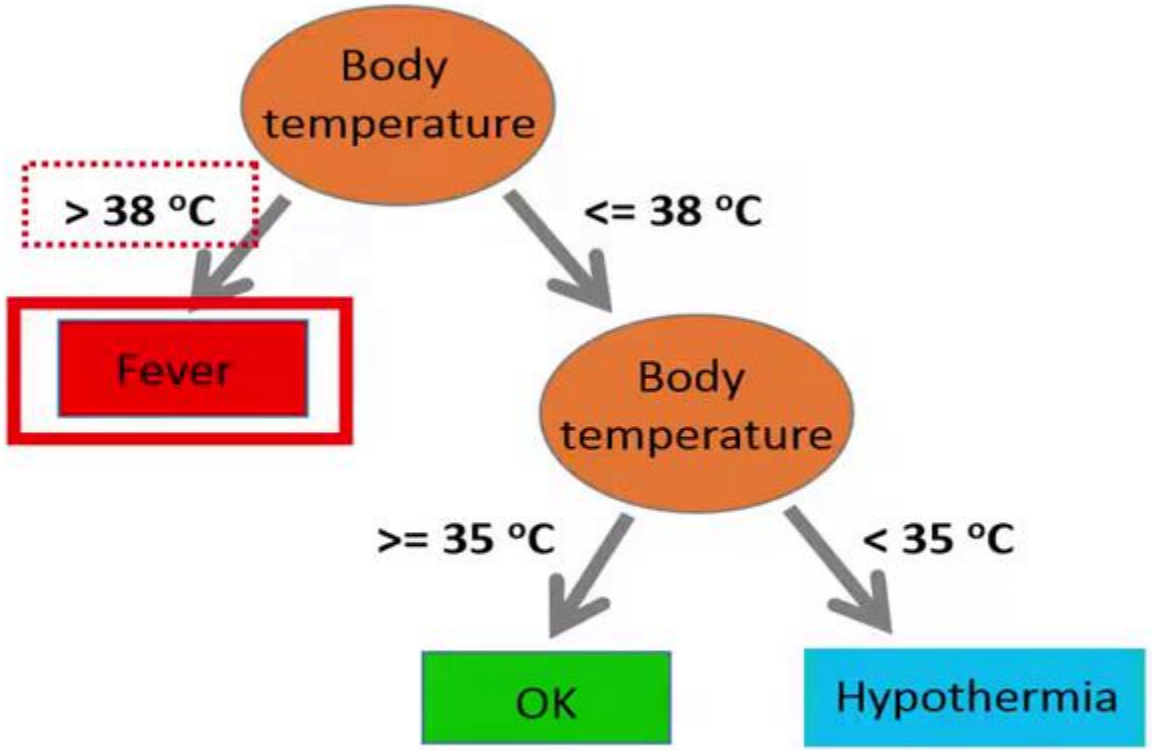
Decision tree





This tree tells us that if a person has a body temperature that is greater than 38,

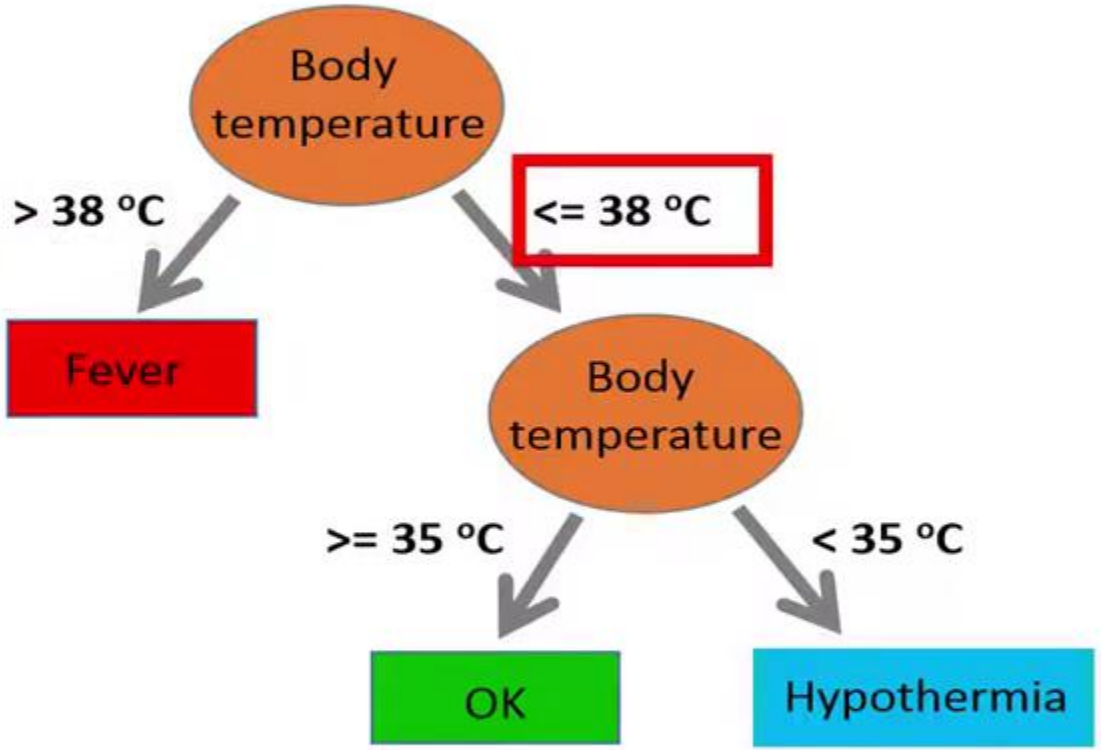
Decision tree





the person is classified as having fever.

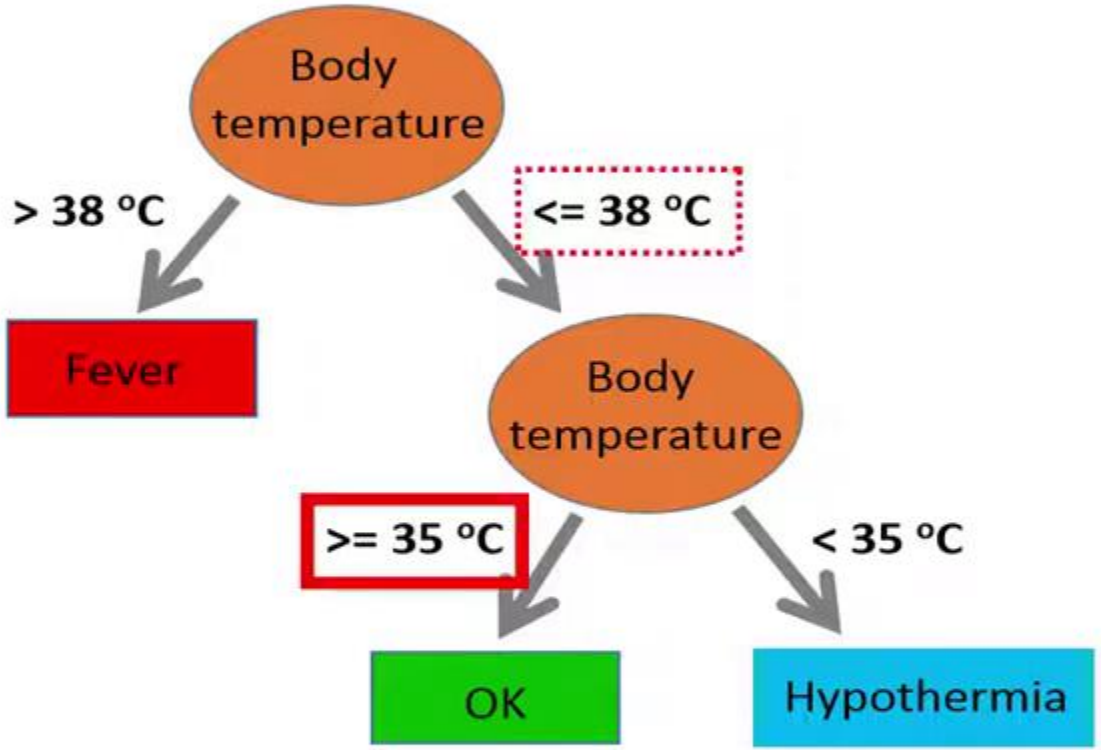
Decision tree

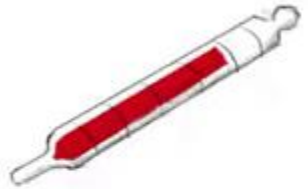




If the person has a body temperature less than, or equal to, 38,

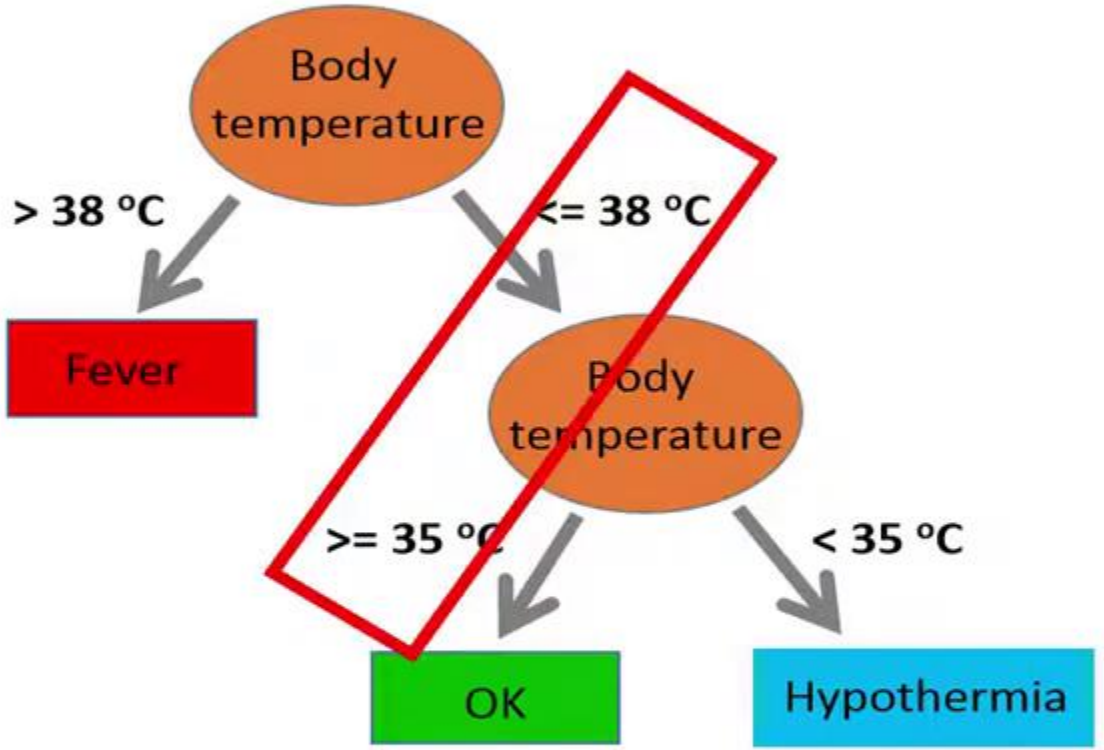
Decision tree

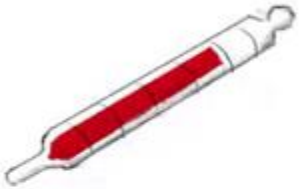




but greater, or equal to 35,

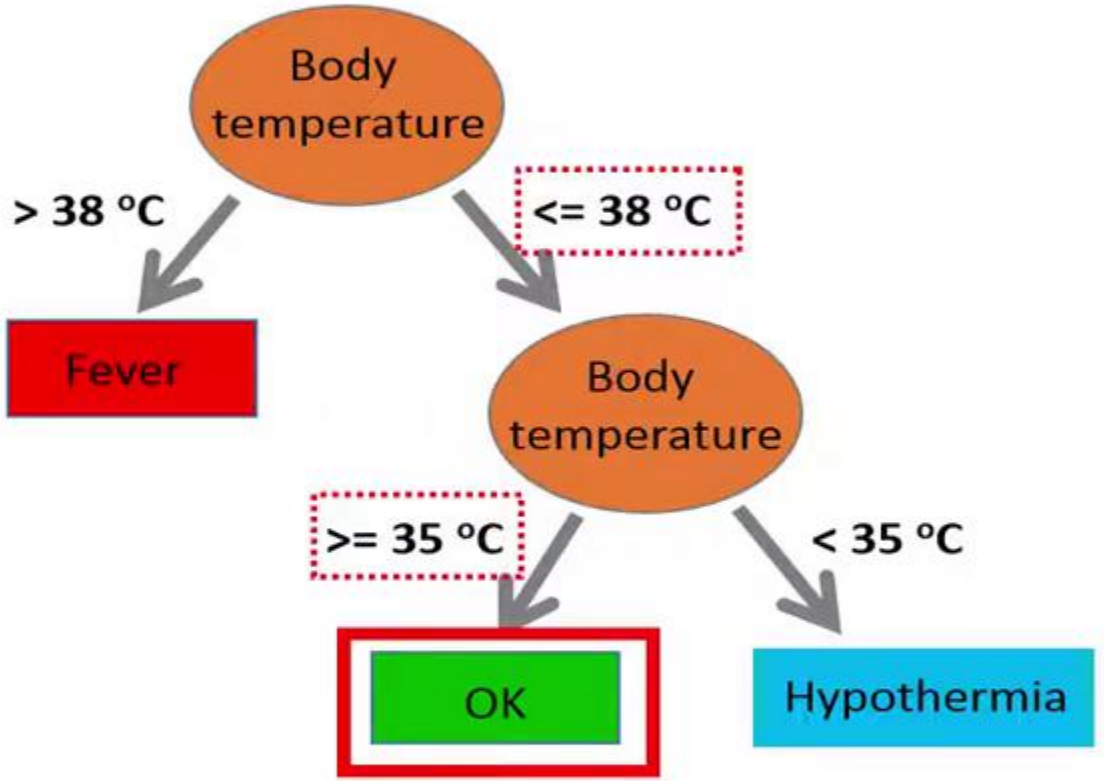
Decision tree

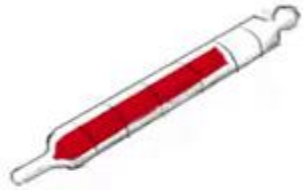




which means that the person's body temperature is between 35 and 38,

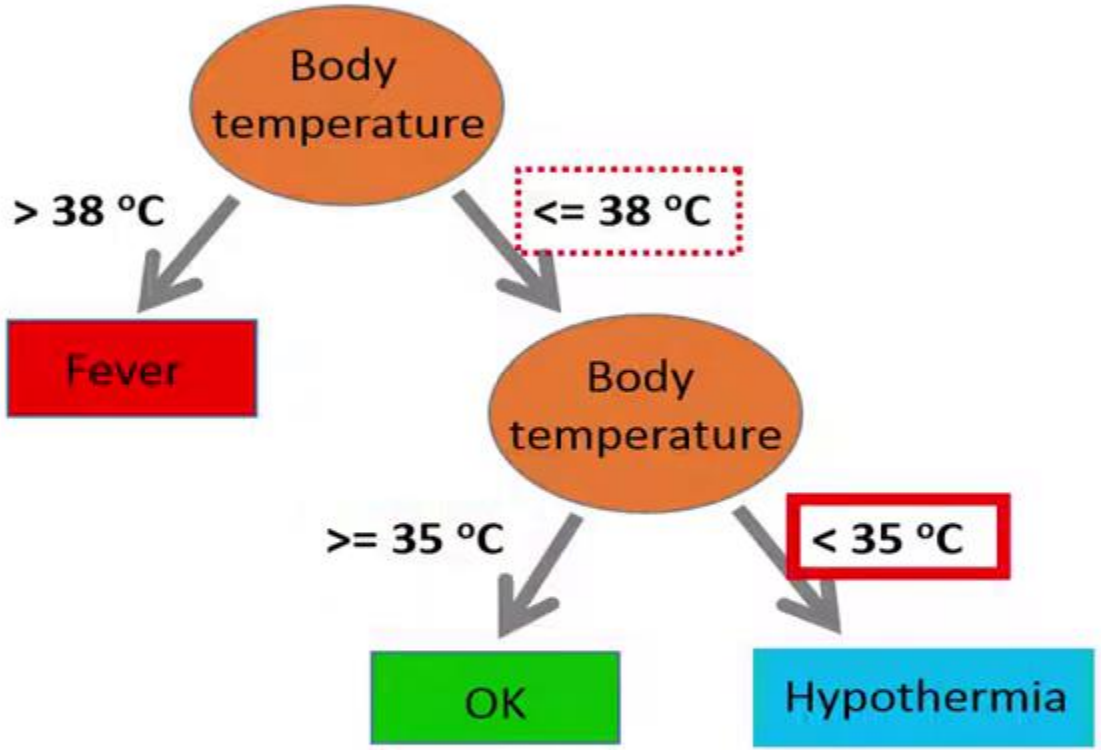
Decision tree

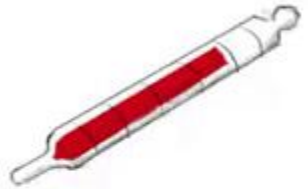




the person has neither hypothermia nor fever.

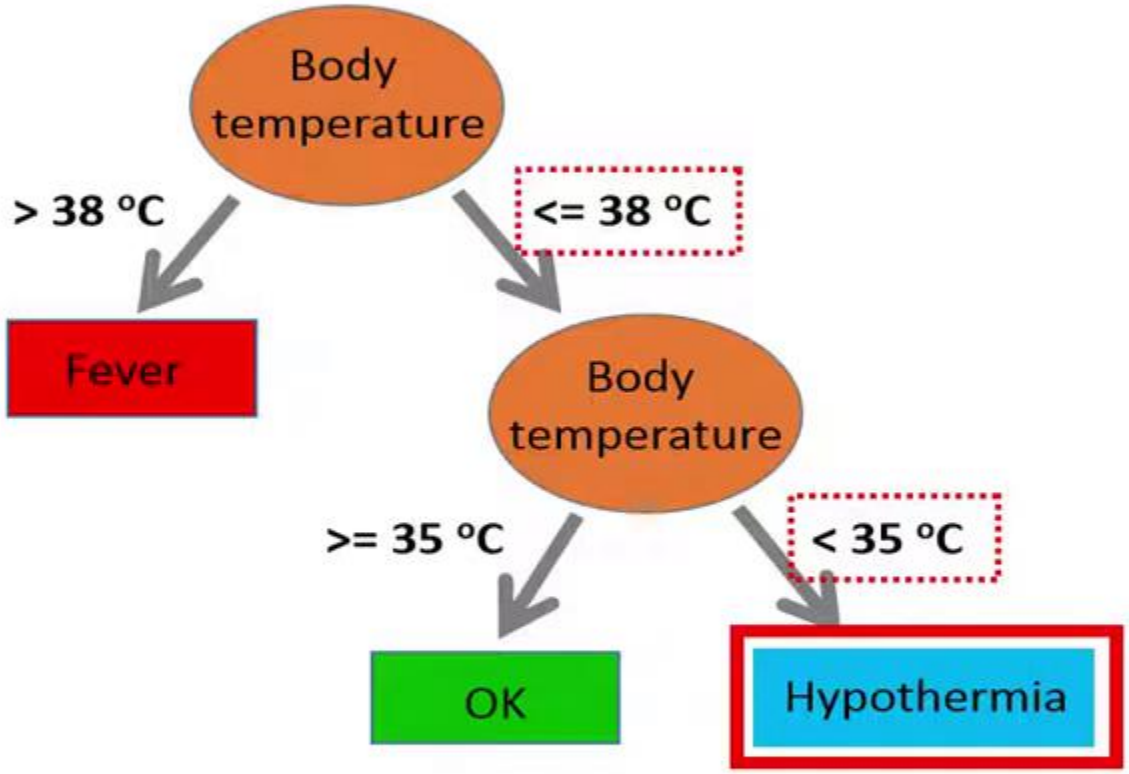
Decision tree

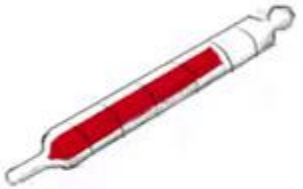




If the body temperature is below 35,

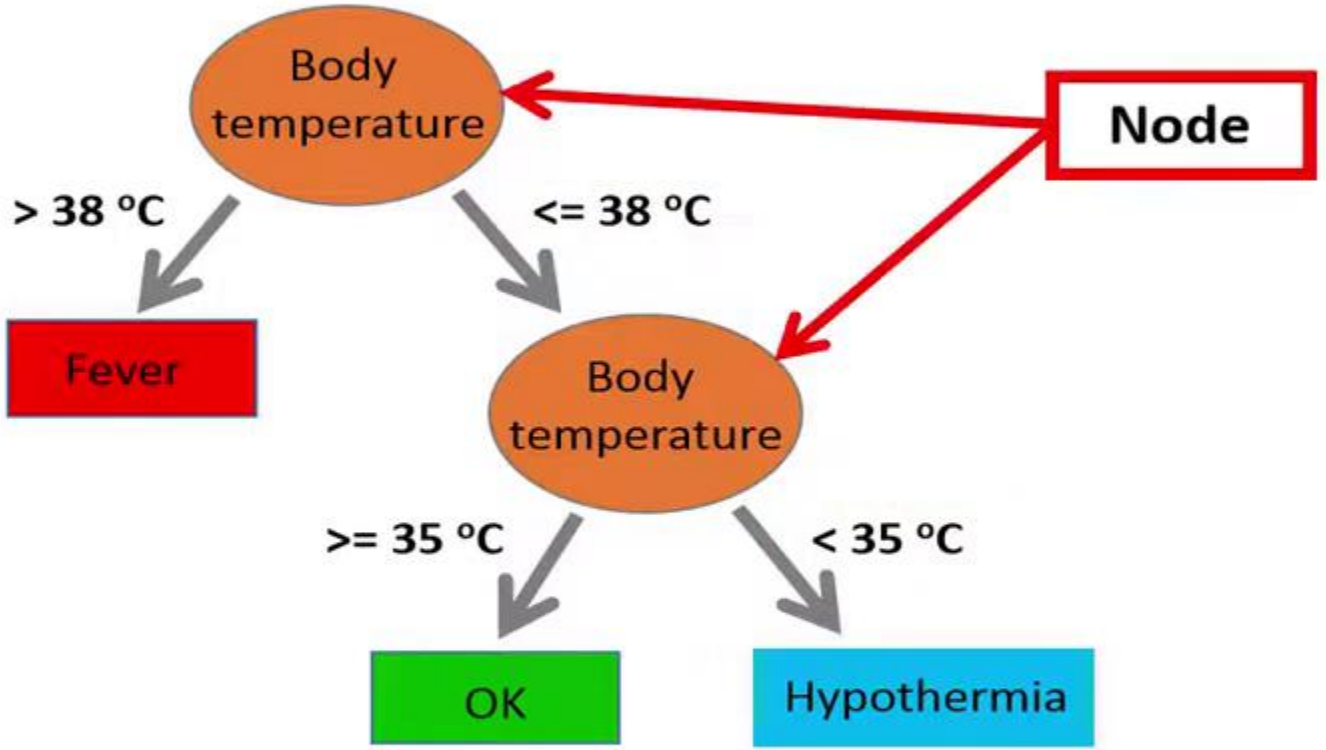
Decision tree





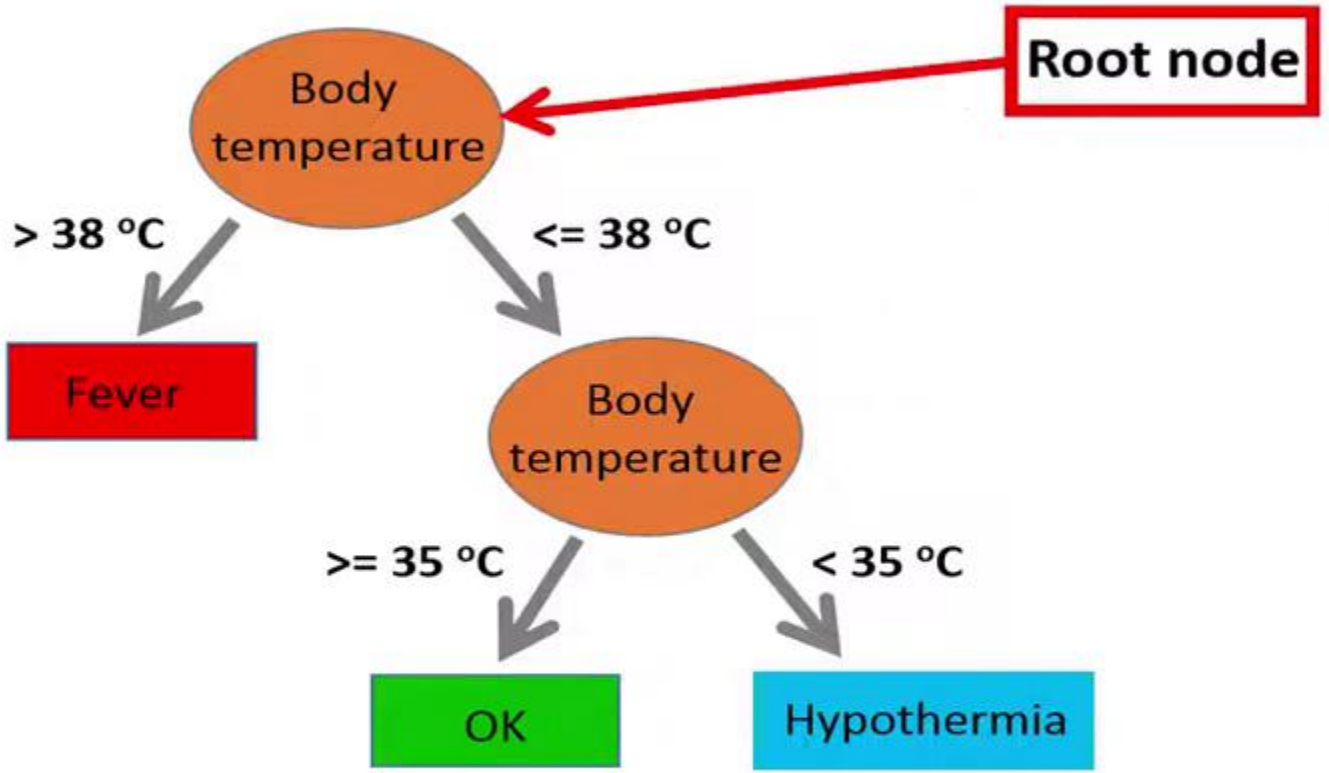
the person is classified as having hypothermia because the body temperature is too low.

Decision tree



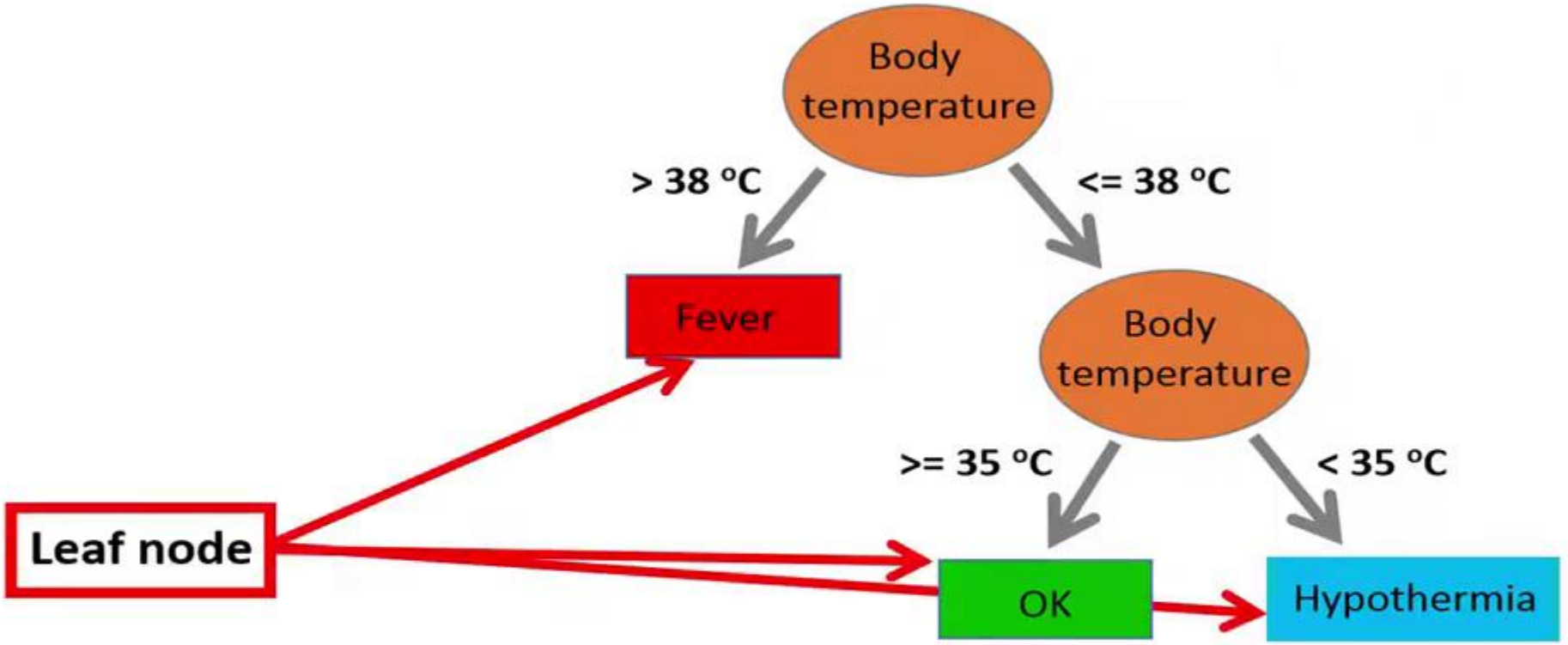
A classification tree consists of nodes, which represent a choice between alternatives based on the value of the measured variable.

Decision tree



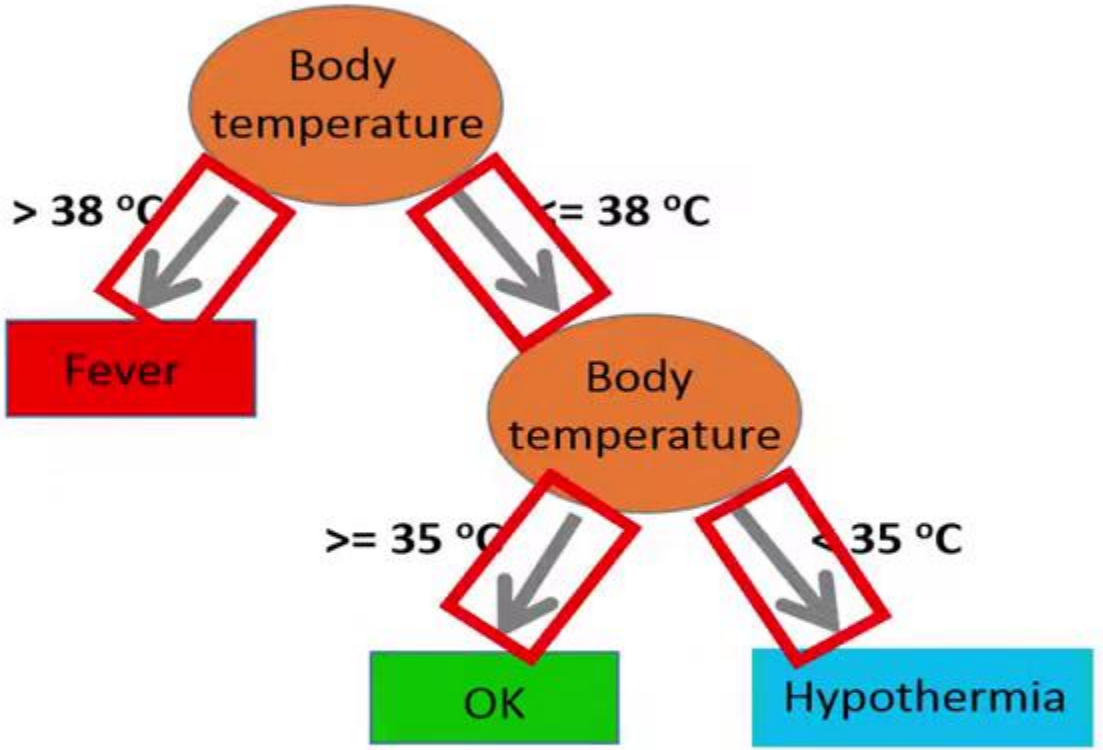
The node at the top of the tree is called the root node.

Decision tree



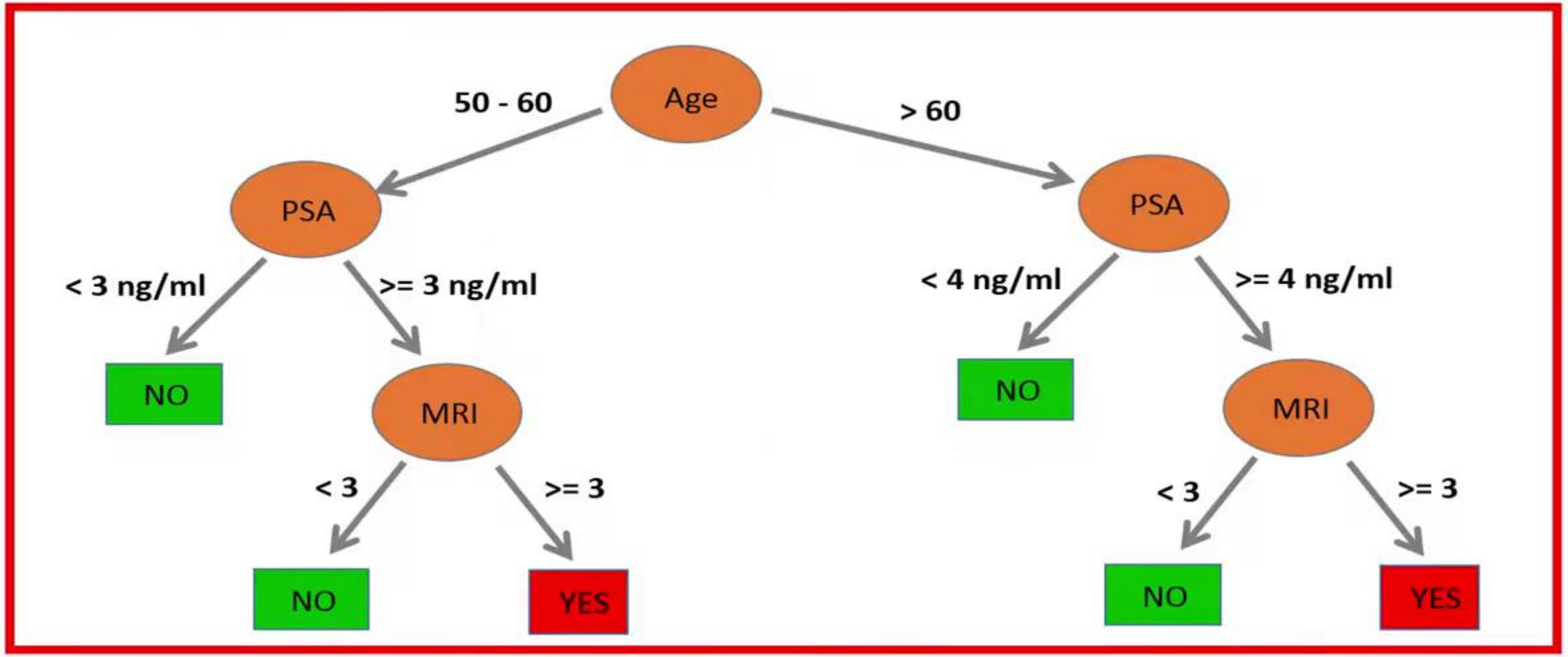
Leaf nodes represent the decision.

Decision tree



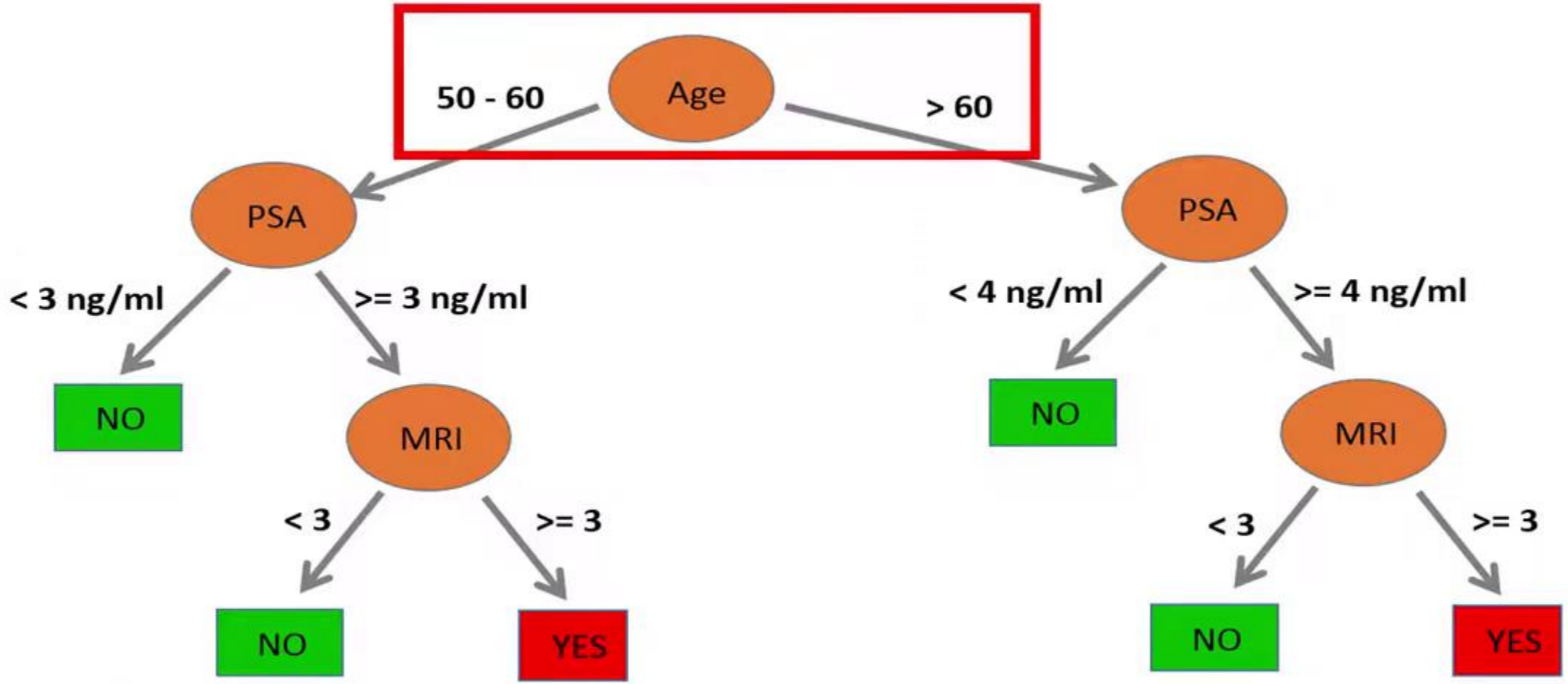
The arrows are called branches or edges.

Decision tree



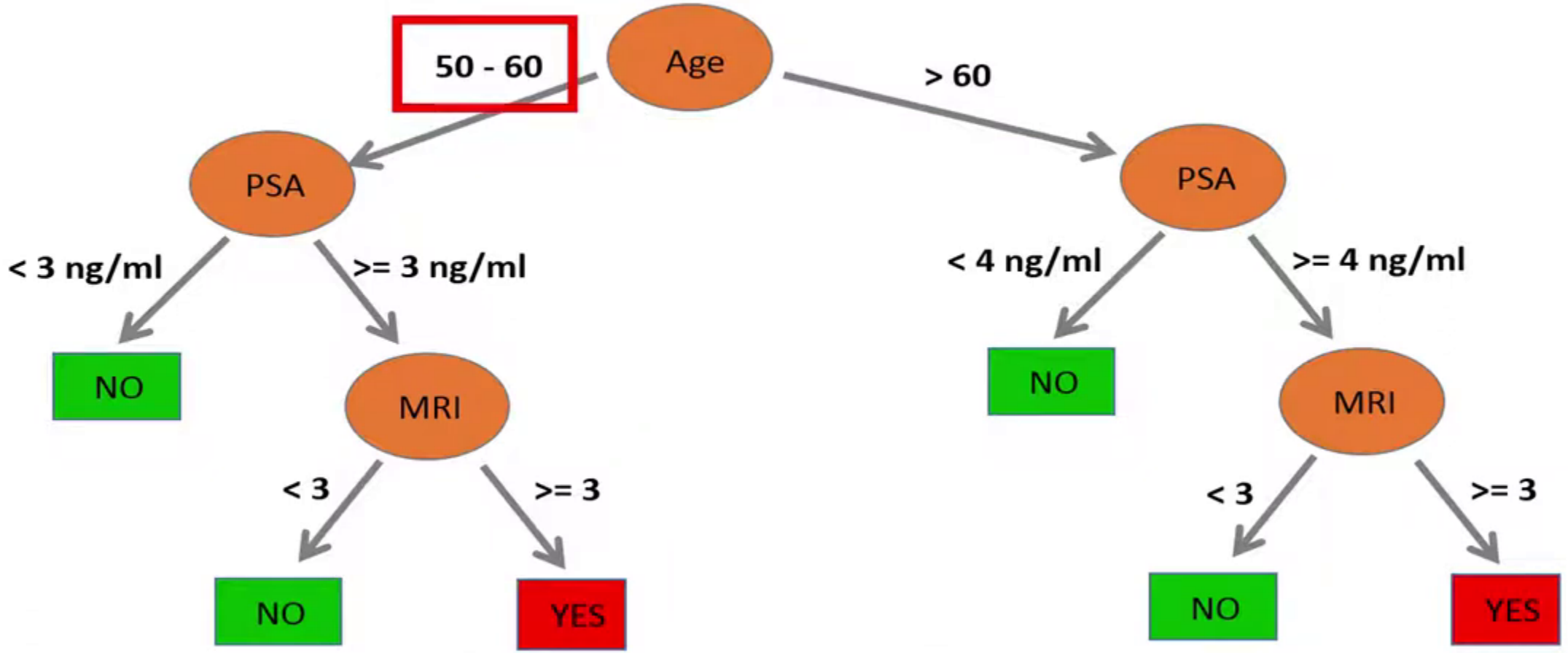
We will now have a look at a larger fictive decision tree that could be used to determine if a biopsy of the prostate is necessary to determine if a person has prostate cancer or not.

Decision tree



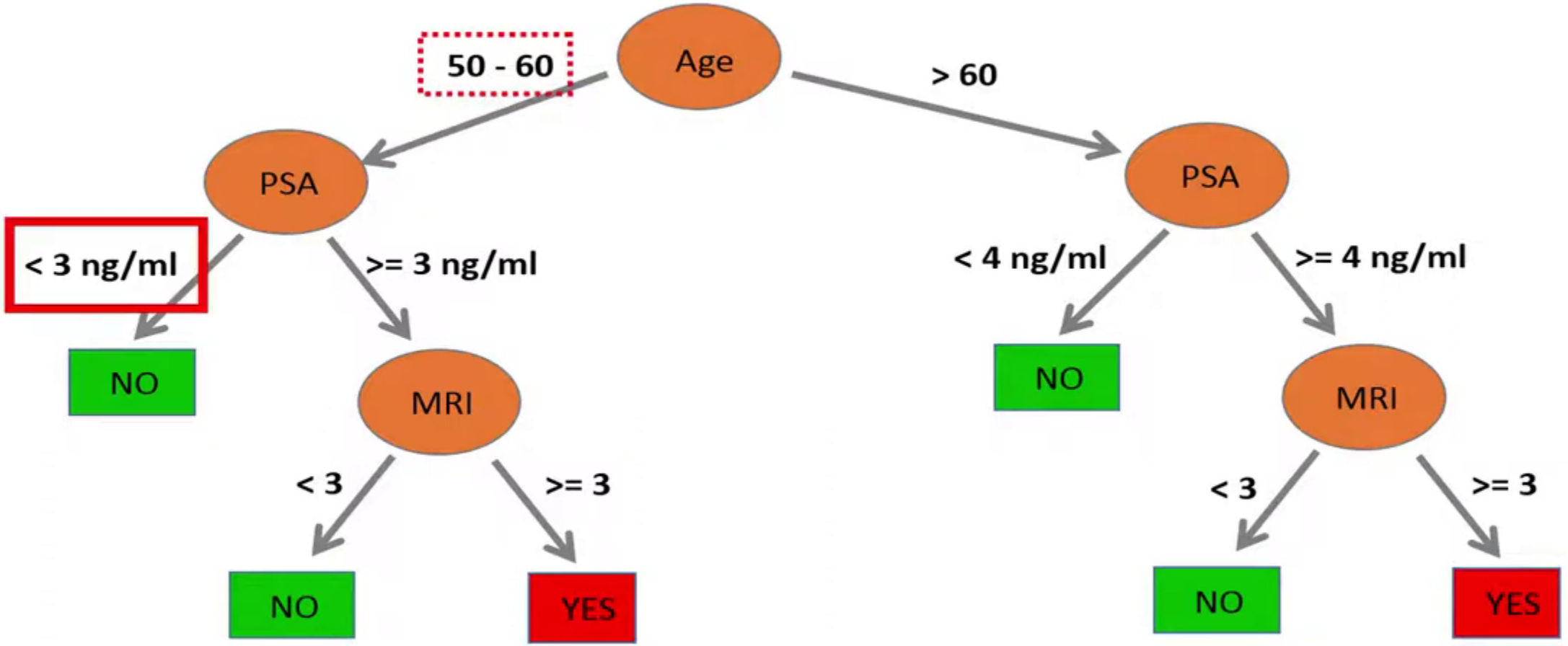
The following decision tree can be used for prostate cancer screening of older men. The aim is to determine if the person should undergo a biopsy of the prostate or not. Such a biopsy can then be used to detect tumor cells.

Decision tree



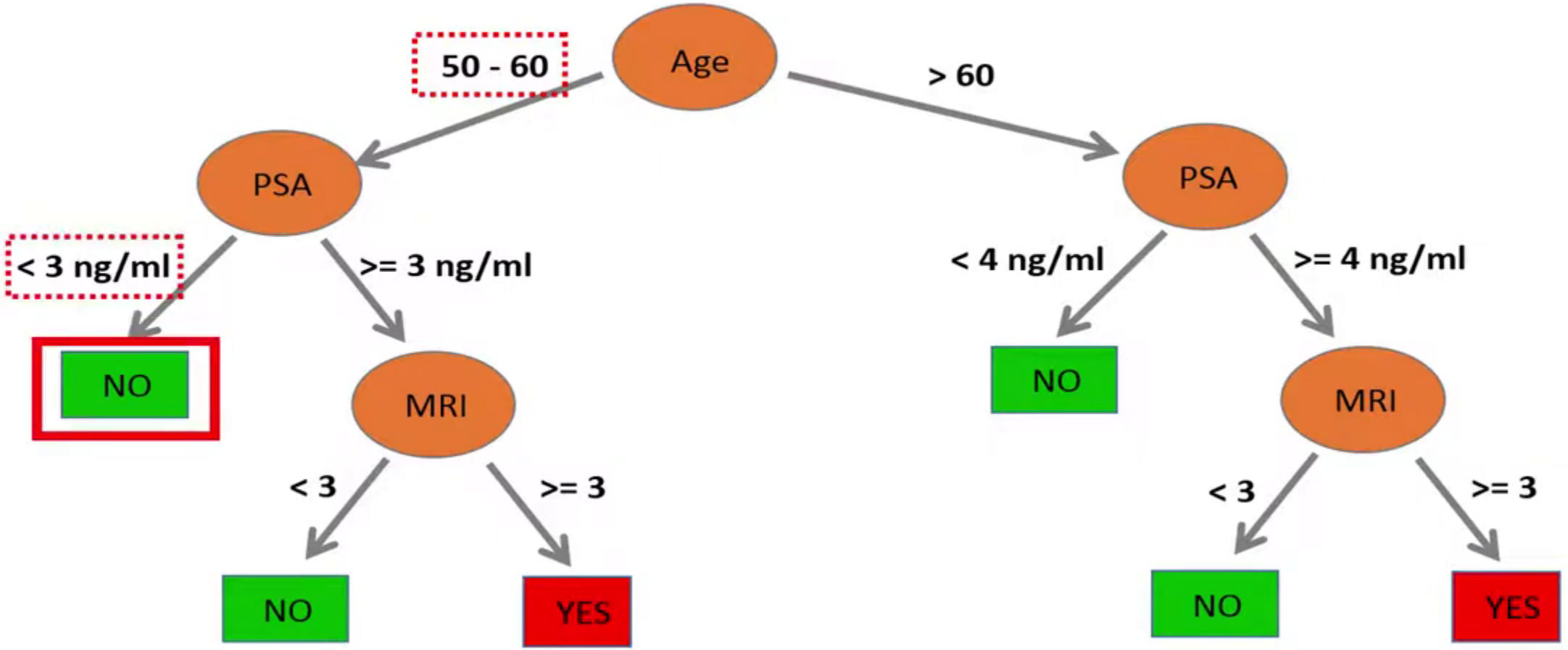
If the person is in the age between 50 and 60,

Decision tree



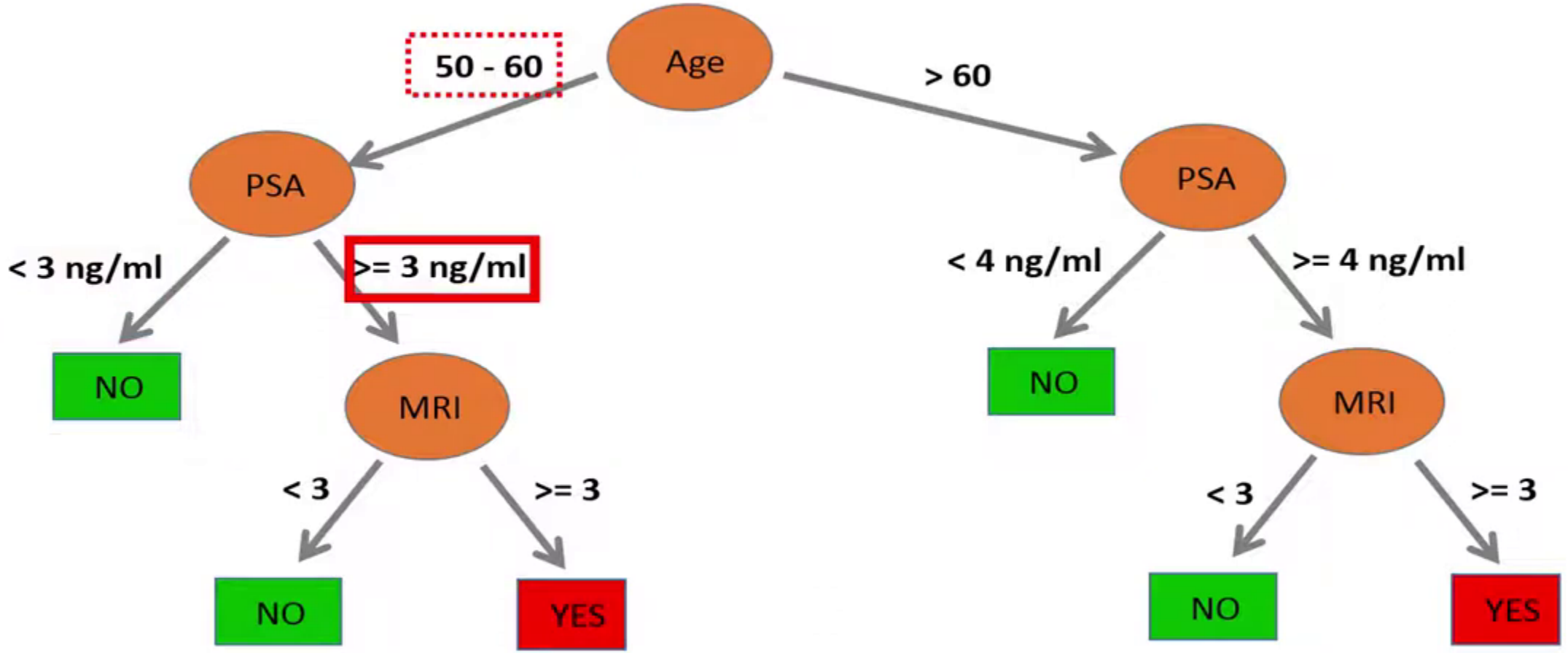
and has a prostate-specific antigen level, or a PSA level, in the blood that is less than 3 ng/ml,

Decision tree



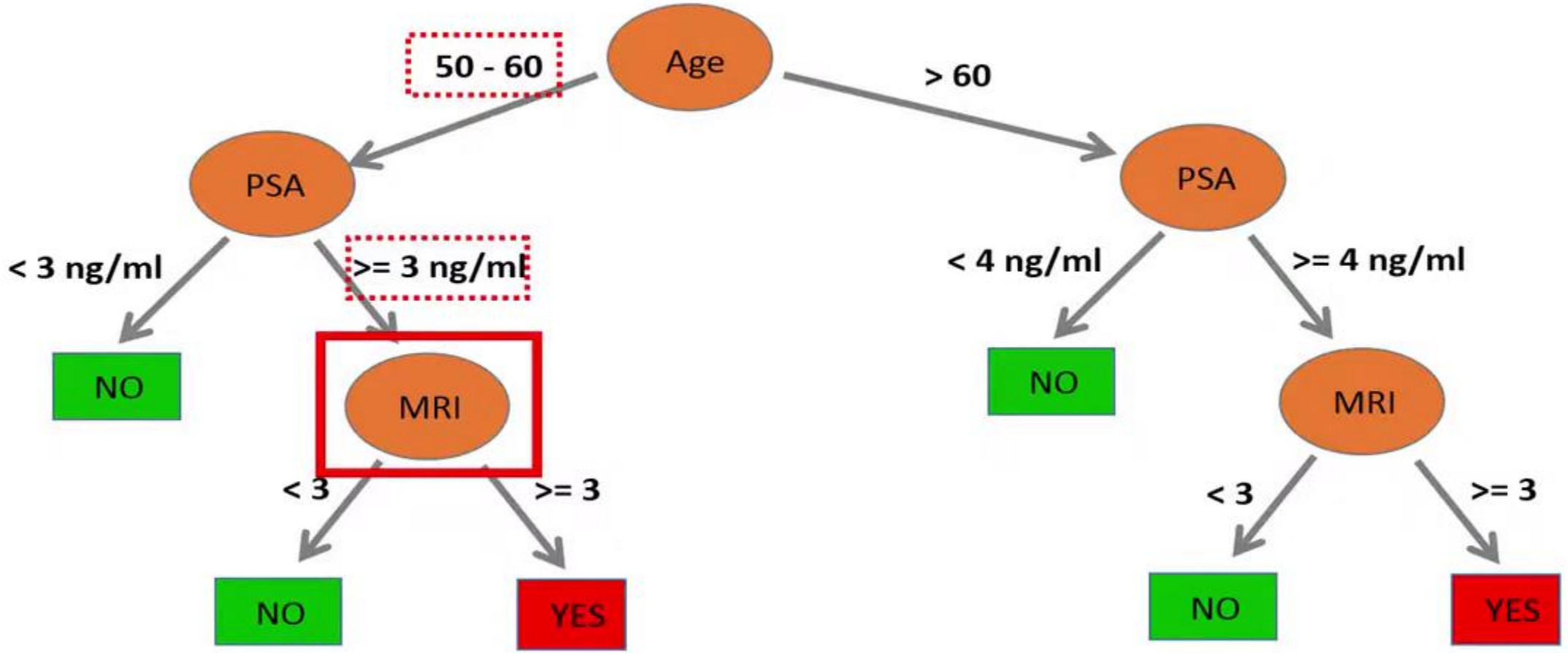
we expect that the person does not have prostate cancer and no further testing is done.

Decision tree



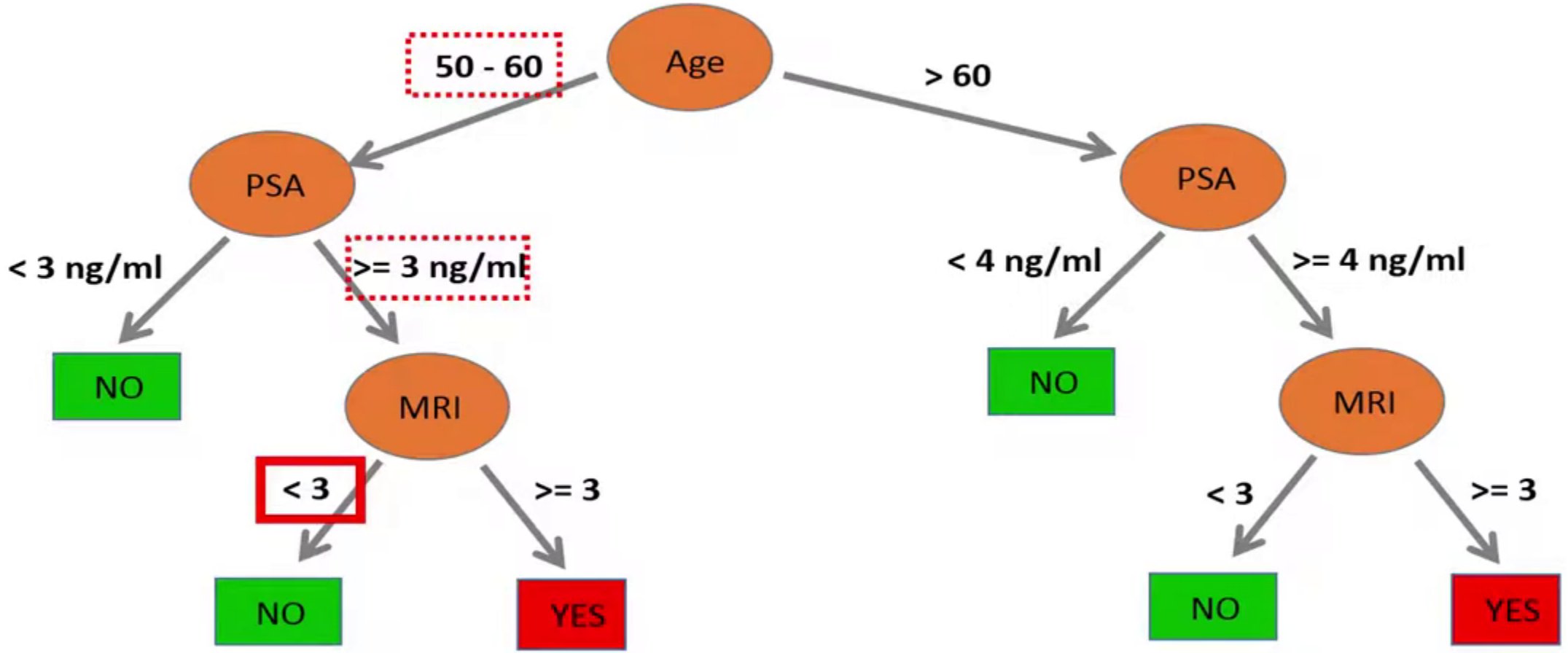
However, if the PSA level is equal to or greater than 3,

Decision tree



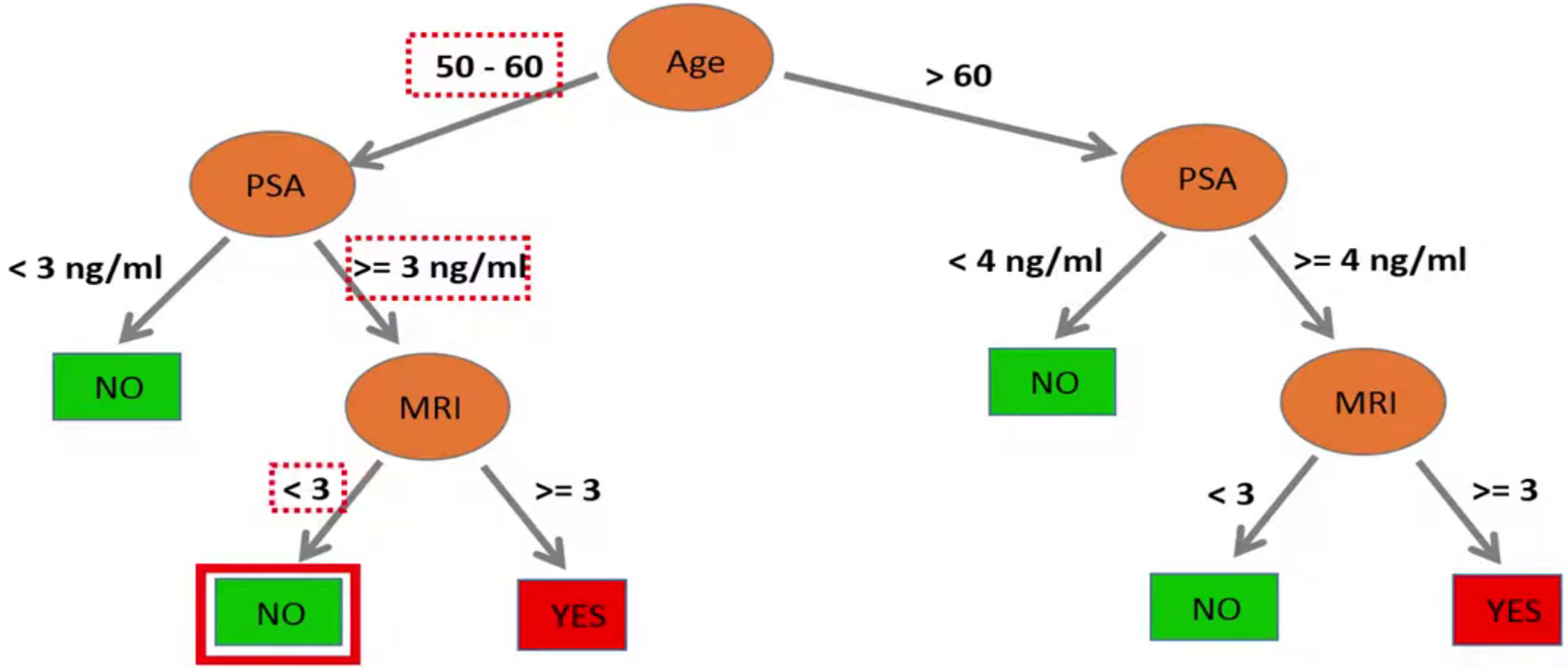
an MRI scan of the prostate is done, which generates a score from one to five. A score less than 3 indicates that it is unlikely that the person has prostate cancer.

Decision tree



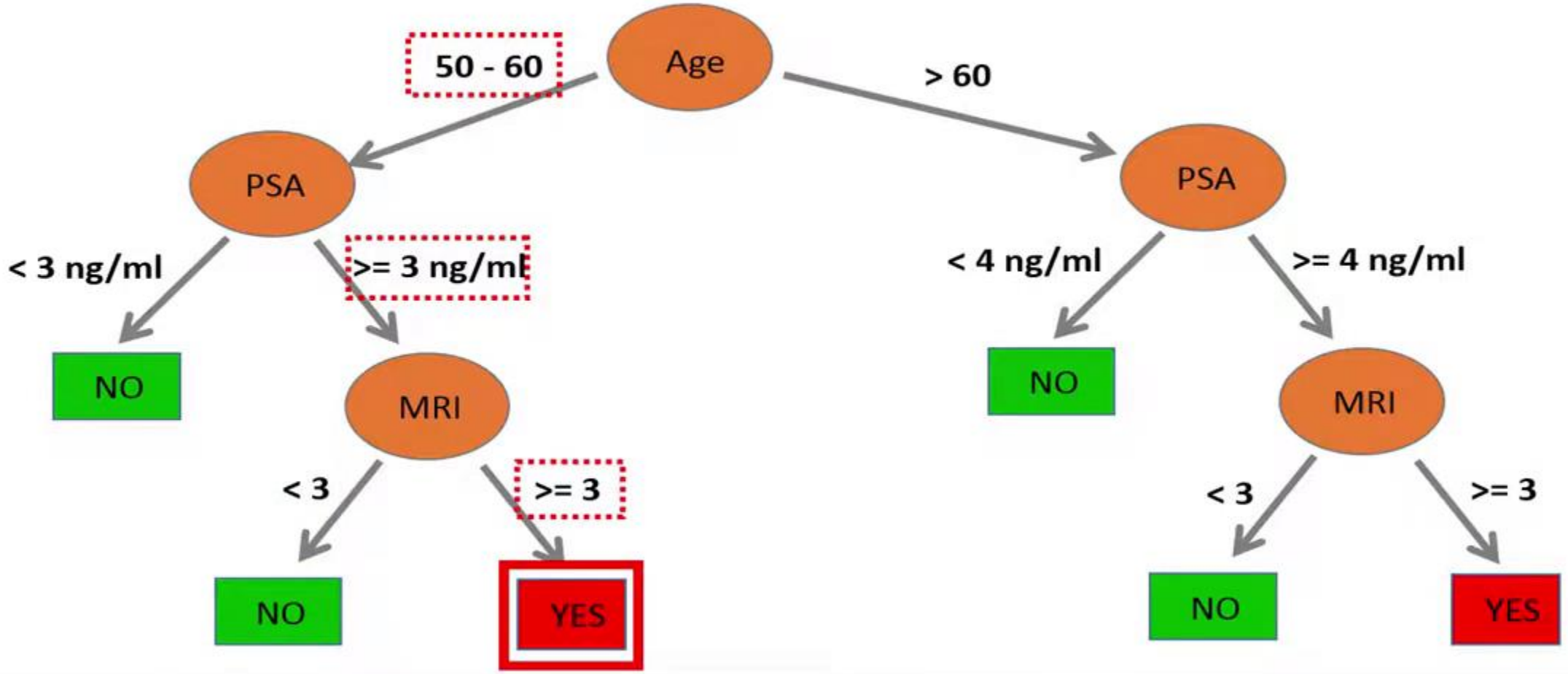
If a person has a score less than 3,

Decision tree



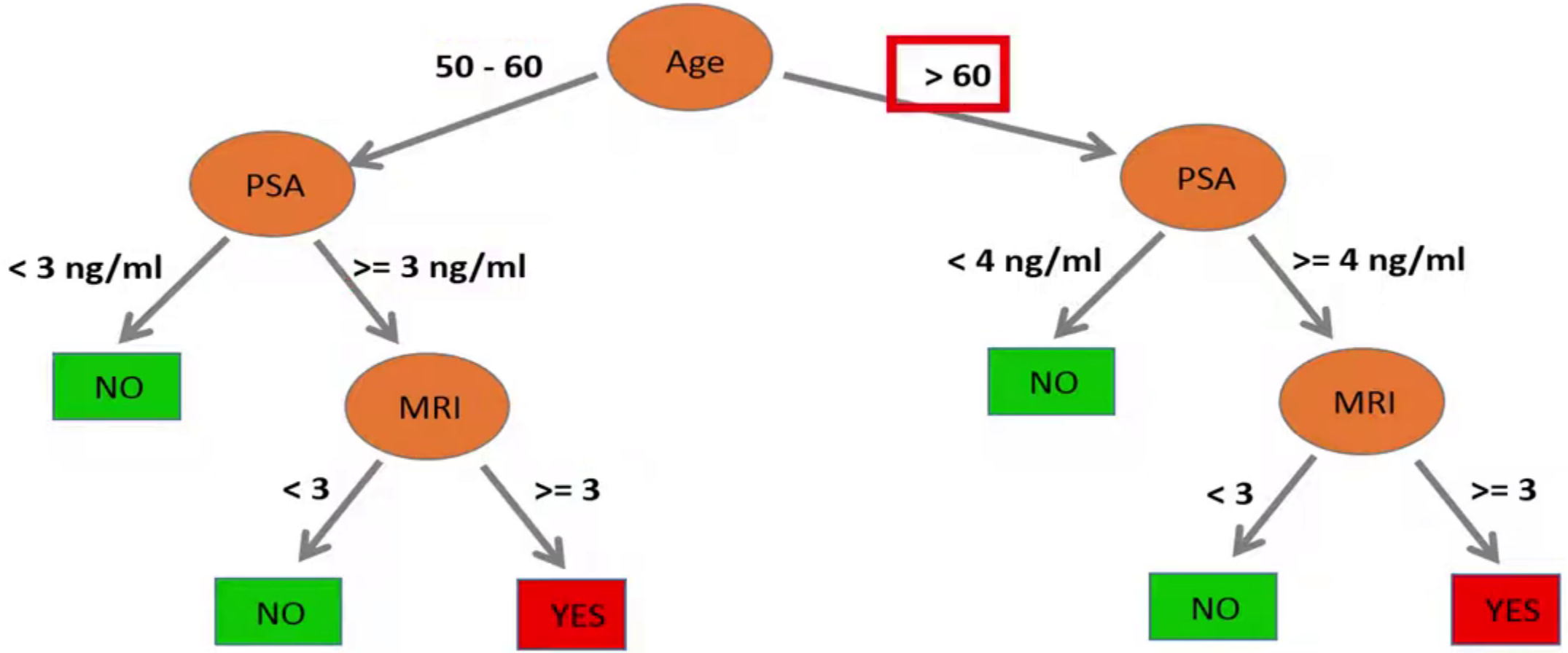
no biopsy is therefore taken from the prostate.

Decision tree



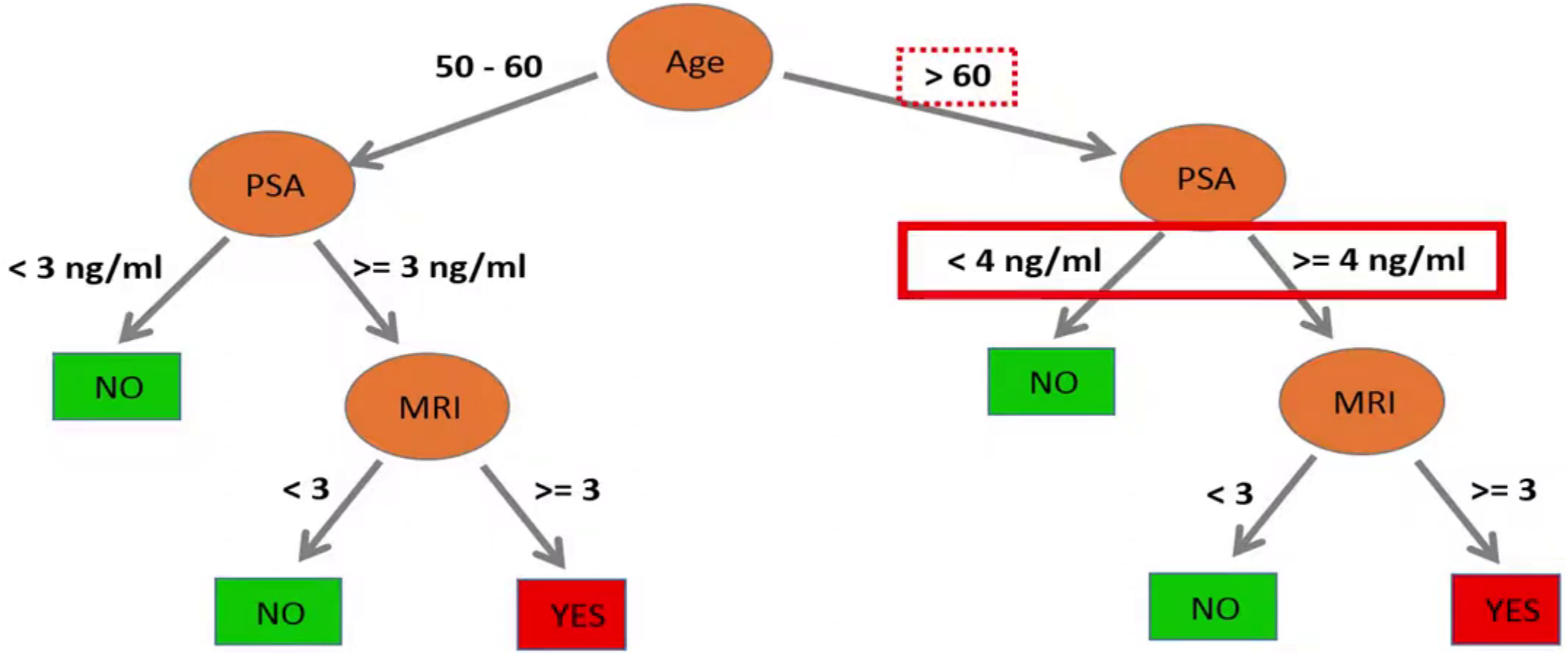
In comparison, if the MRI score is equal to or greater than 3, it is quite likely that the person has prostate cancer. The person should therefore undergo a biopsy to confirm this.

Decision tree



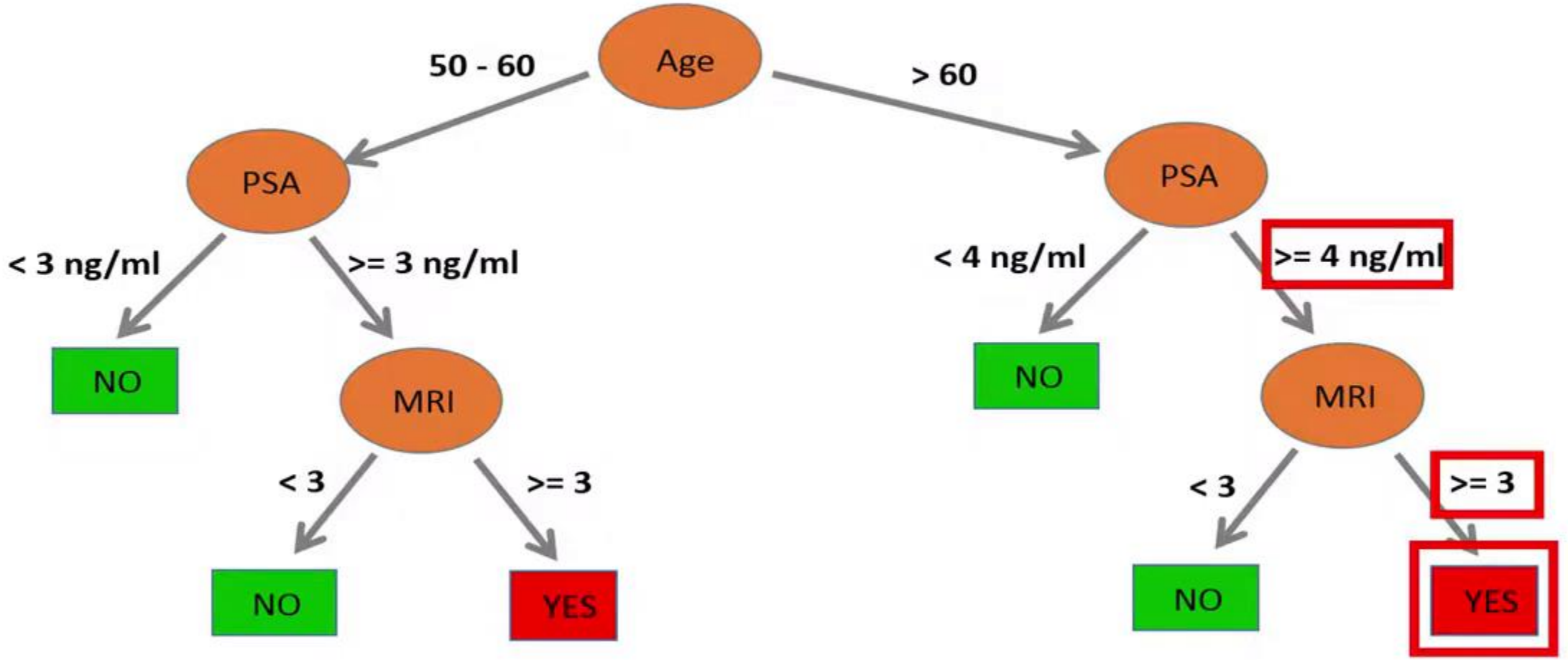
For men that are older than 60 years of age,

Decision tree



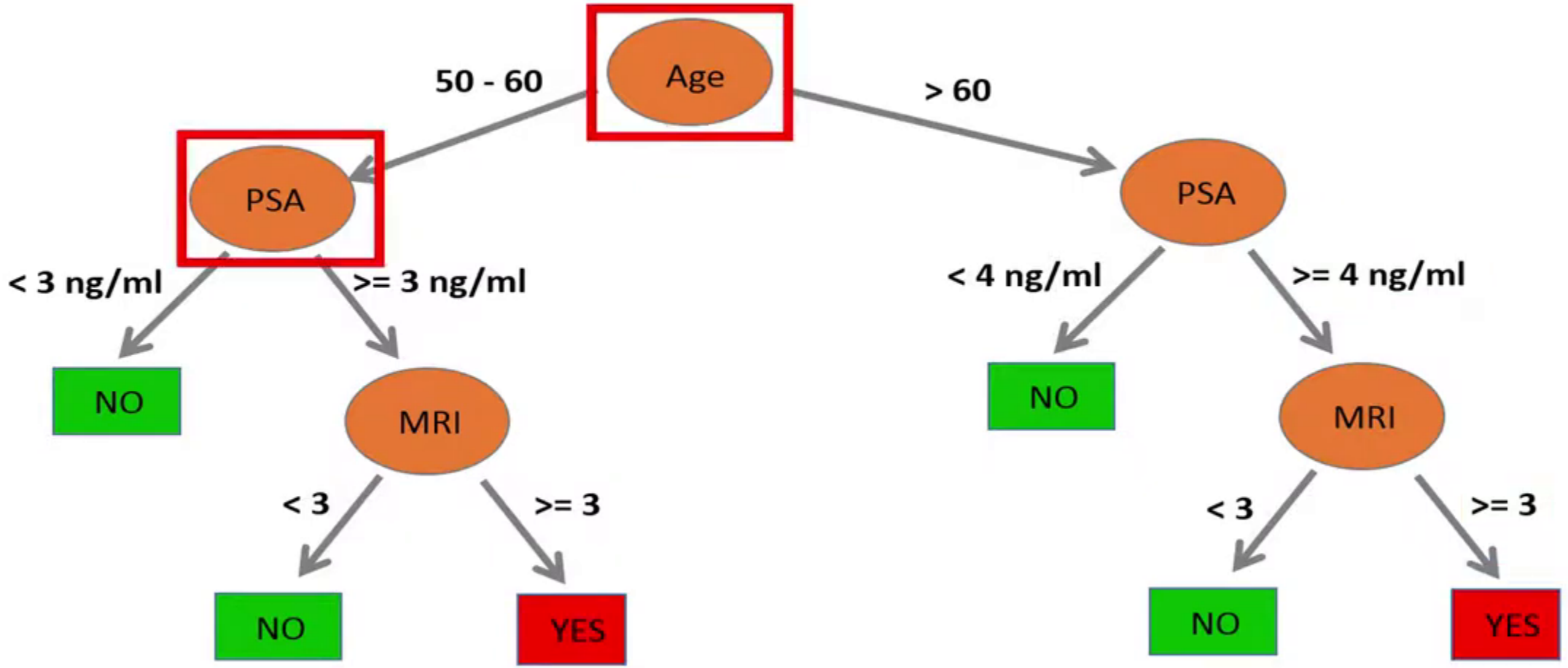
the cutoff value for the PSA level is a bit higher because the PSA level tends to increase during aging.

Decision tree



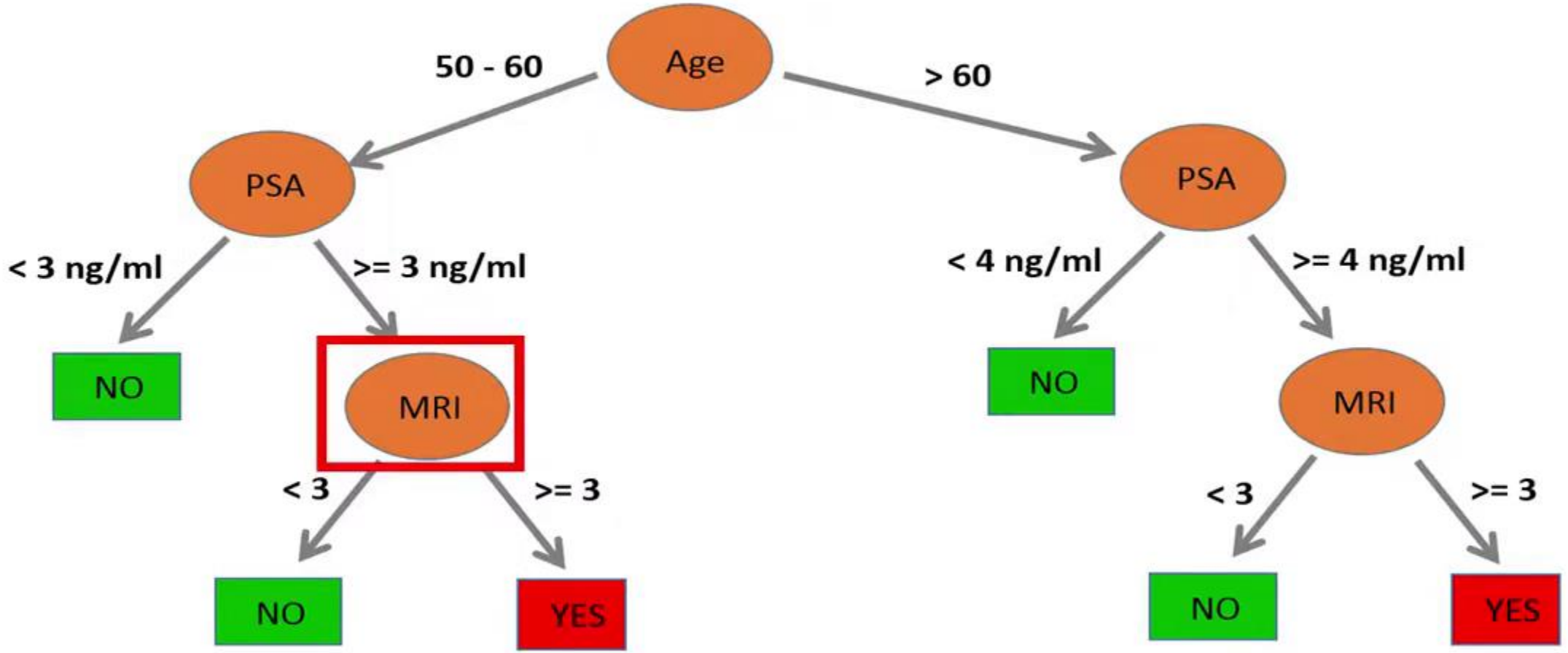
Older men should therefore have a PSA level that is greater than 4, and an MRI score of three or more before a biopsy is recommended.

Decision tree



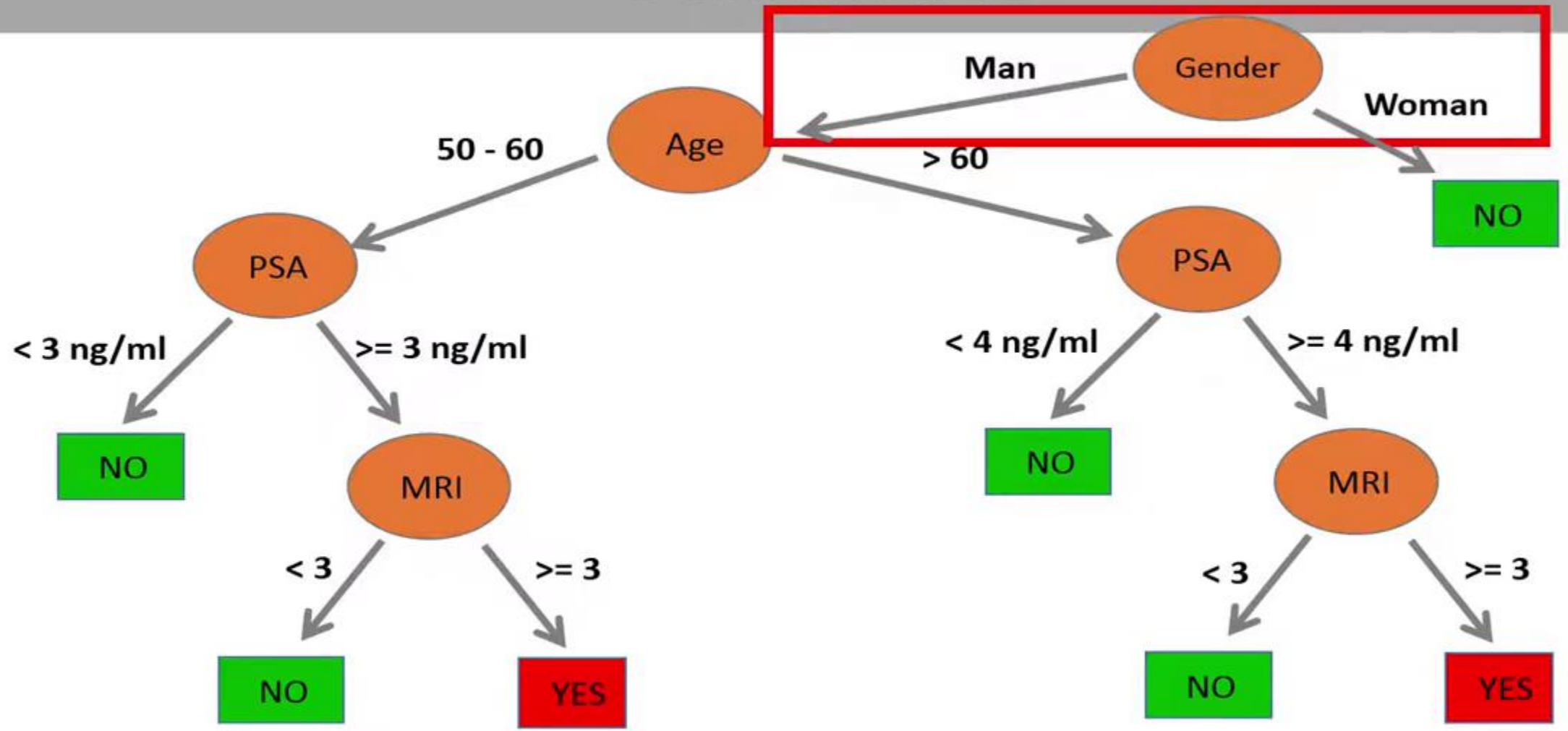
Note that the variables age and PSA are on a numerical scale,

Decision tree



whereas the variable MRI has an ordinal scale. Decision trees work just fine with variables on both numeric and categorical scales.

Decision tree



We can even have a variable on a nominal scale. This is one advantage with decision trees because most other classification methods do not work well with categorical variables.

How to construct a decision tree

We will now learn how to construct a decision tree.

How to construct a decision tree



To construct a decision tree, we need some training data where we know the class of each observation. Let's say that we would like to create a decision tree that can predict if a man has prostate cancer or not based on a PSA test.

How to construct a decision tree

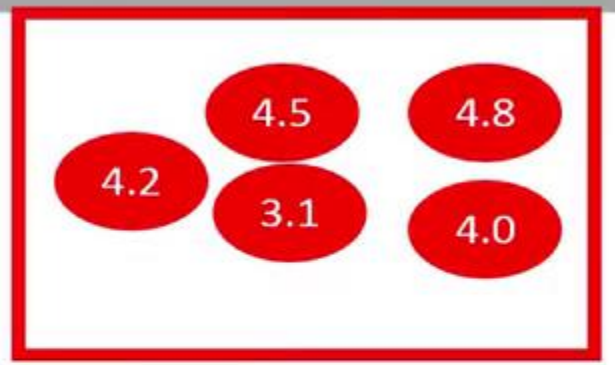




These four individuals are known to be healthy. The numbers inside the circles show the PSA level in the blood of these four individuals.

How to construct a decision tree





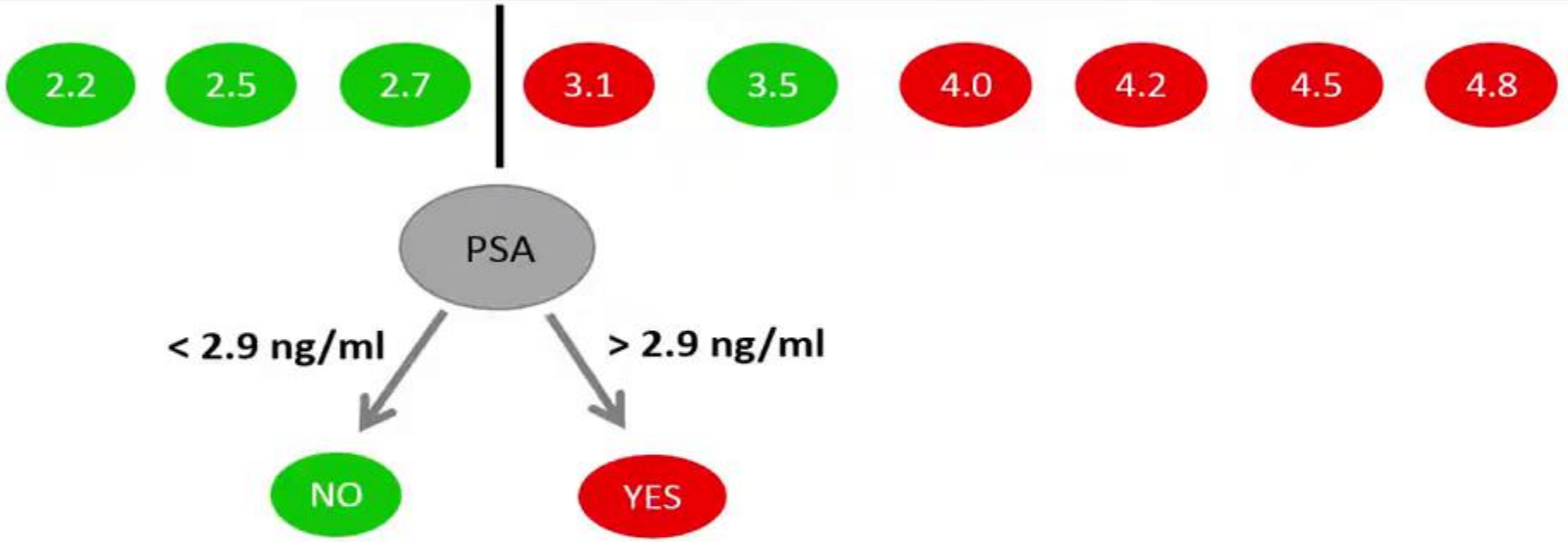
And these are five individuals that are known to have prostate cancer. Note that the PSA level is a bit higher in these patients compared to the healthy controls.

How to construct a decision tree



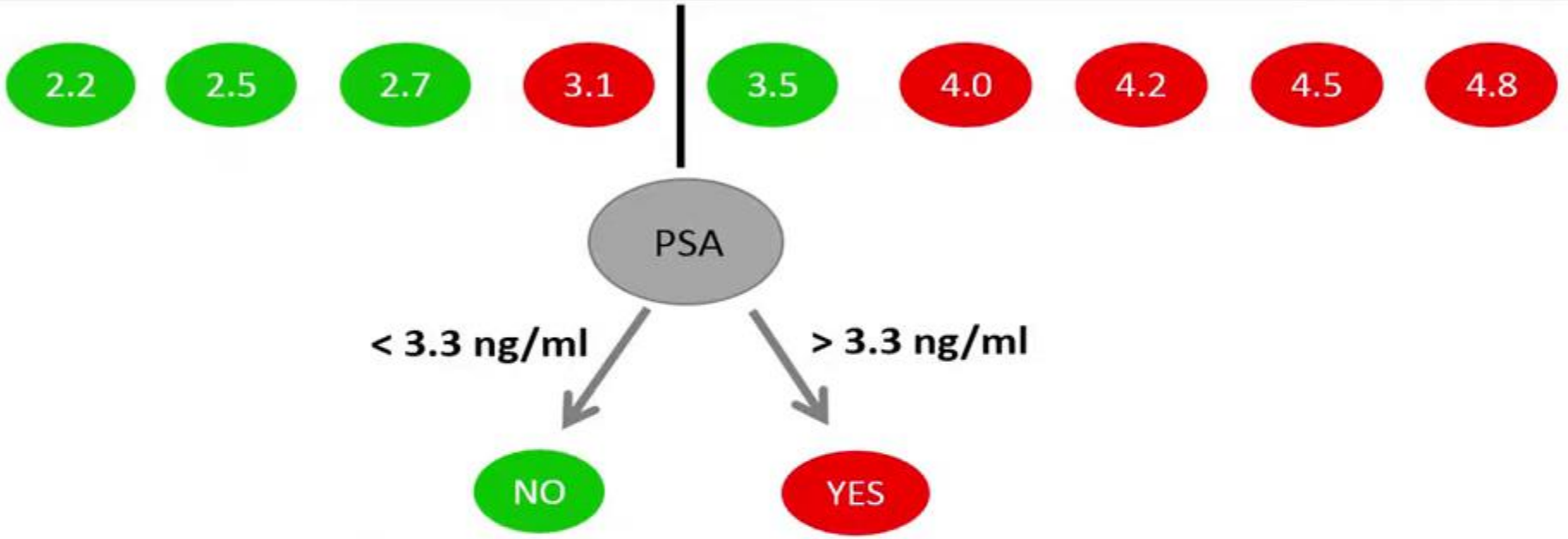
We first begin to sort the individuals based on their PSA concentrations. Next, we should find a good cutoff value that separates the two groups.

How to construct a decision tree



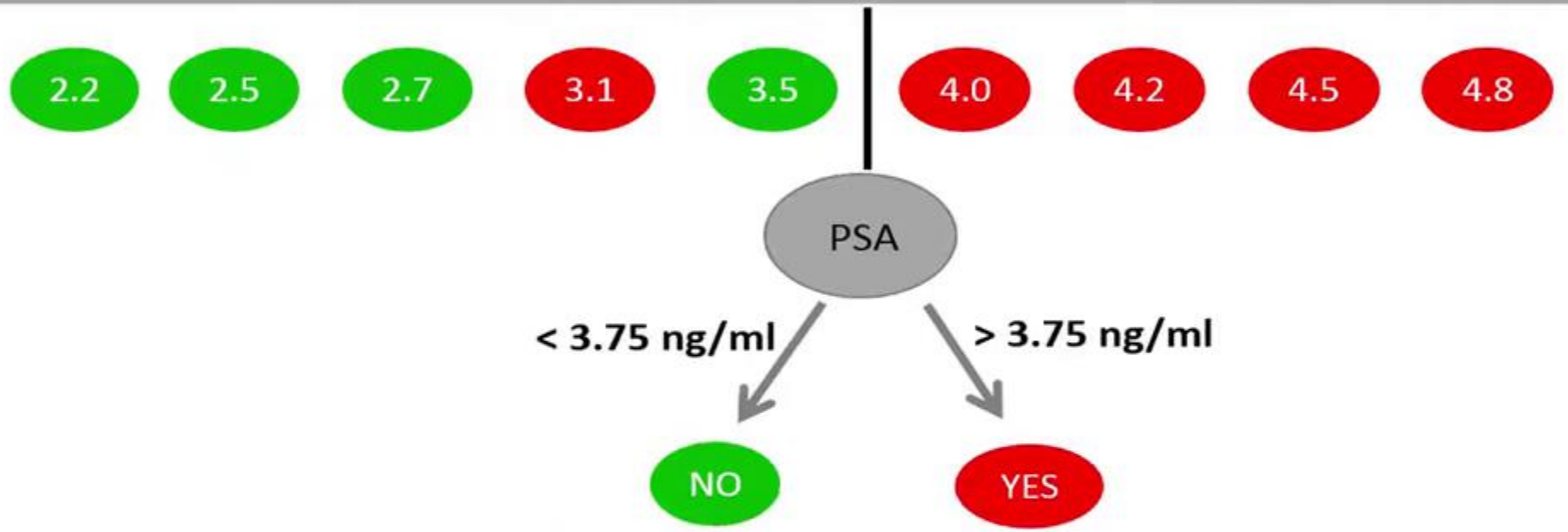
However, where should we set the cutoff to predict if someone has prostate cancer or not? Maybe a cutoff value at 2.9,

How to construct a decision tree



or at 3.3,

How to construct a decision tree



or at 3.75?

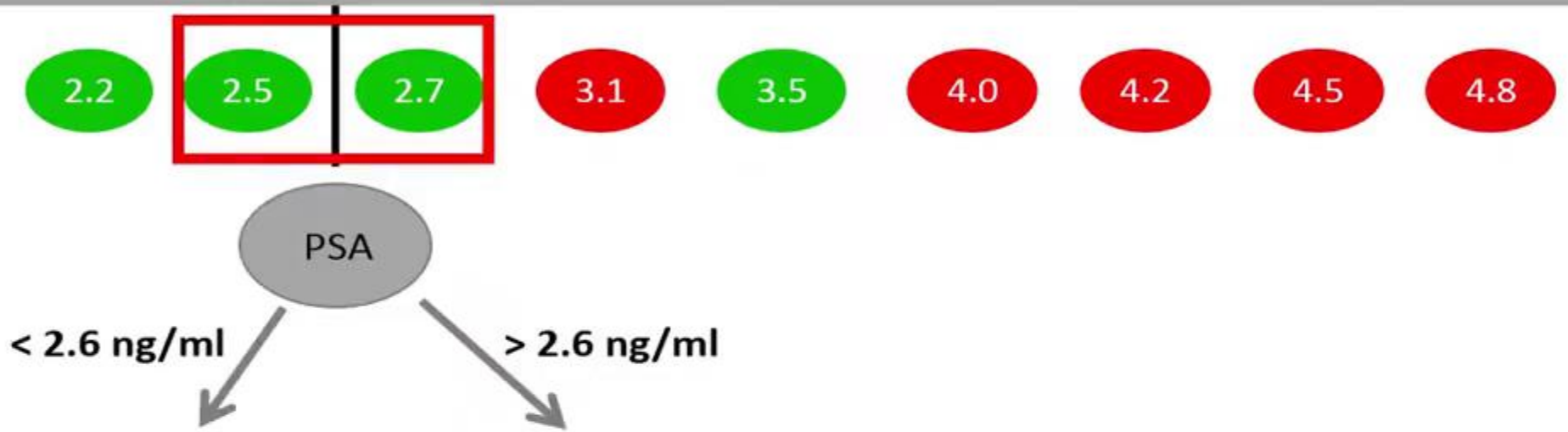
How to construct a decision tree



Gini index

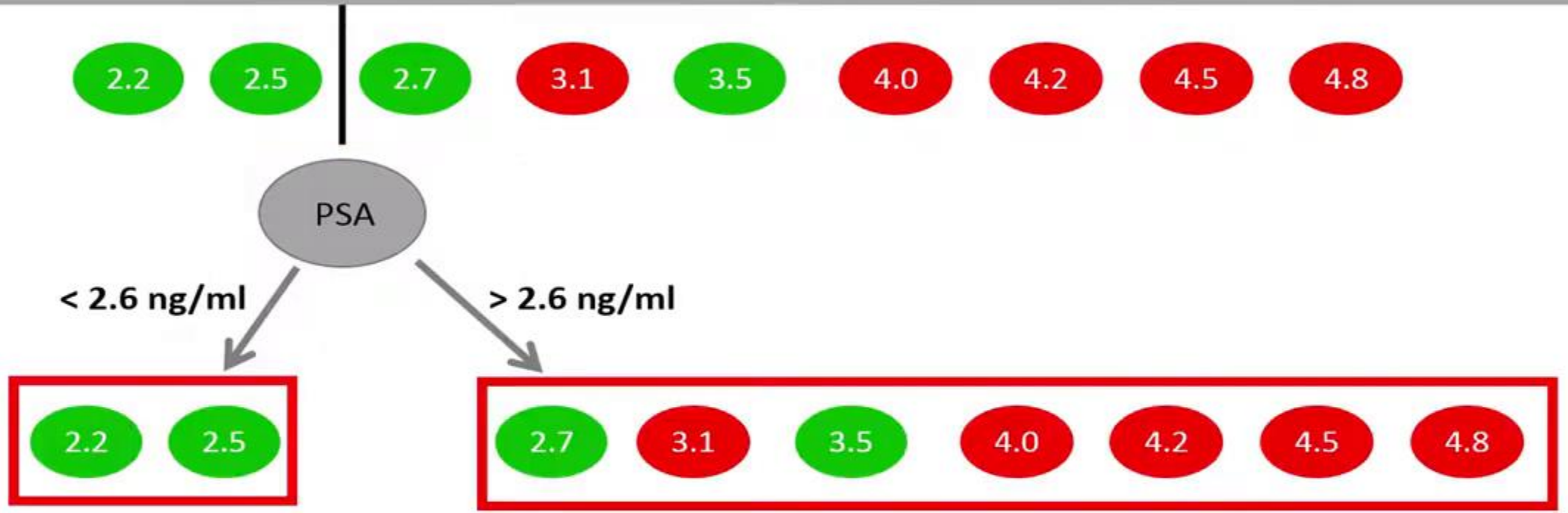
One way to determine the cutoff value is to use the so-called Gini index, which is a measure of impurity. Since we want as pure groups as possible, we should select a split that results in as low Gini index as possible.

How to construct a decision tree



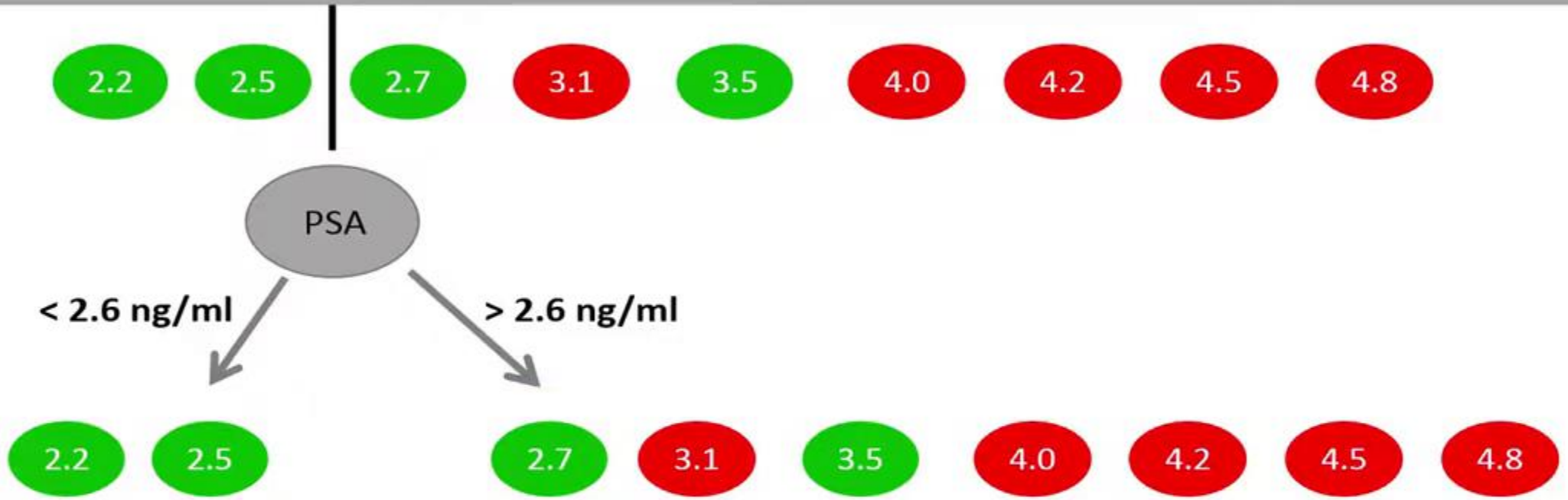
If we would split the data here, we should select a cutoff value of 2.6 because that is the mean value of the two closest data points for this split.

How to construct a decision tree



A cutoff of 2.6 will split the data into these two groups.

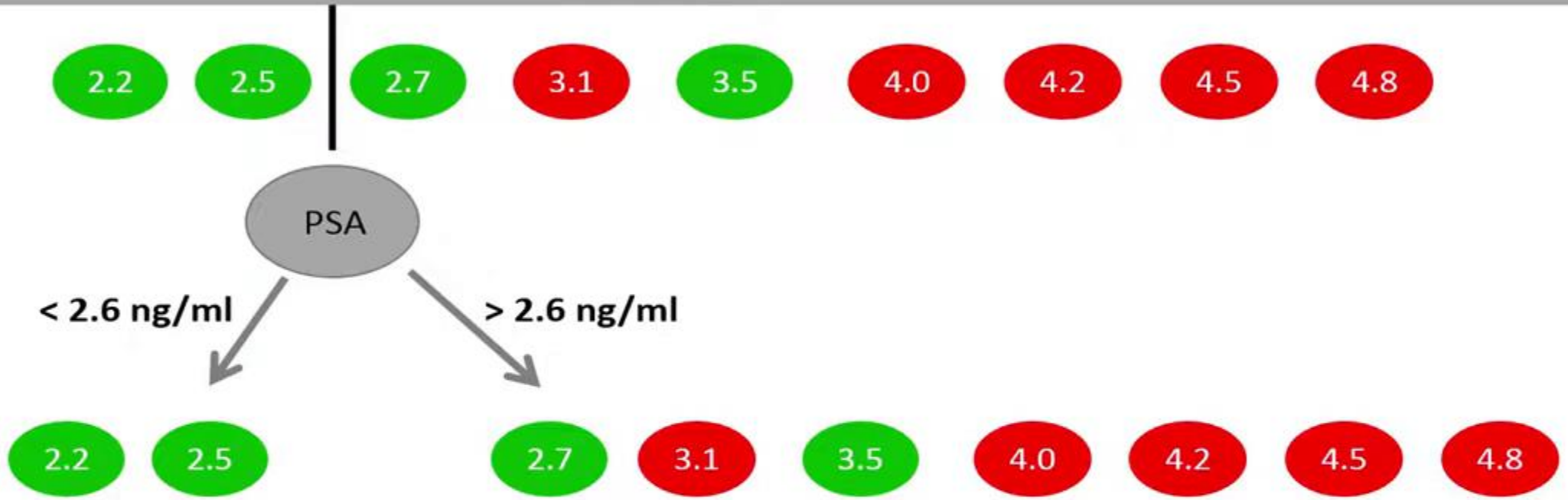
How to construct a decision tree



$$\text{Gini}(T) = 1 - \sum_{j=1}^{n} p_j^2$$

The Gini index is calculated like this,

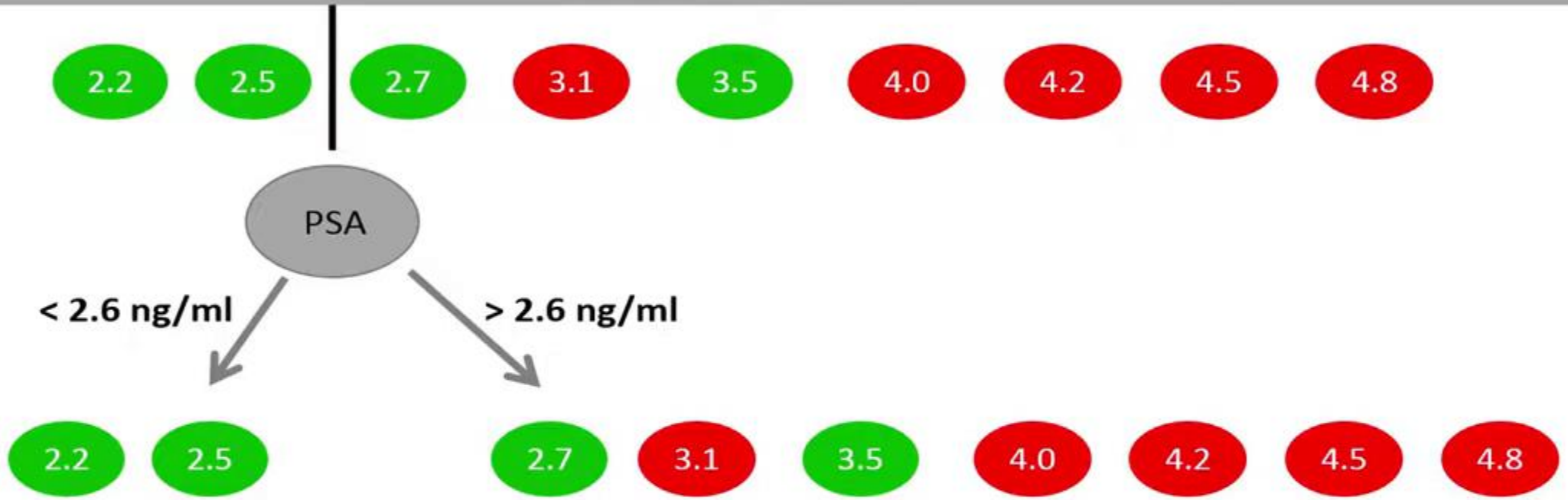
How to construct a decision tree



$$\text{Gini}(T) = 1 - \sum_{j=1}^{n} p_j^2$$

where p is the proportion of each class.

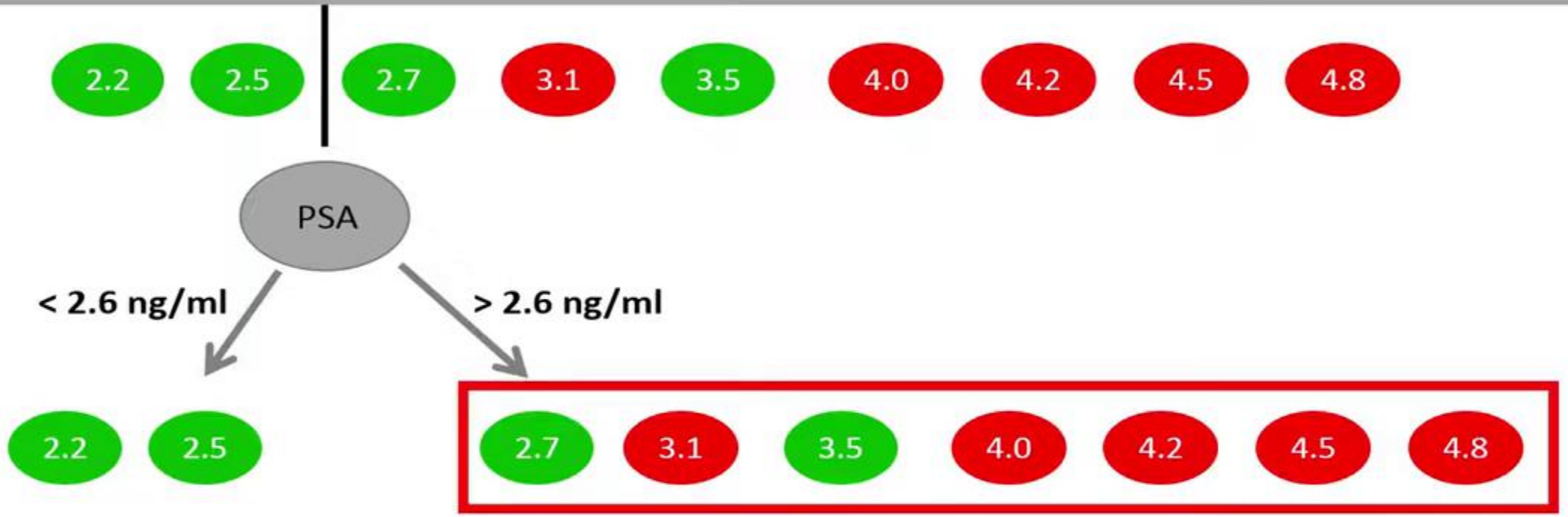
How to construct a decision tree



$$\text{Gini(T)} = 1 - \sum_{j=1}^{n} p_j^2$$

Since we have two classes in this example, prostate cancer and healthy, we should sum the squared proportion of these two classes.

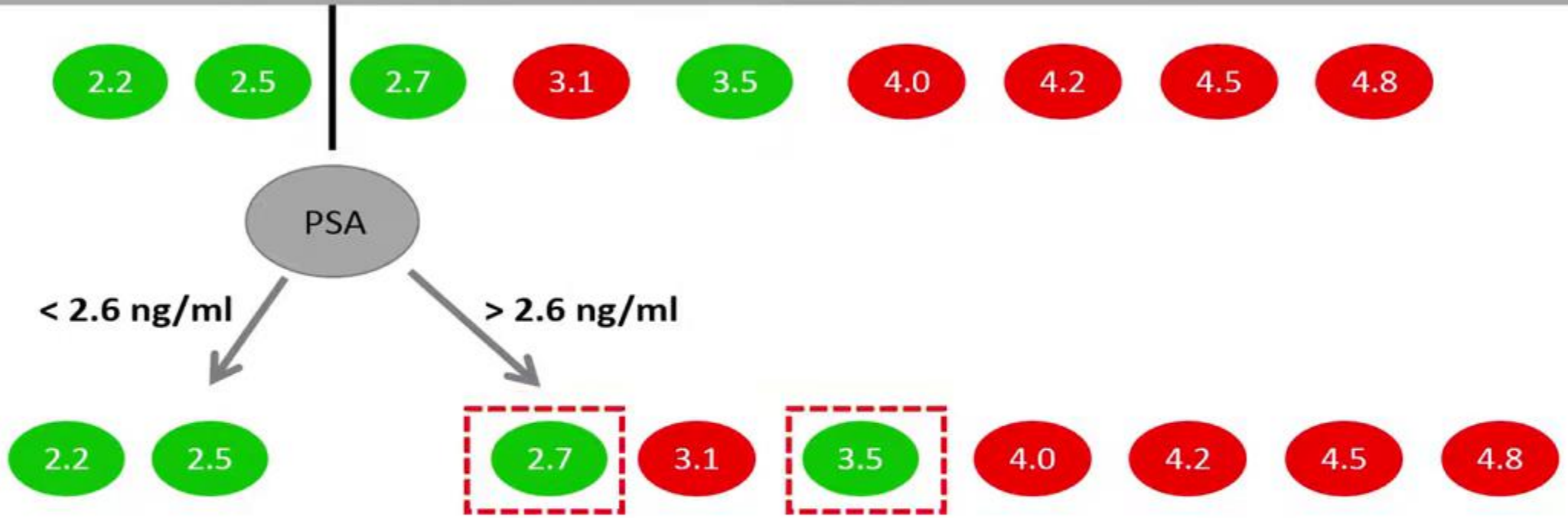
How to construct a decision tree



$$\text{Gini(T)} = 1 - \sum_{j=1}^{n} p_j^2$$

Let's calculate the Gini index of the following subgroup.

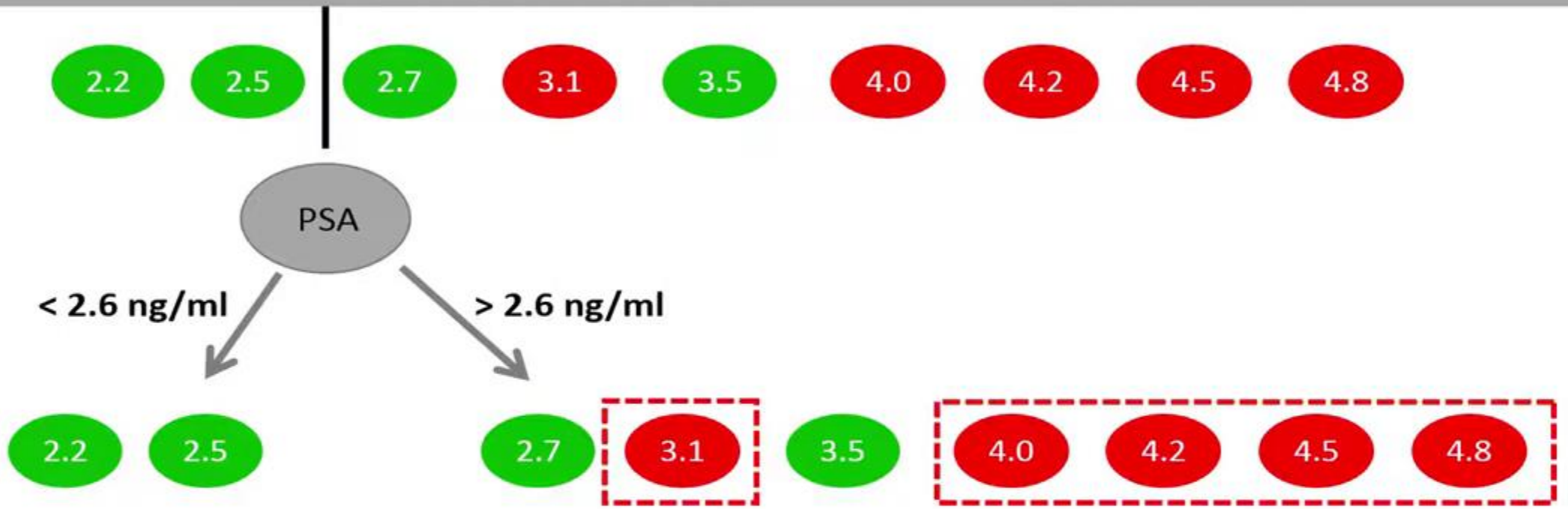
How to construct a decision tree



$$\text{Gini}(T_2) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((2/7)^2 + (5/7)^2 \right) = 0.41$$

Two out of the seven individuals in this subgroup are healthy individuals,

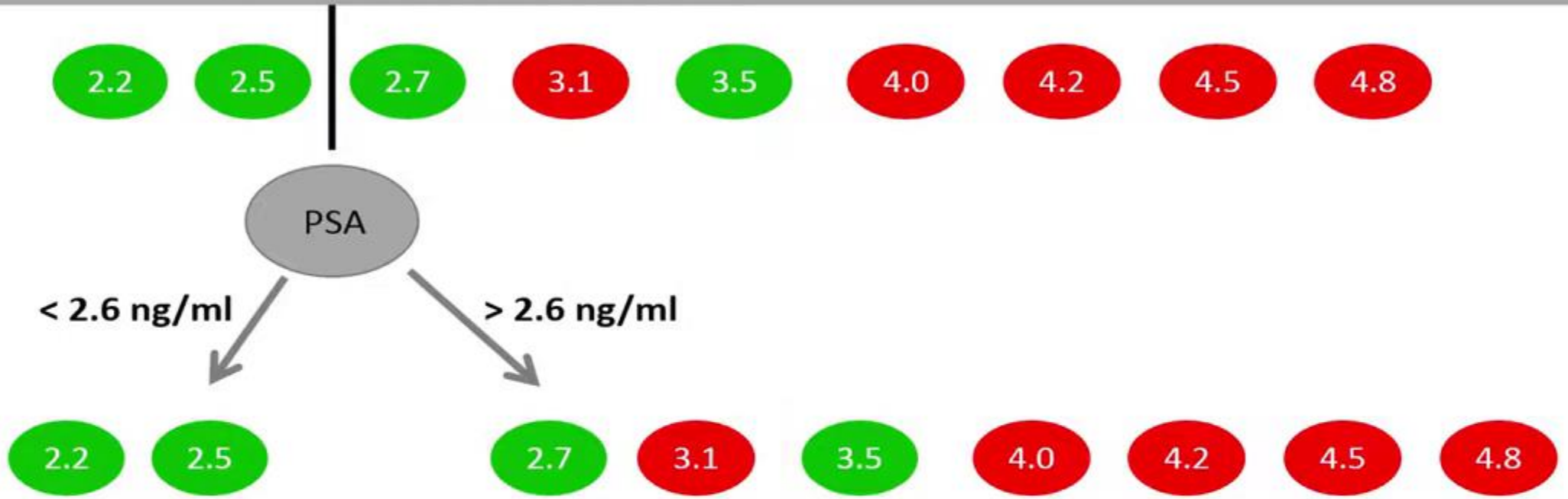
How to construct a decision tree



$$\text{Gini}(T_2) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((2/7)^2 + (5/7)^2 \right) = 0.41$$

whereas five out of seven individuals are patients with prostate cancer.

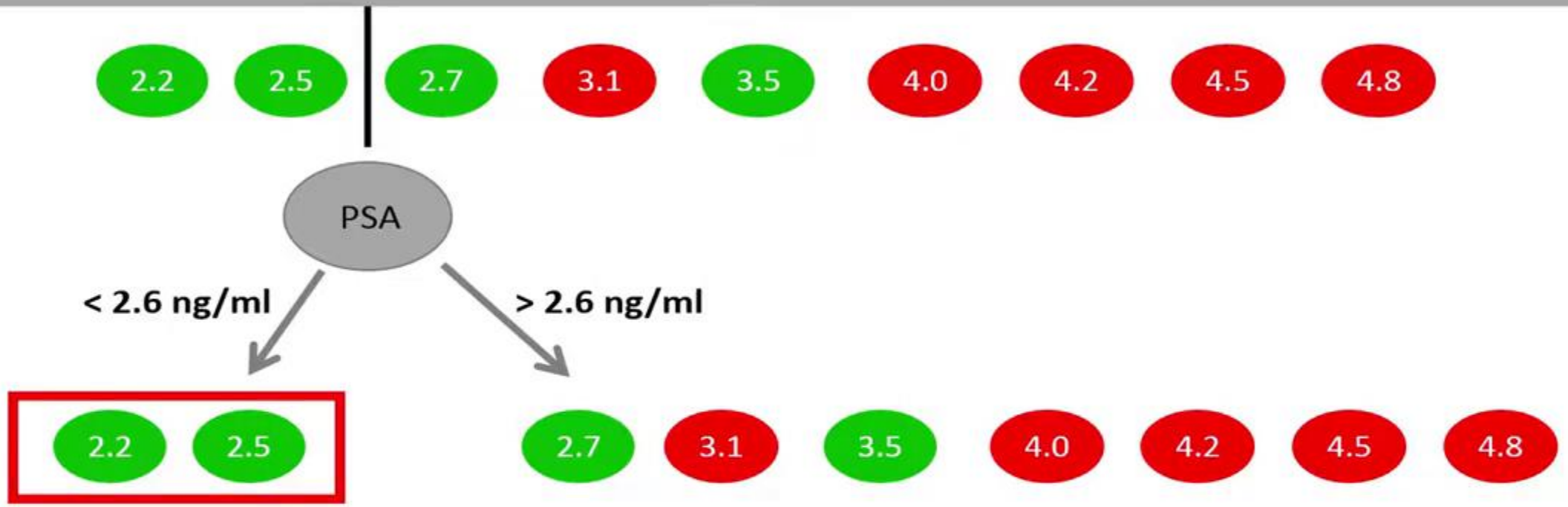
How to construct a decision tree



$$\text{Gini}(T_2) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((2/7)^2 + (5/7)^2\right) = 0.41$$

This gives us a Gini index of about 0.41.

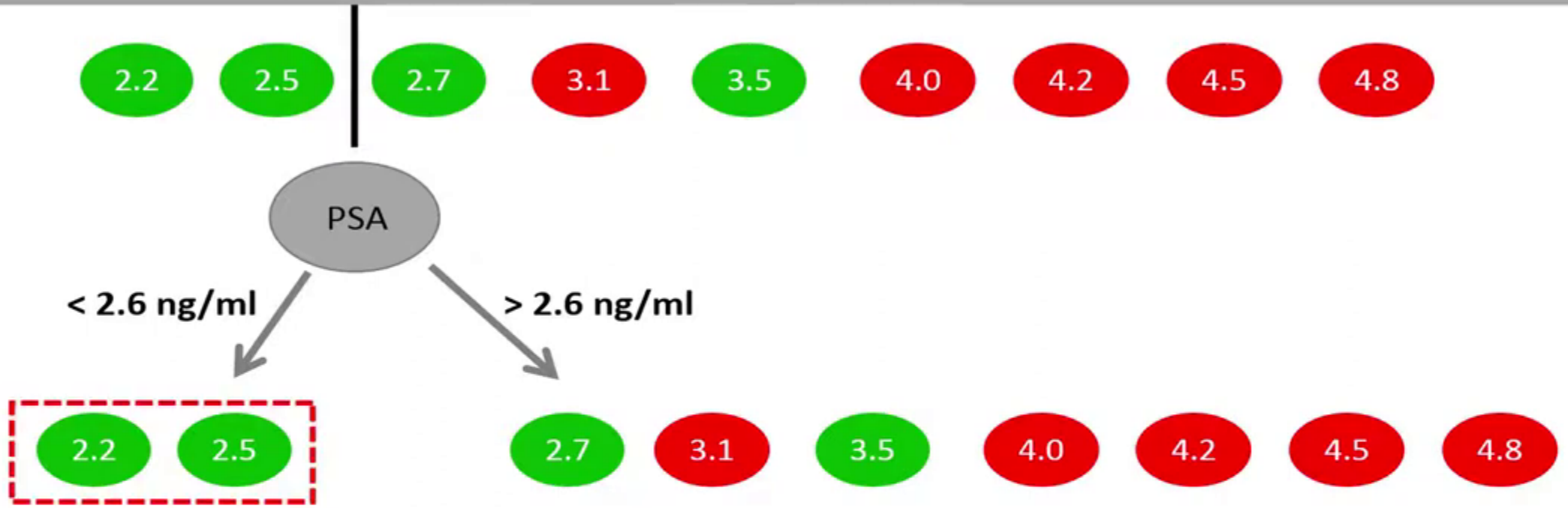
How to construct a decision tree



$$\text{Gini}(T_1) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((2/2)^2 + (0/2)^2\right) = 0$$

If we do the corresponding calculations for this subgroup,

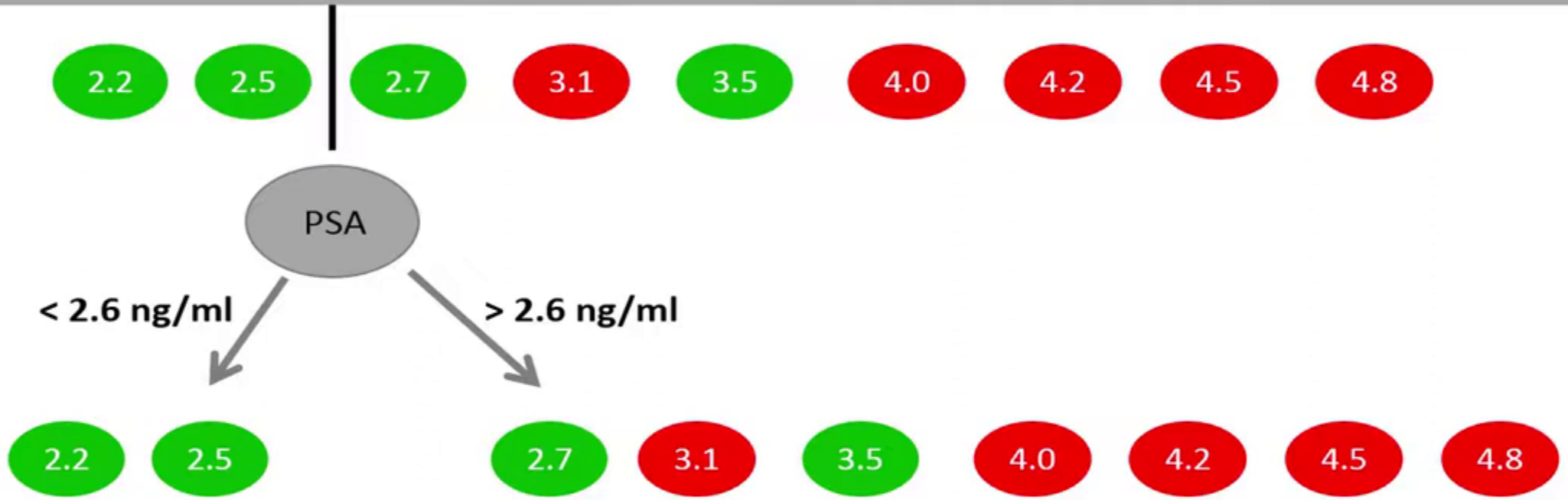
How to construct a decision tree



$$\text{Gini}(T_1) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((2/2)^2 + (0/2)^2\right) = 0$$

we see that the Gini index is equal to zero, which means that we have 100% purity or 0% impurity.

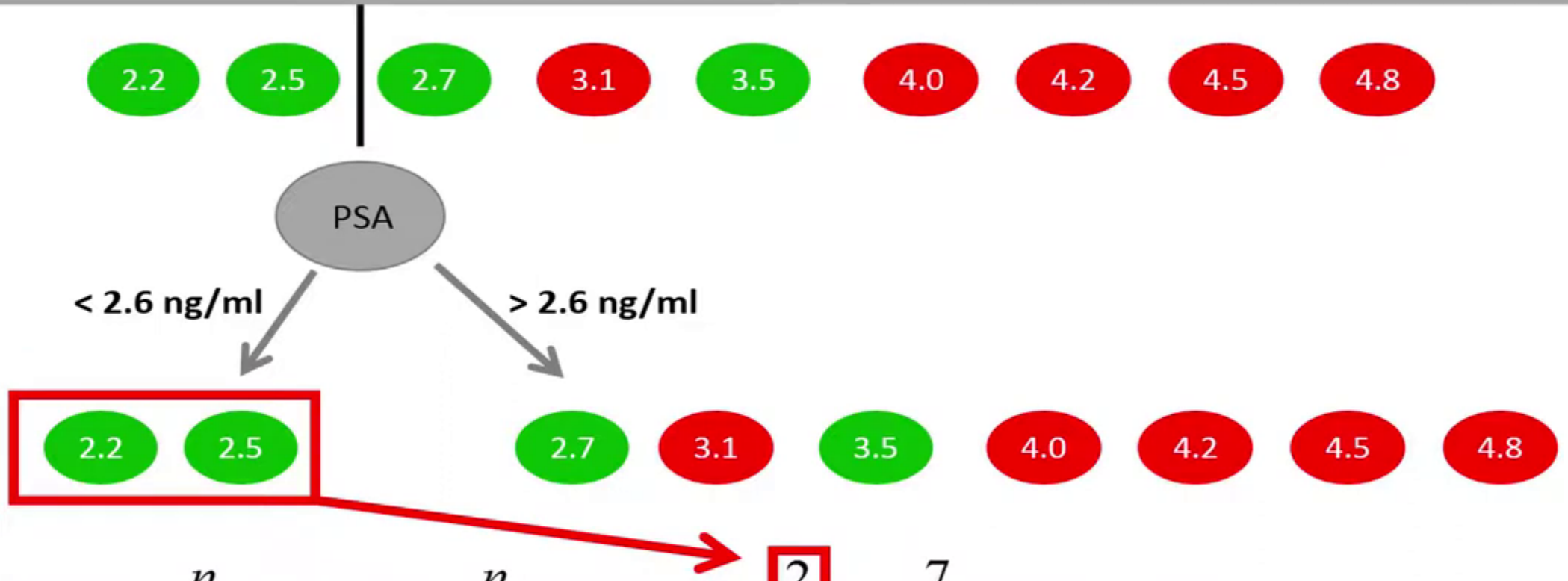
How to construct a decision tree



$$\text{Gini}(T) = \frac{n_1}{N}\text{Gini}(T_1) + \frac{n_2}{N}\text{Gini}(T_2) = \frac{2}{9}\cdot 0 + \frac{7}{9}\cdot 0.41 = 0.32$$

Then we calculate the weighted average of these two Gini indexes.

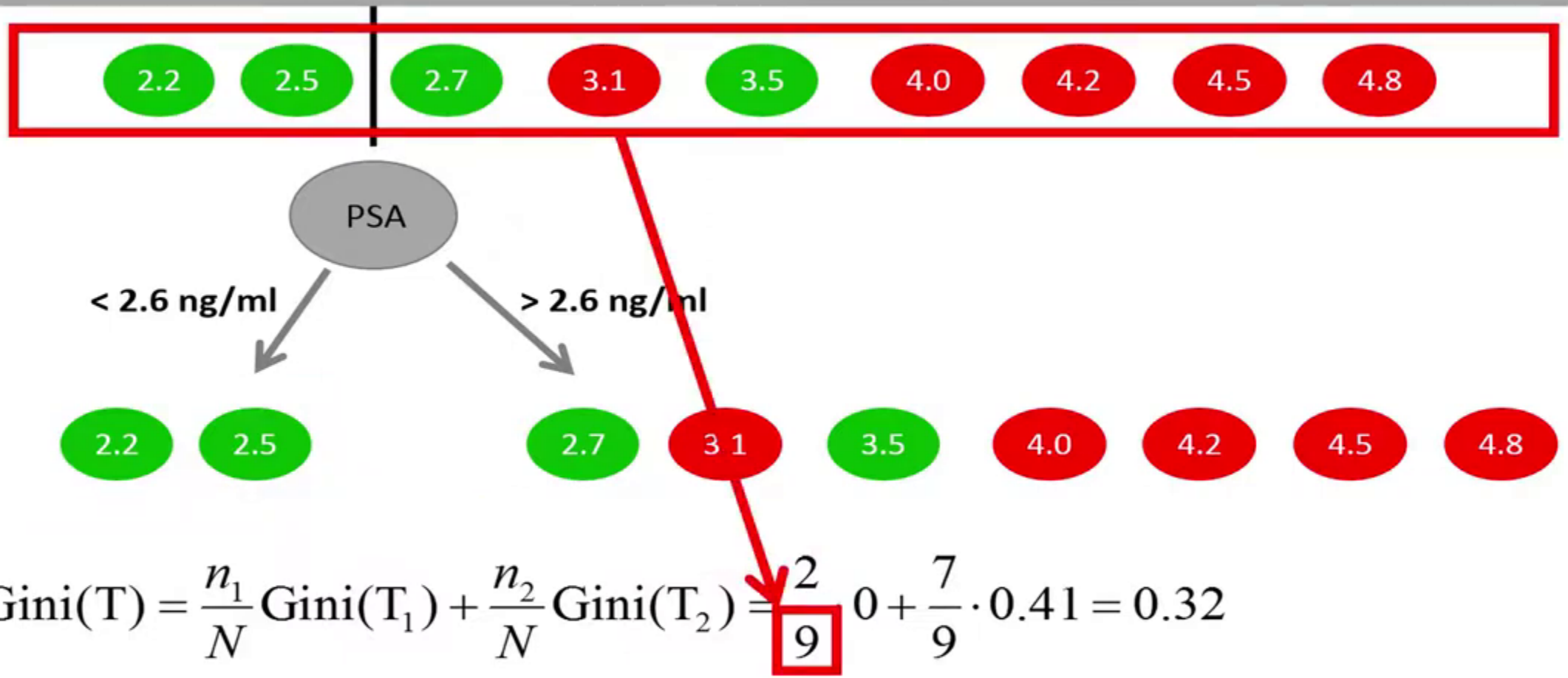
How to construct a decision tree



$$\text{Gini(T)} = \frac{n_1}{N}\text{Gini}(\text{T}_1) + \frac{n_2}{N}\text{Gini}(\text{T}_2) = \frac{2}{9} \cdot 0 + \frac{7}{9} \cdot 0.41 = 0.32$$

Since this subgroup consists of two data points, we multiply the Gini index by two

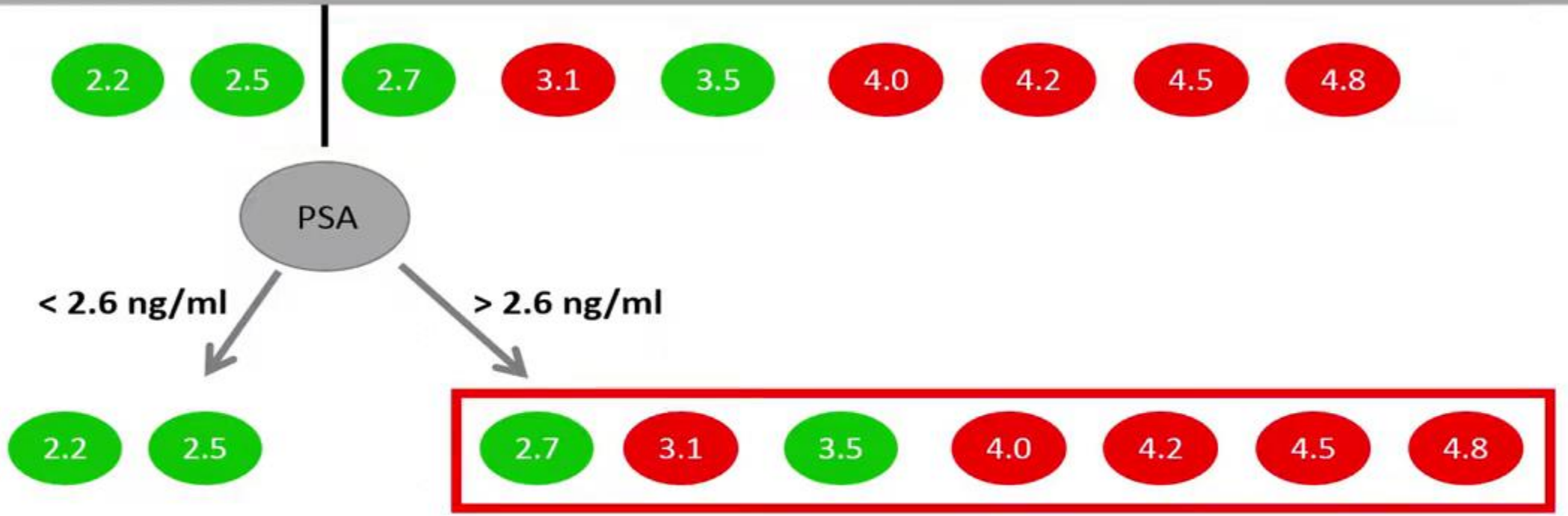
How to construct a decision tree



$$\text{Gini(T)} = \frac{n_1}{N}\text{Gini}(\text{T}_1) + \frac{n_2}{N}\text{Gini}(\text{T}_2) = \frac{2}{9}\,0 + \frac{7}{9}\cdot 0.41 = 0.32$$

over nine, since we have nine data points in total for this split.

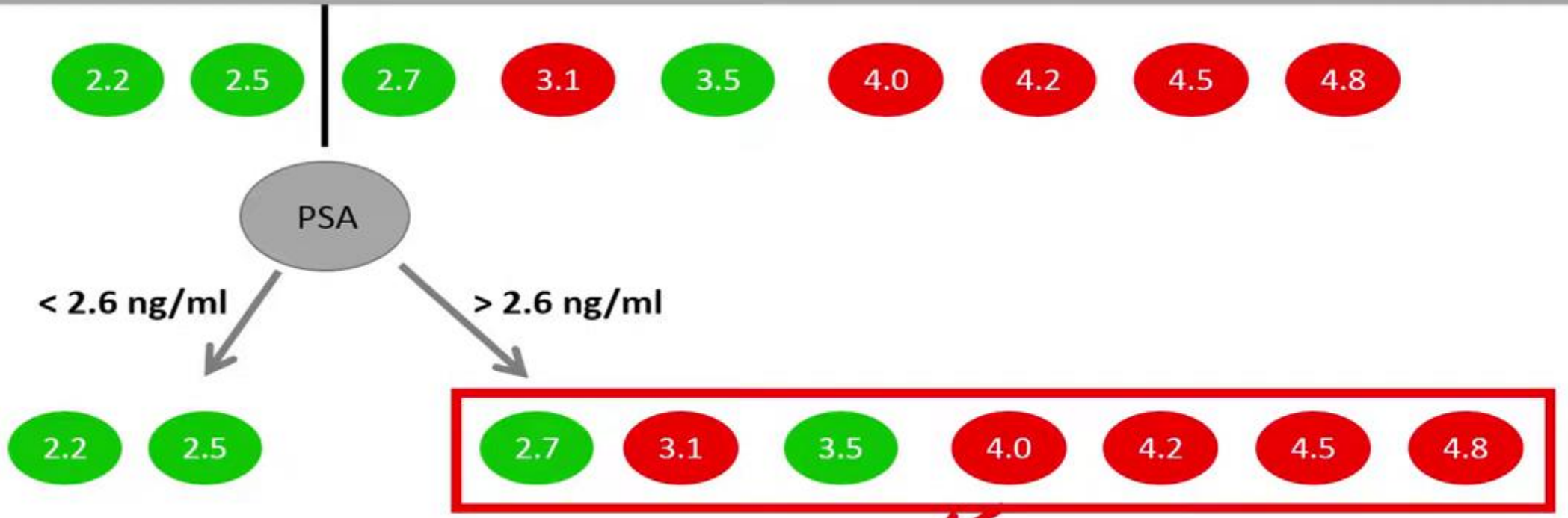
How to construct a decision tree



$$\text{Gini(T)} = \frac{n_1}{N}\text{Gini}(\text{T}_1) + \frac{n_2}{N}\text{Gini}(\text{T}_2) = \frac{2}{9}\cdot 0 + \frac{7}{9}\cdot 0.41 = 0.32$$

And since this subgroup consists of seven data points,

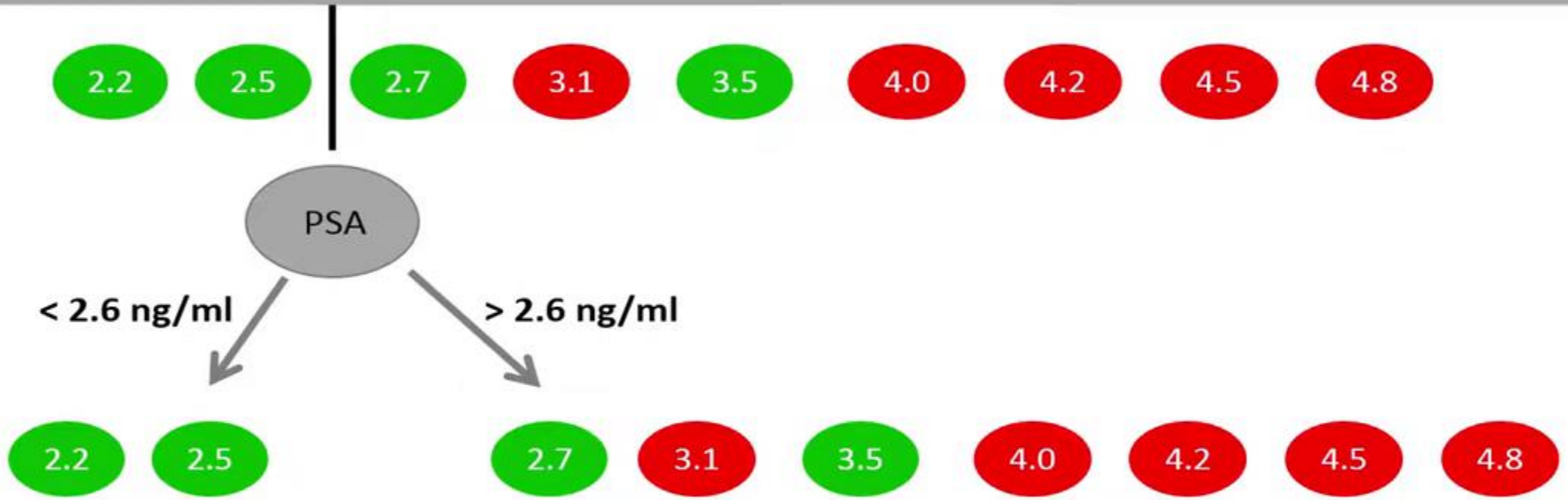
How to construct a decision tree



$$\text{Gini(T)} = \frac{n_1}{N}\text{Gini}(\text{T}_1) + \frac{n_2}{N}\text{Gini}(\text{T}_2) = \frac{2}{9}\cdot 0 + \frac{7}{9}\cdot 0.41 = 0.32$$

we multiply the Gini index by seven over nine.

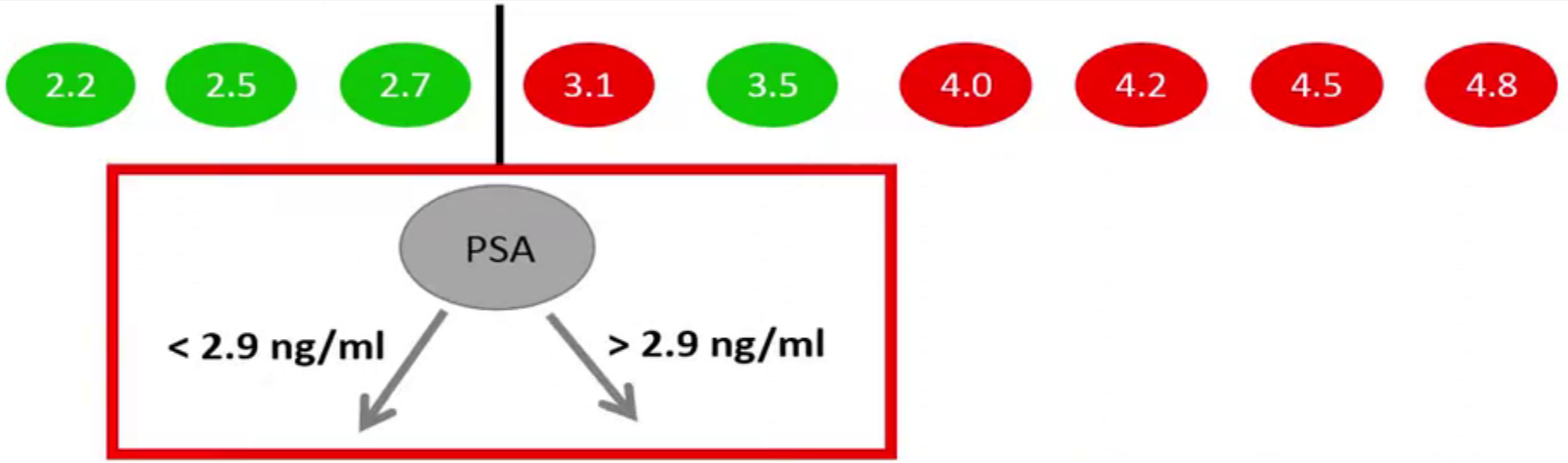
How to construct a decision tree



$$\text{Gini(T)} = \frac{n_1}{N}\text{Gini}(\text{T}_1) + \frac{n_2}{N}\text{Gini}(\text{T}_2) = \frac{2}{9}\cdot 0 + \frac{7}{9}\cdot 0.41 = \boxed{0.32}$$

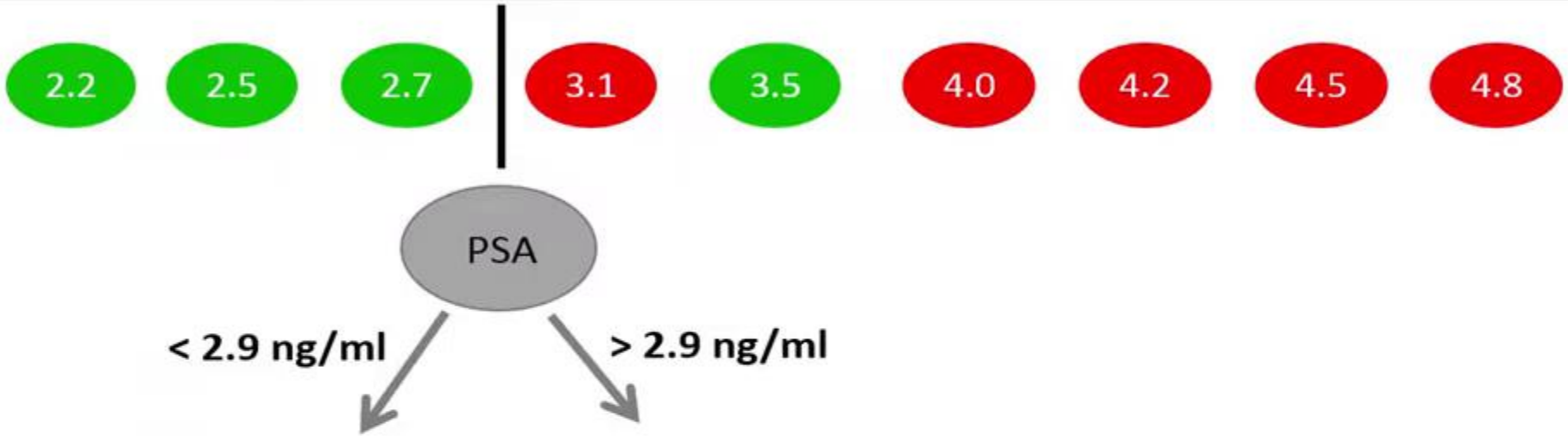
By using a cutoff value of 2.6, the Gini index is equal to about 0.32.

How to construct a decision tree



Now, let's use a cutoff value of 2.9.

How to construct a decision tree



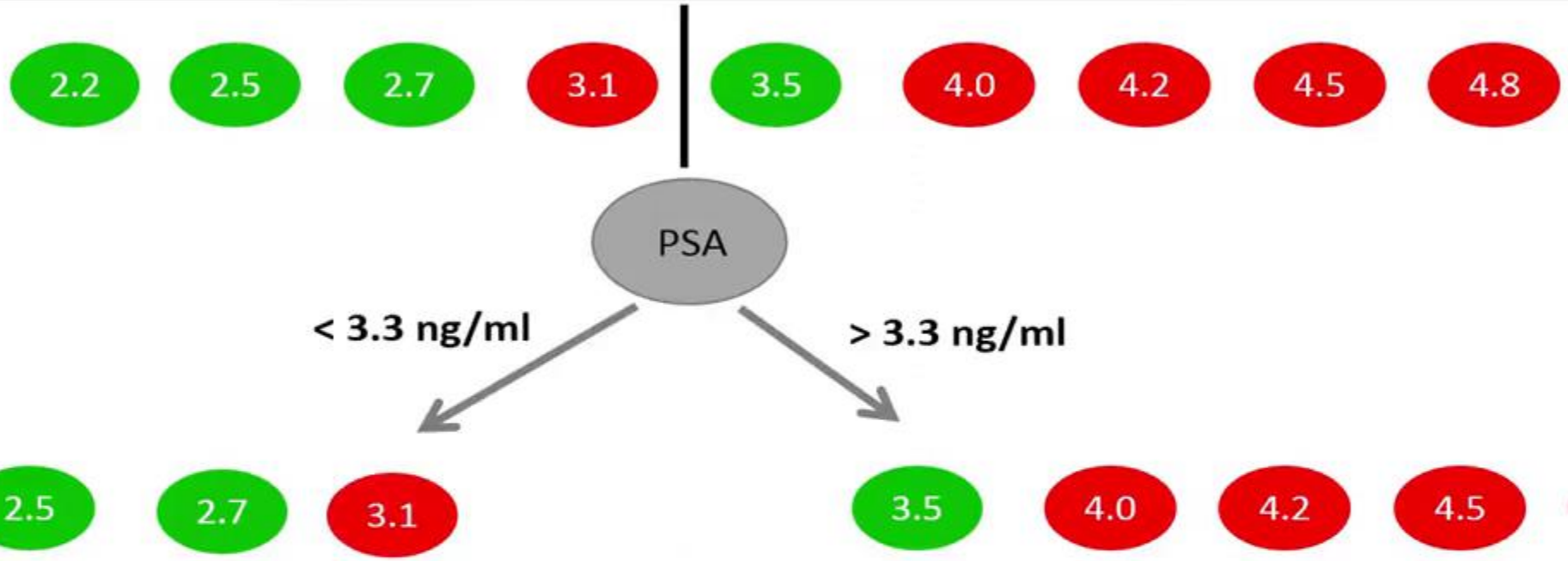
$$\text{Gini}(T_1) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((3/3)^2 + (0/3)^2\right) = 0$$

$$\text{Gini}(T_2) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((1/6)^2 + (5/6)^2\right) = 0.28$$

$$\text{Gini}(T) = \frac{n_1}{N}\text{Gini}(T_1) + \frac{n_2}{N}\text{Gini}(T_2) = \frac{3}{9}\cdot 0 + \frac{6}{9}\cdot 0.28 = \boxed{0.19}$$

This results in a higher purity because the Gini index has been reduced to 0.19.

How to construct a decision tree



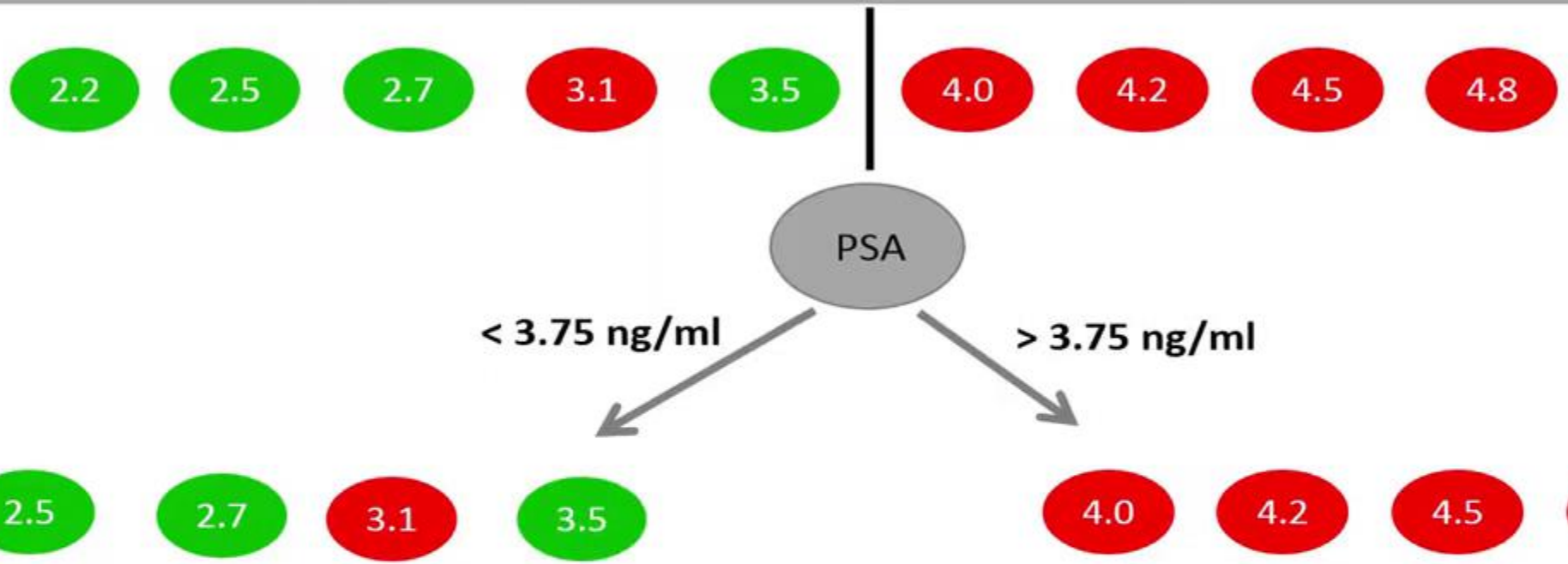
$$\text{Gini}(T_1) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((3/4)^2 + (1/4)^2\right) = 0.375$$

$$\text{Gini}(T_2) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((1/5)^2 + (4/5)^2\right) = 0.32$$

$$\text{Gini}(T) = \frac{n_1}{N}\text{Gini}(T_1) + \frac{n_2}{N}\text{Gini}(T_2) = \frac{4}{9}\cdot 0.375 + \frac{5}{9}\cdot 0.32 = 0.34$$

A cutoff of 3.3 results in a Gini index of 0.34,

How to construct a decision tree



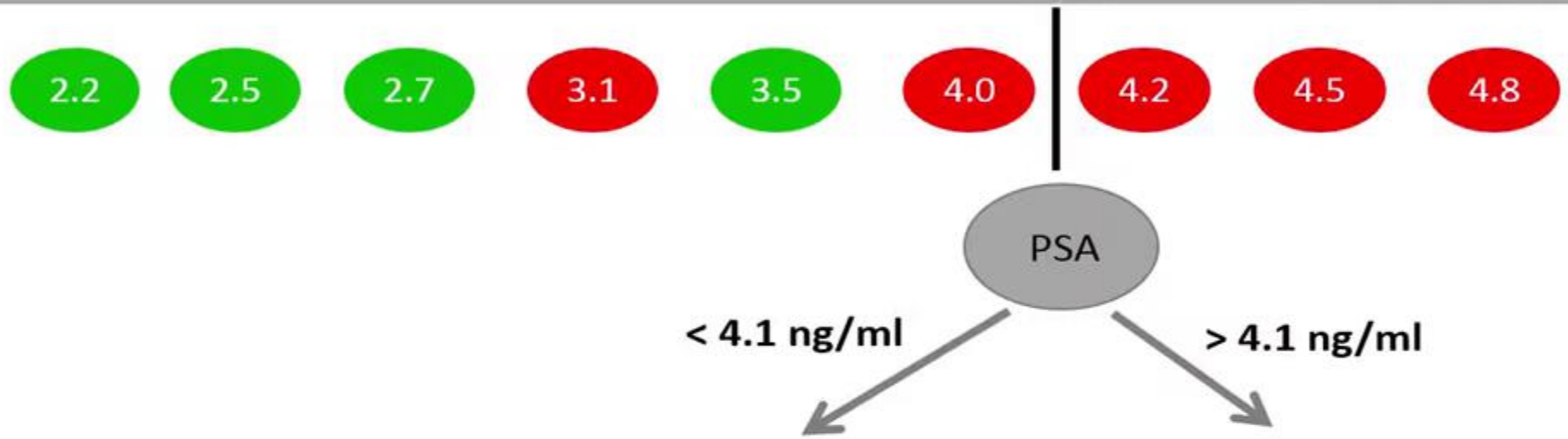
$$\text{Gini}(T_1) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((4/5)^2 + (1/5)^2\right) = 0.32$$

$$\text{Gini}(T_2) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((0/4)^2 + (4/4)^2\right) = 0$$

$$\text{Gini}(T) = \frac{n_1}{N}\text{Gini}(T_1) + \frac{n_2}{N}\text{Gini}(T_2) = \frac{5}{9}\cdot 0.32 + \frac{4}{9}\cdot 0 = \boxed{0.18}$$

and a cutoff value of 3.75 results in a Gini index of 0.18,

How to construct a decision tree







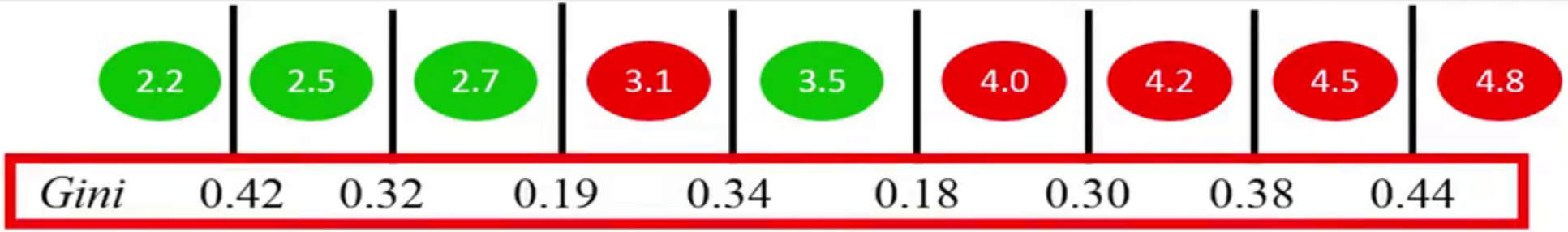
$$\text{Gini}(T_1) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((4/6)^2 + (2/6)^2\right) = 0.44$$

$$\text{Gini}(T_2) = 1 - \sum_{j=1}^{n} p_j^2 = 1 - \left((0/3)^2 + (3/3)^2\right) = 0$$

$$\text{Gini}(T) = \frac{n_1}{N}\text{Gini}(T_1) + \frac{n_2}{N}\text{Gini}(T_2) = \frac{6}{9} \cdot 0.44 + \frac{3}{9} \cdot 0 = \boxed{0.30}$$

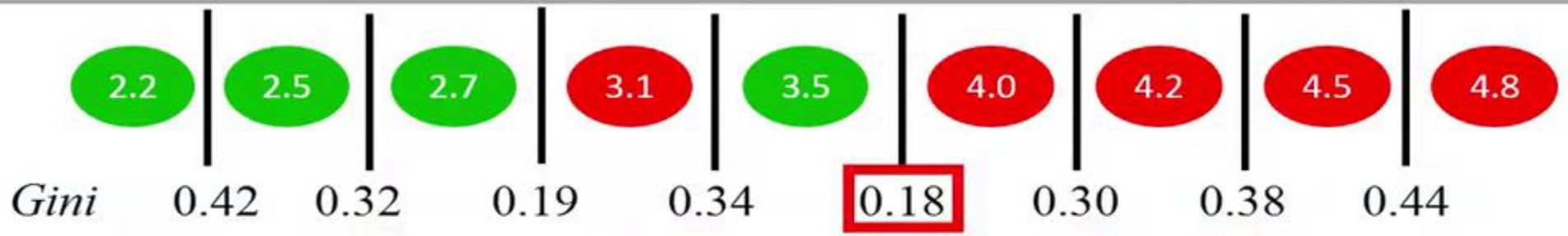
and so forth.

How to construct a decision tree



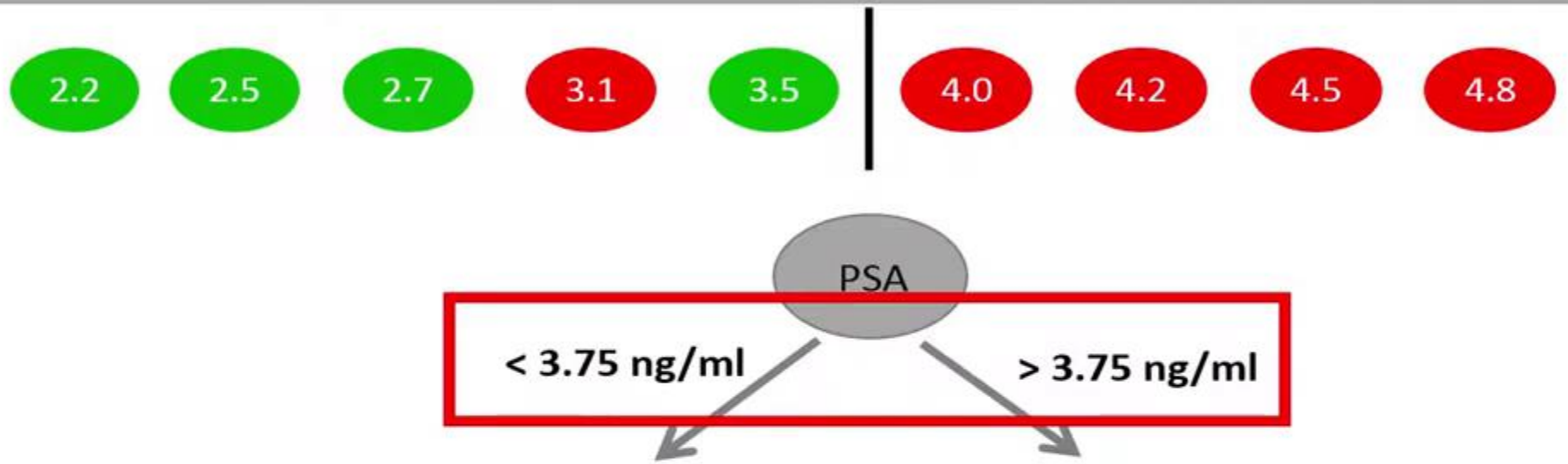
After we have calculated the Gini index for all possible splits,

How to construct a decision tree



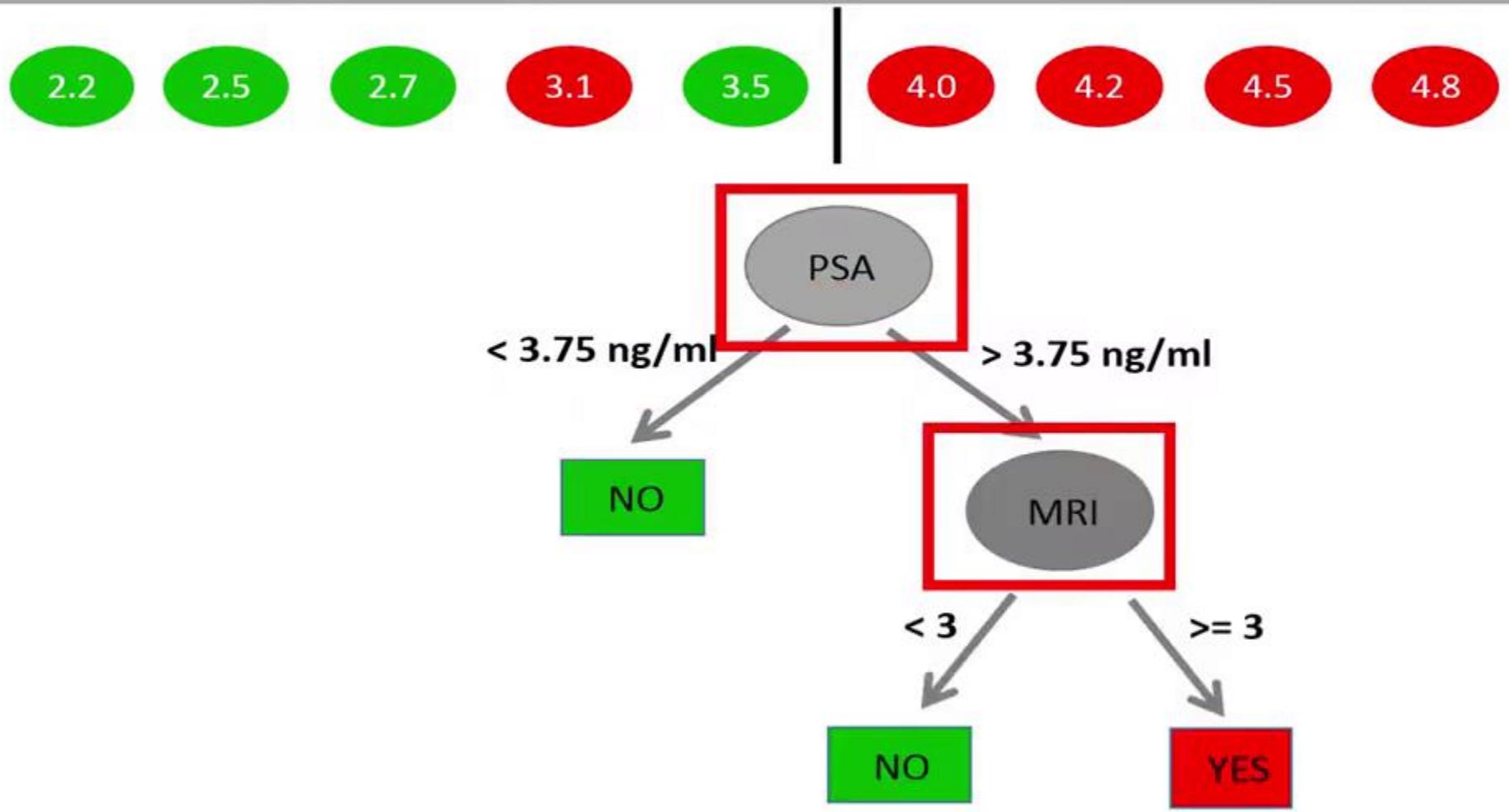
we select the split that results in the lowest Gini index, which is the one that generates the purest groups.

How to construct a decision tree



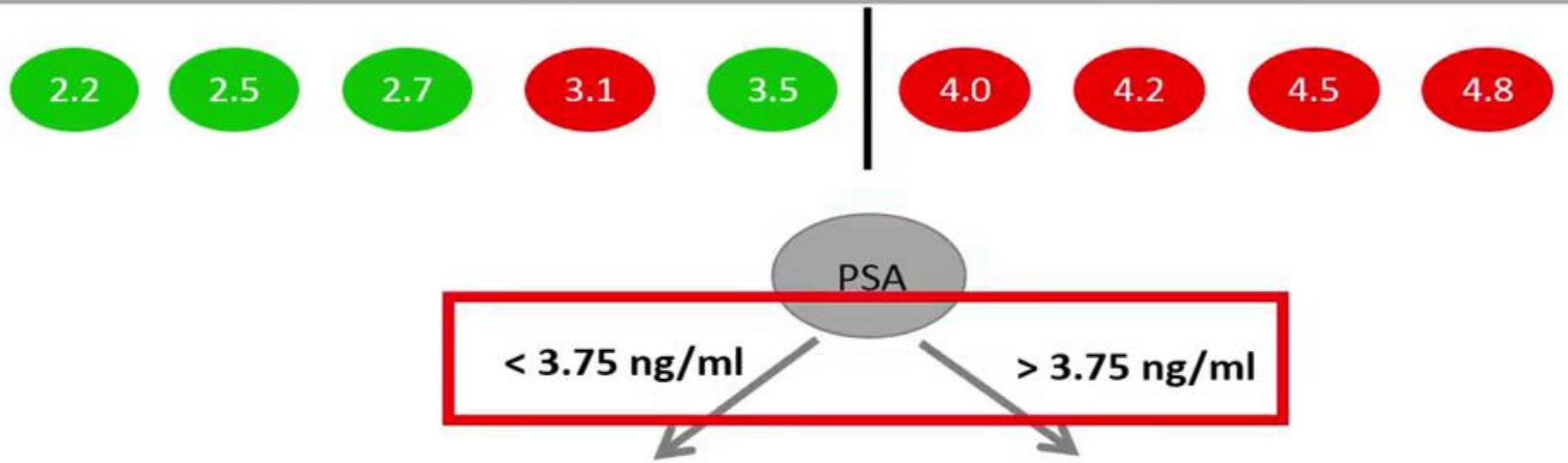
This is the reason why we should select a cutoff value of 3.75 for this data set.

How to construct a decision tree



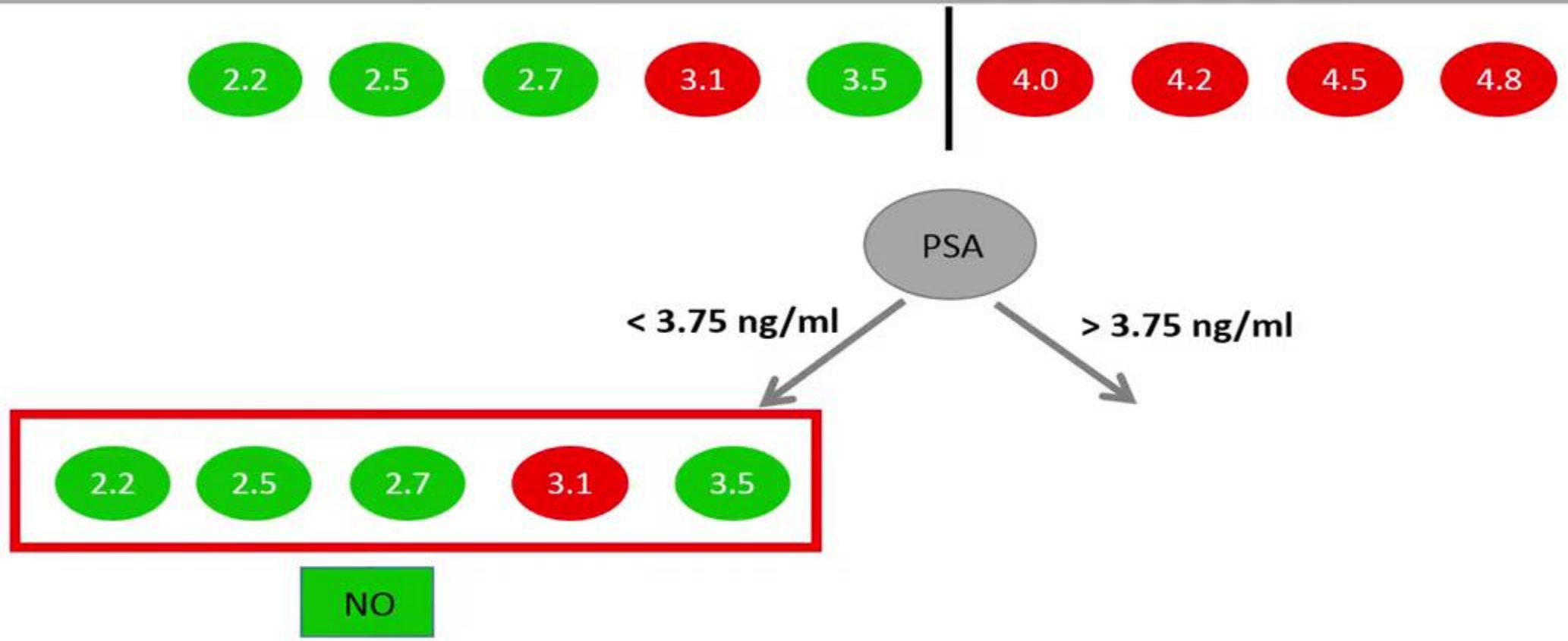
If we have two or more variables,

Accuracy of the classification



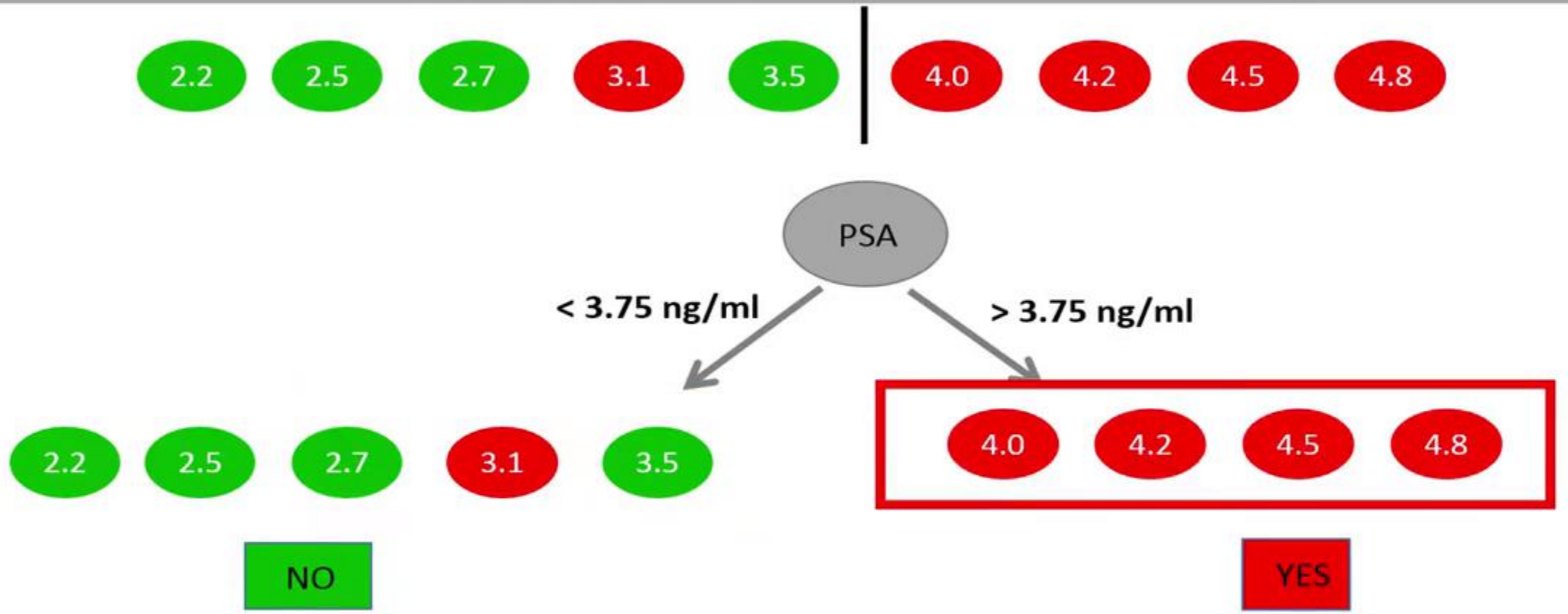
If we would use the following cutoff value,

Accuracy of the classification



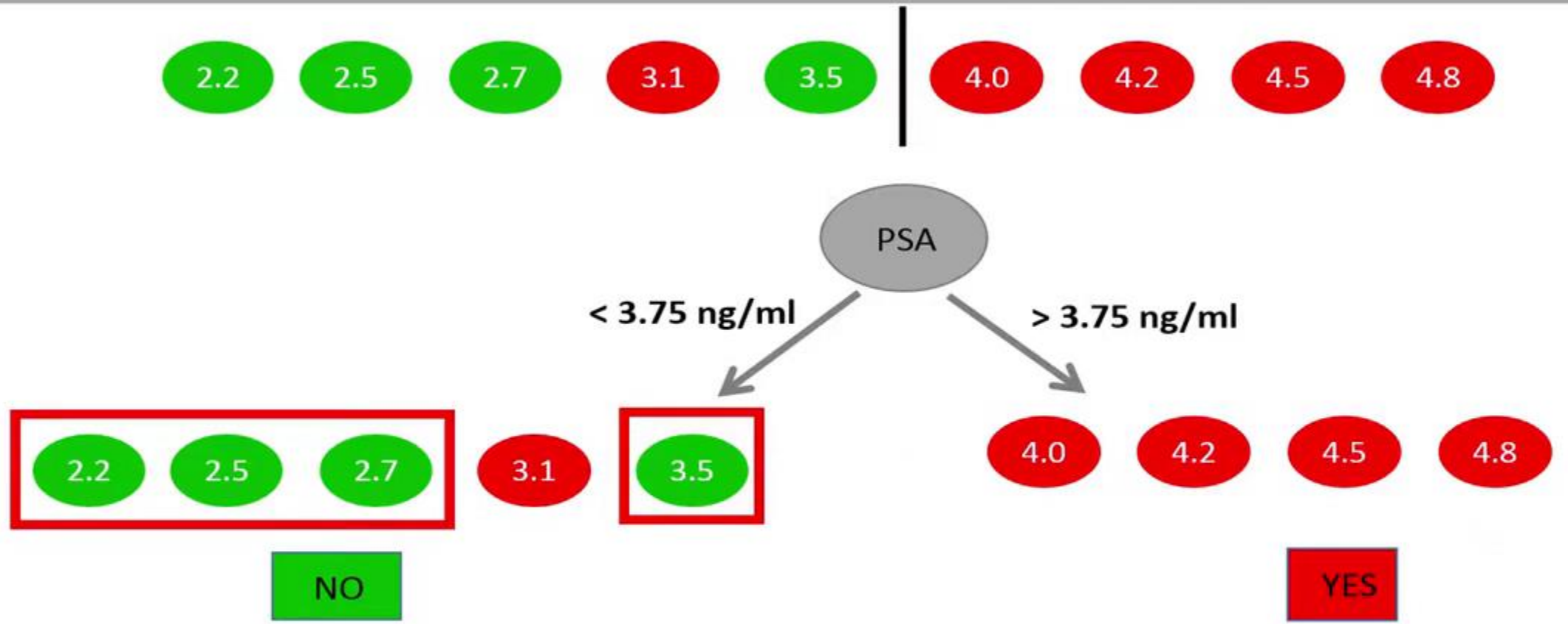
we would predict that these individuals do not have prostate cancer, because the majority of the observations in this leaf node are healthy individuals,

Accuracy of the classification



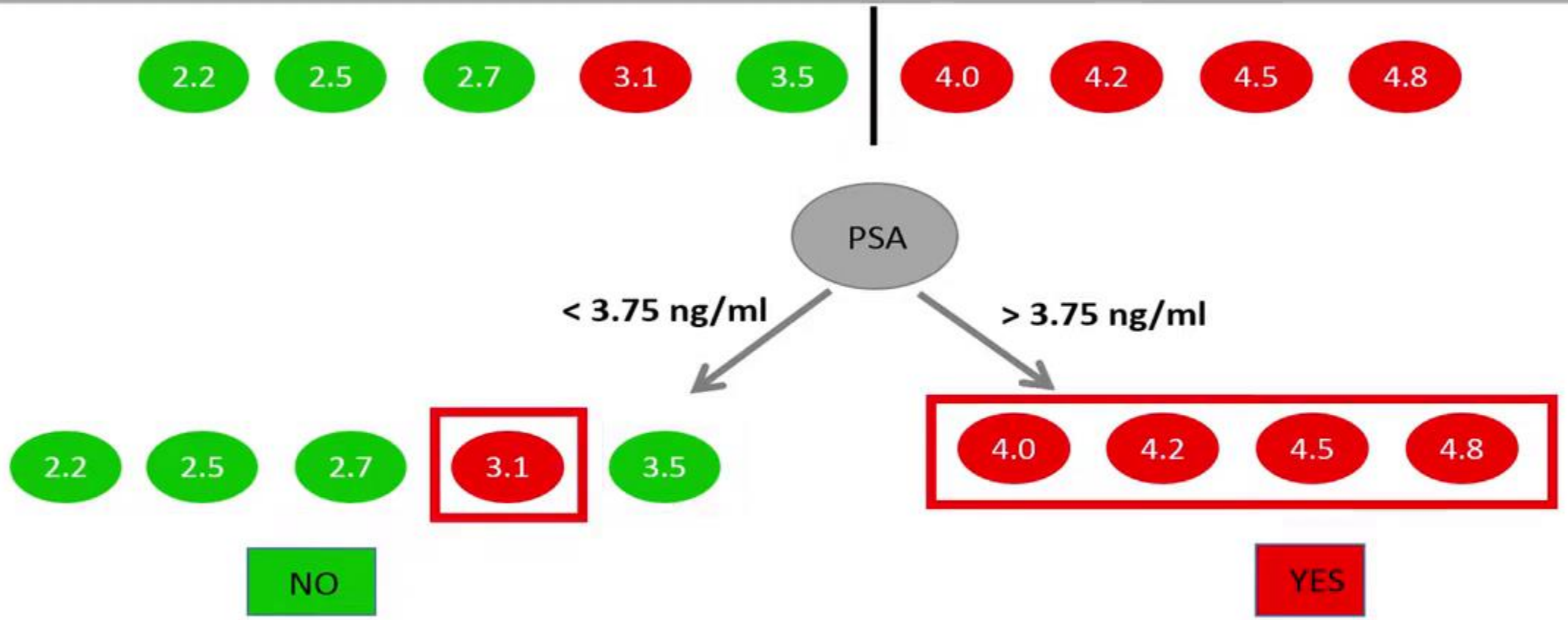
whereas these individuals are predicted to have prostate cancer.

Accuracy of the classification



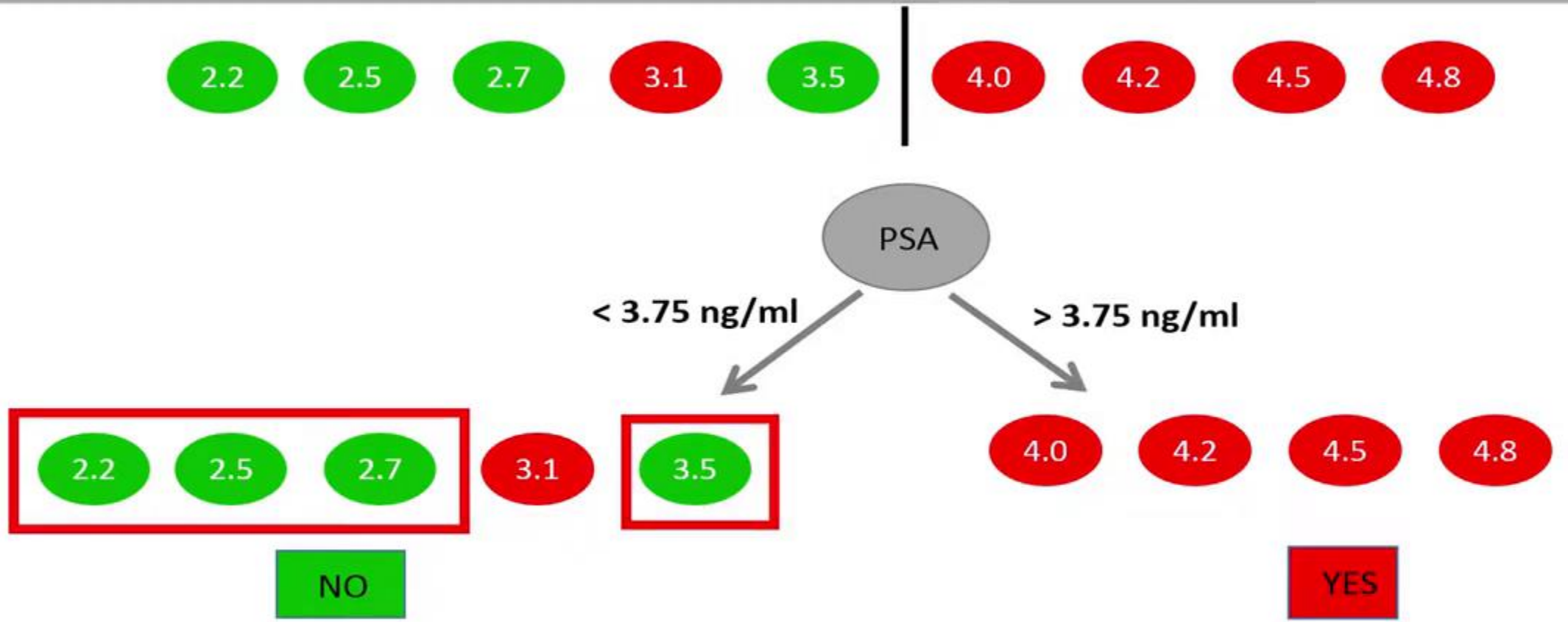
Since we know, based on the training data, that these individuals are actually healthy,

Accuracy of the classification



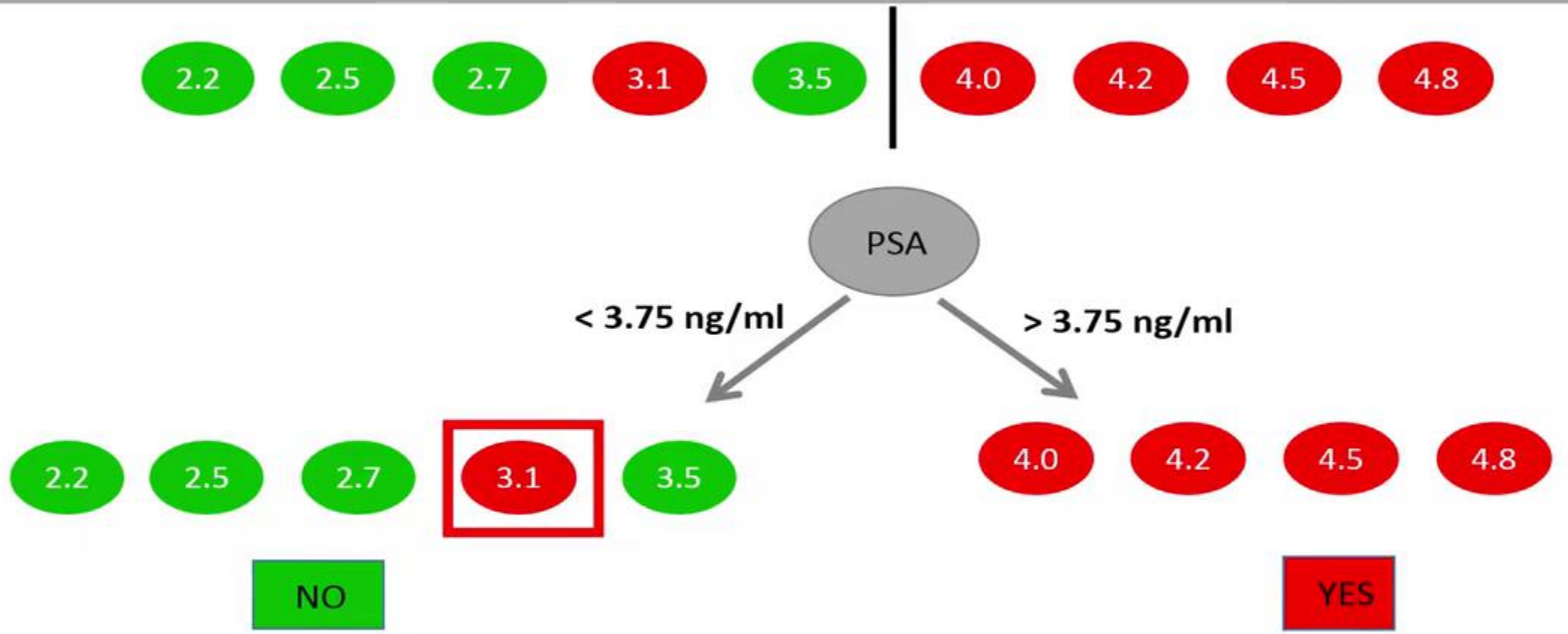
and that these actually have prostate cancer,

Accuracy of the classification



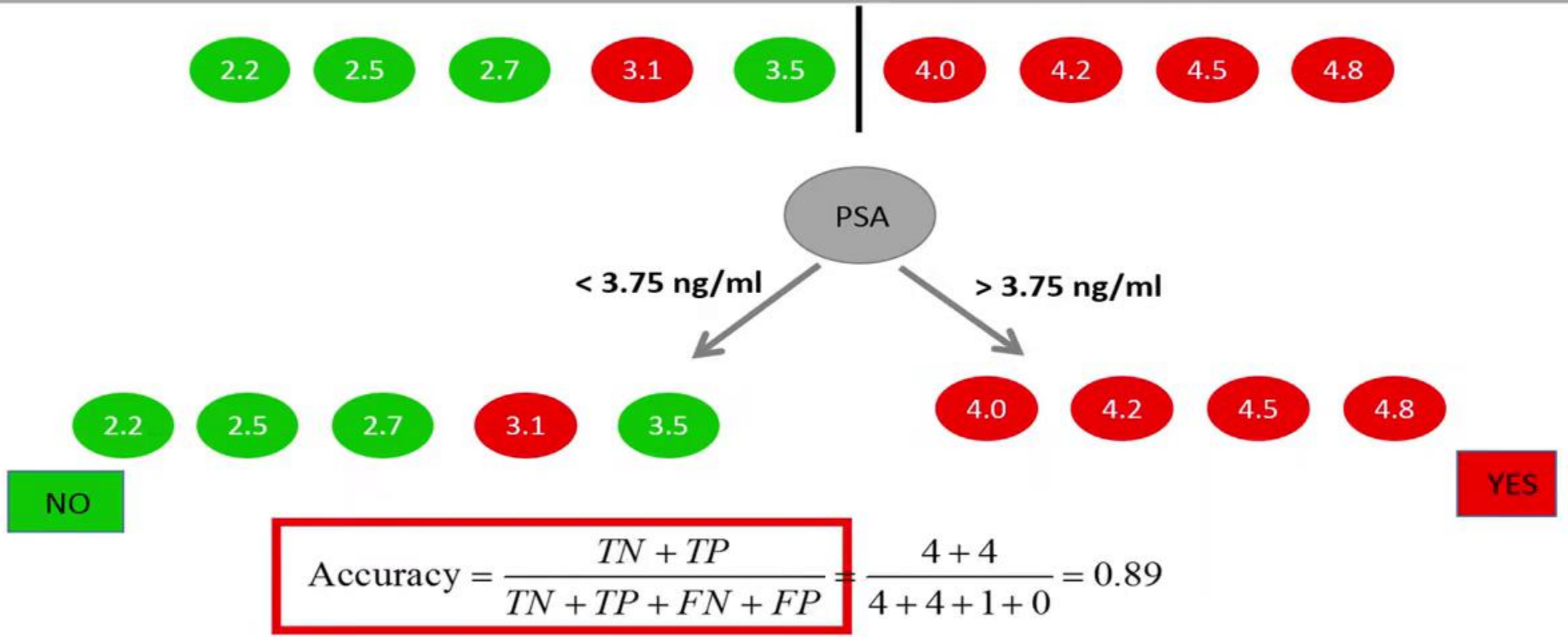
we know that we have four true negative results,

Accuracy of the classification



one false negative,

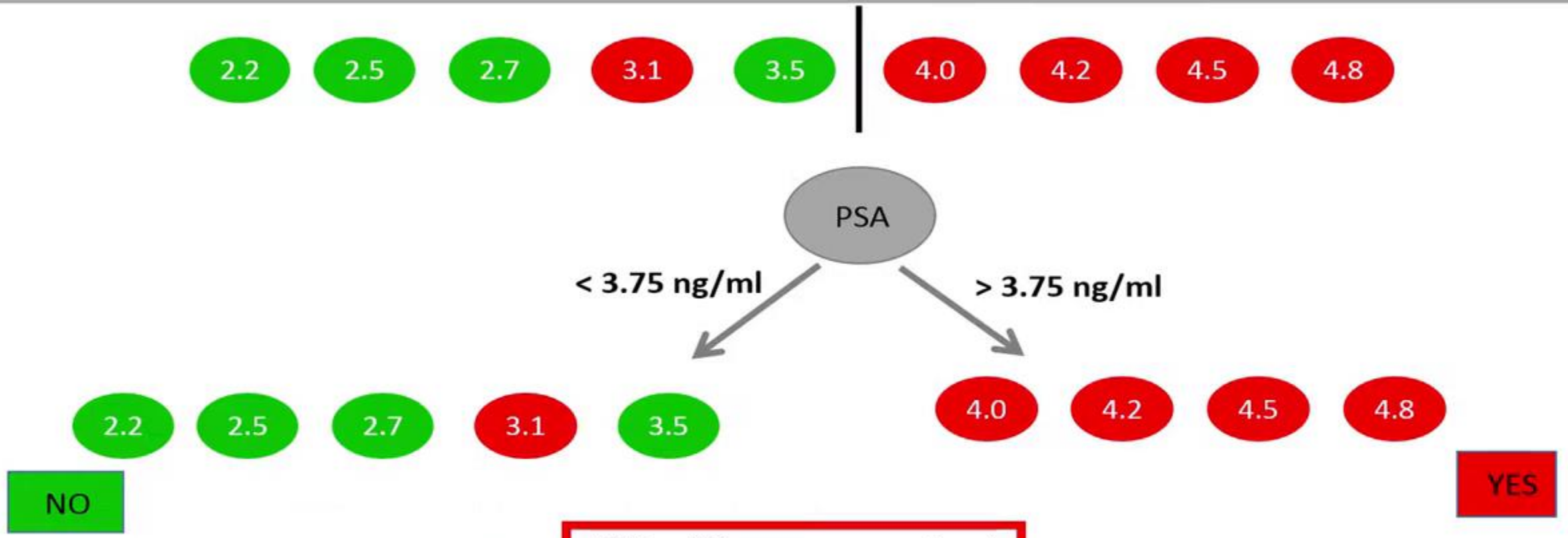
Accuracy of the classification



$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} = \frac{4+4}{4+4+1+0} = 0.89$$

We can therefore calculate the accuracy of this tree based on the training data,

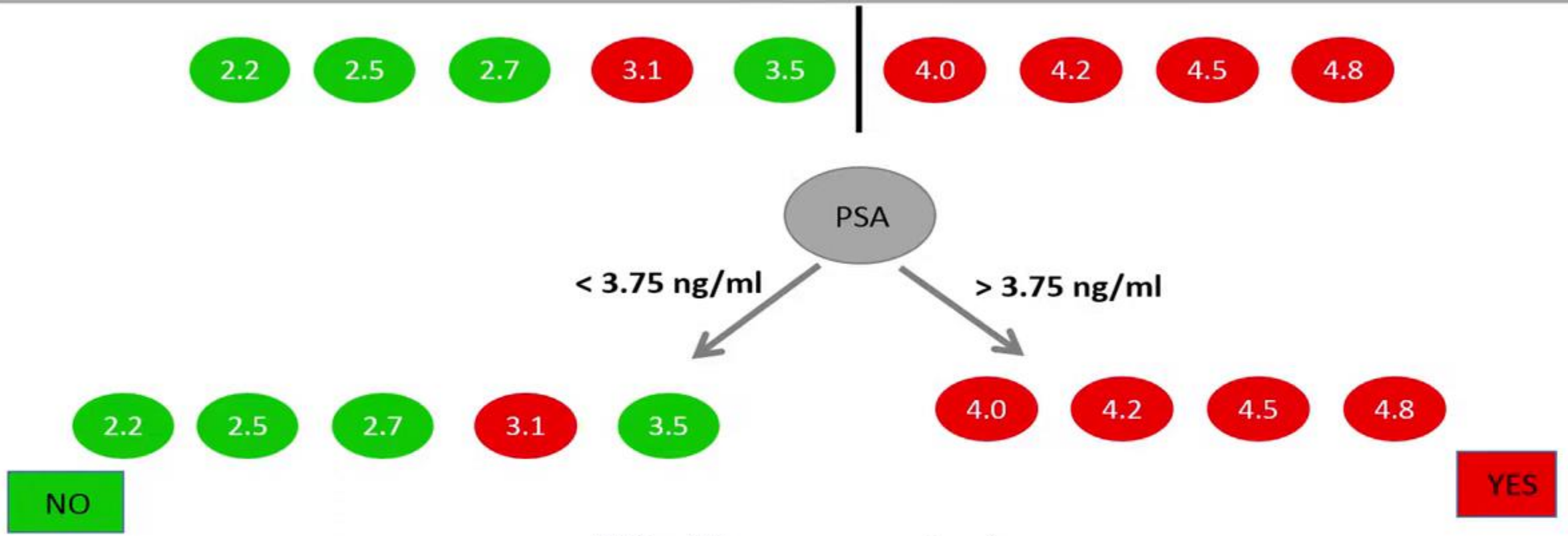
Accuracy of the classification



$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} = \frac{4+4}{4+4+1+0} = 0.89$$

by dividing the number of true negatives and positives,

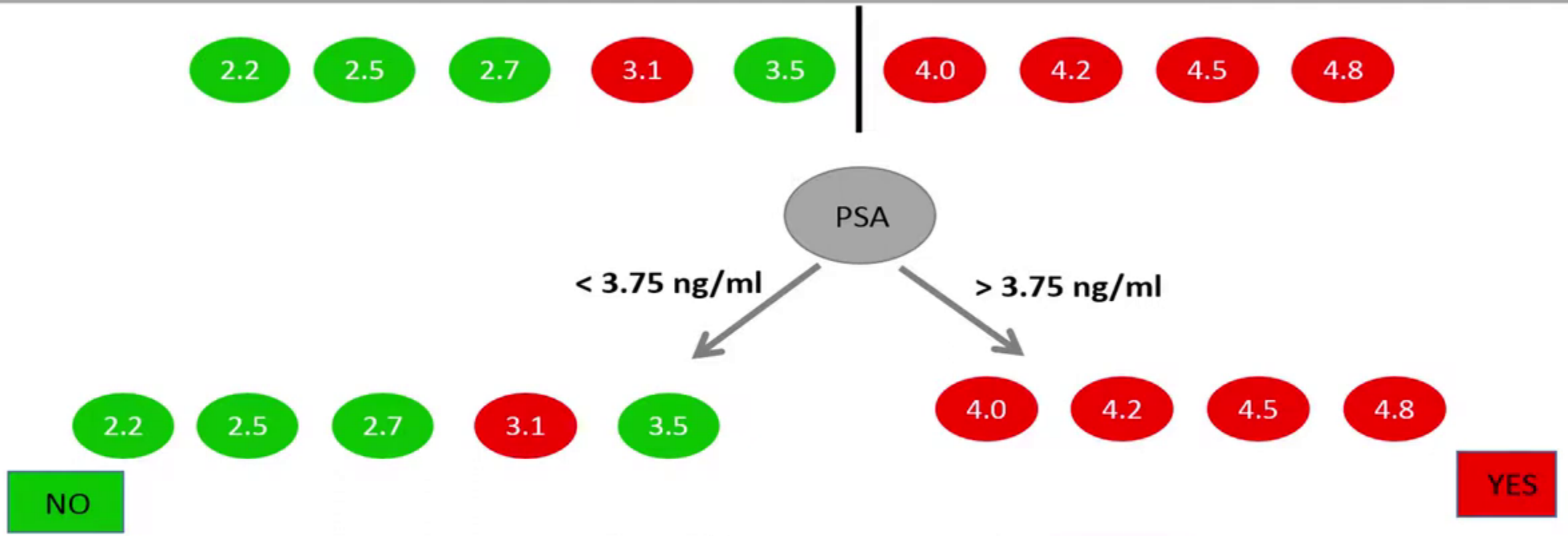
Accuracy of the classification



$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} = \frac{4+4}{4+4+1+0} = 0.89$$

by the total number of observations we have.

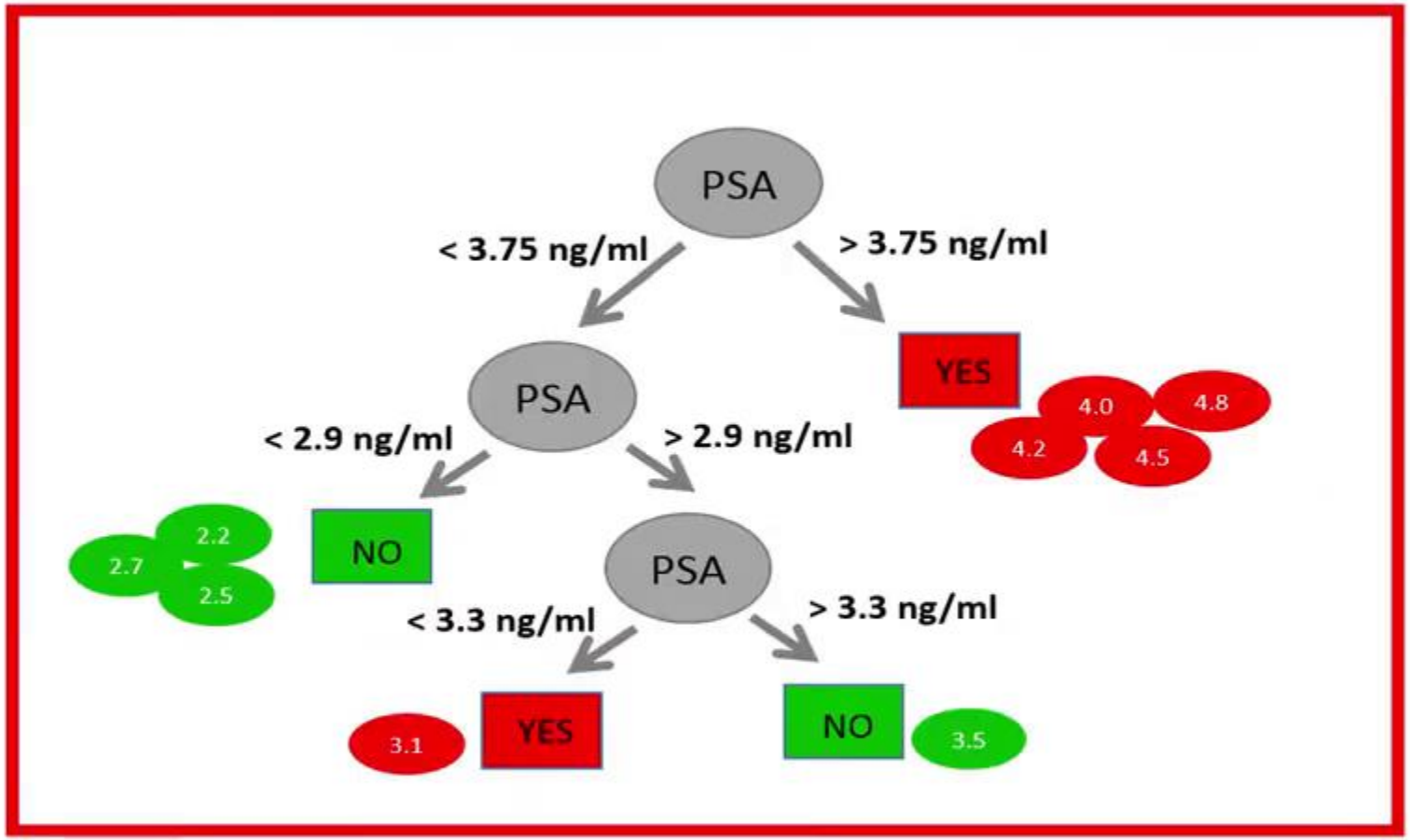
Accuracy of the classification



$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} = \frac{4+4}{4+4+1+0} = 0.89$$

We see that the accuracy of the classifier based on this example data is about 89%.

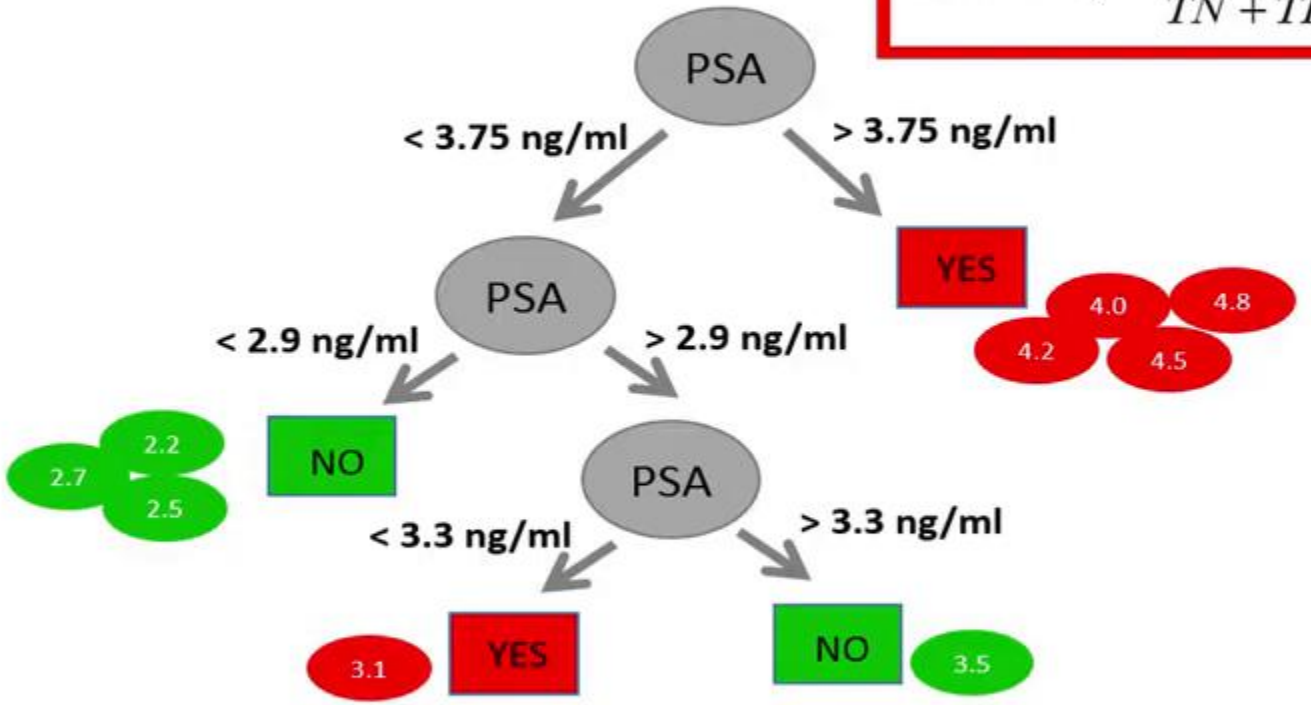
Overfitting



Note that we can continue to grow the tree to get an even better prediction.

Overfitting

$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} = \frac{4+5}{4+5+0+0} = 100\%$$



If we let the tree grow, the tree will always perfectly predict the training data and we will always get an accuracy of 100%. By letting the tree grow too much, we will encounter something that is called overfitting.

Overfitting

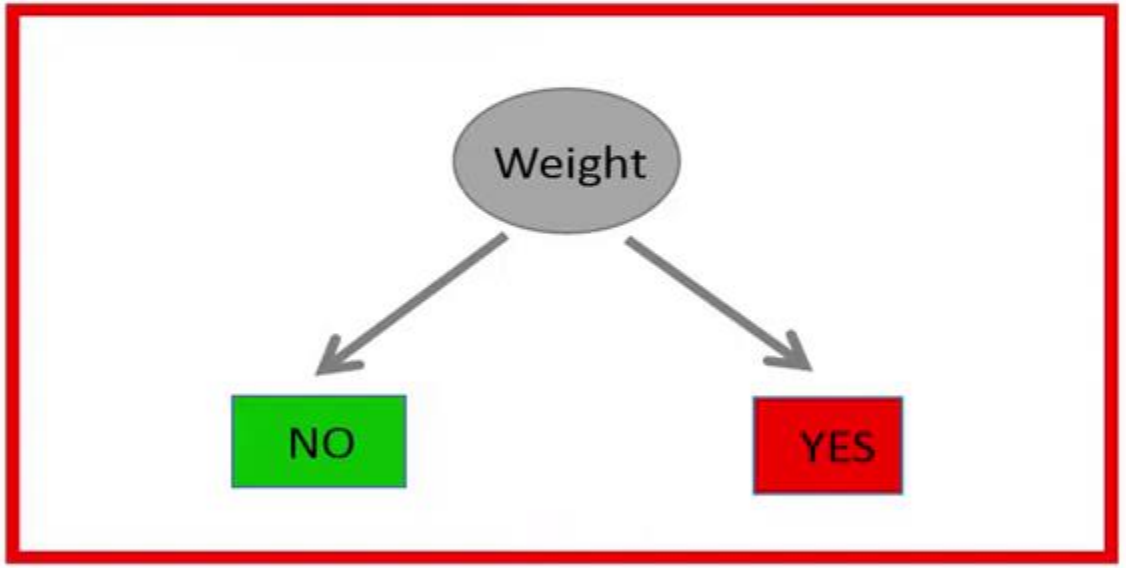




To understand the concept of overfitting, let's say that we also have data on the body weights, in kg, of the nine individuals.

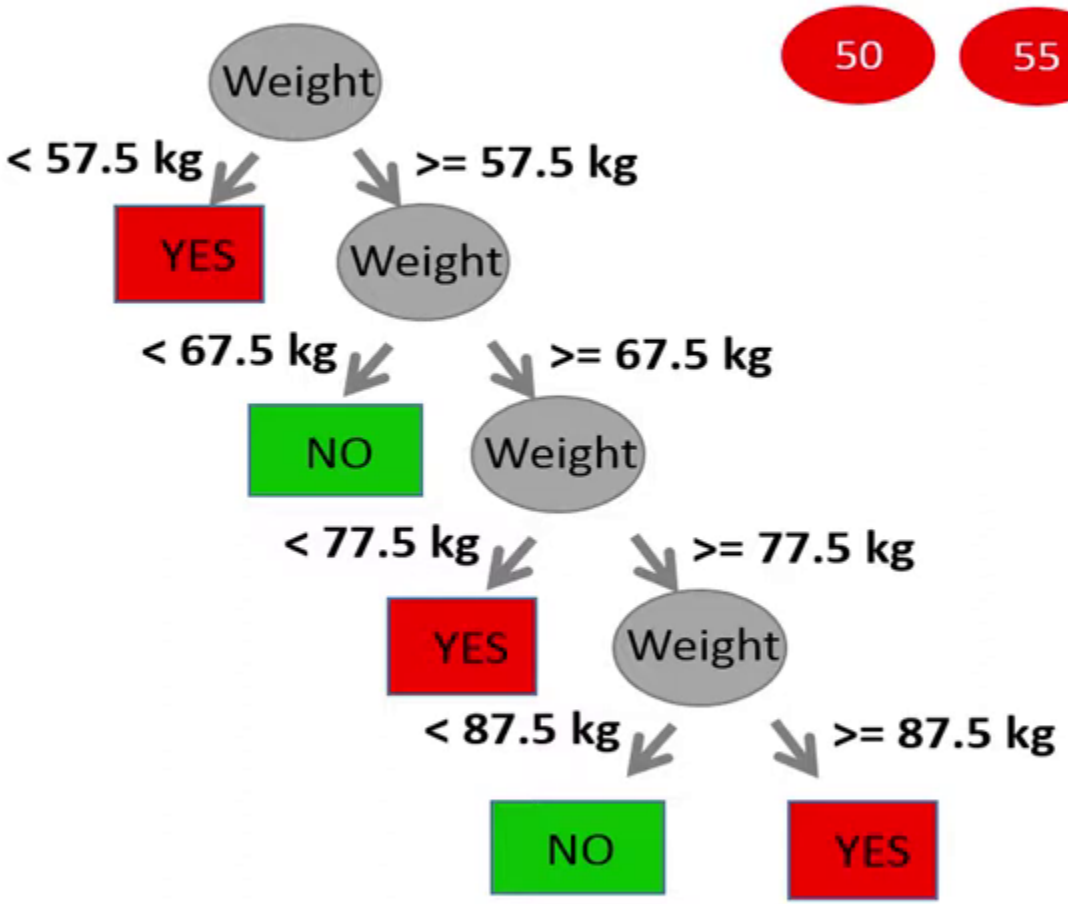
Overfitting





Think about this for a second. Do you think that it would be possible to predict if someone has prostate cancer based on its body weight? That seems impossible, right?

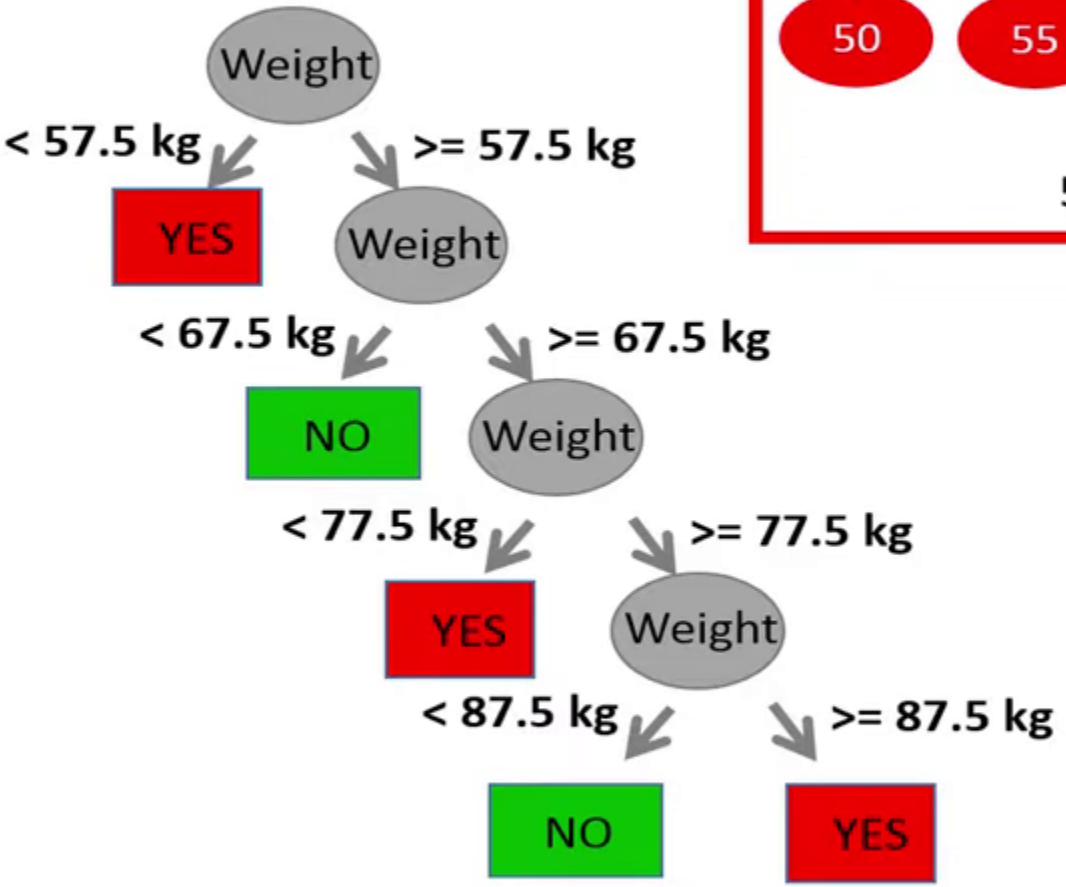
Overfitting

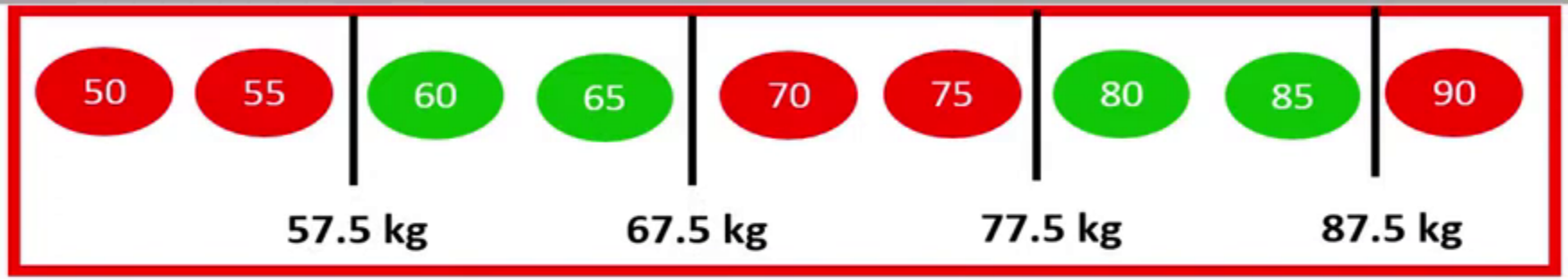




However, a decision tree would have no problem in dividing the groups with 100% purity.

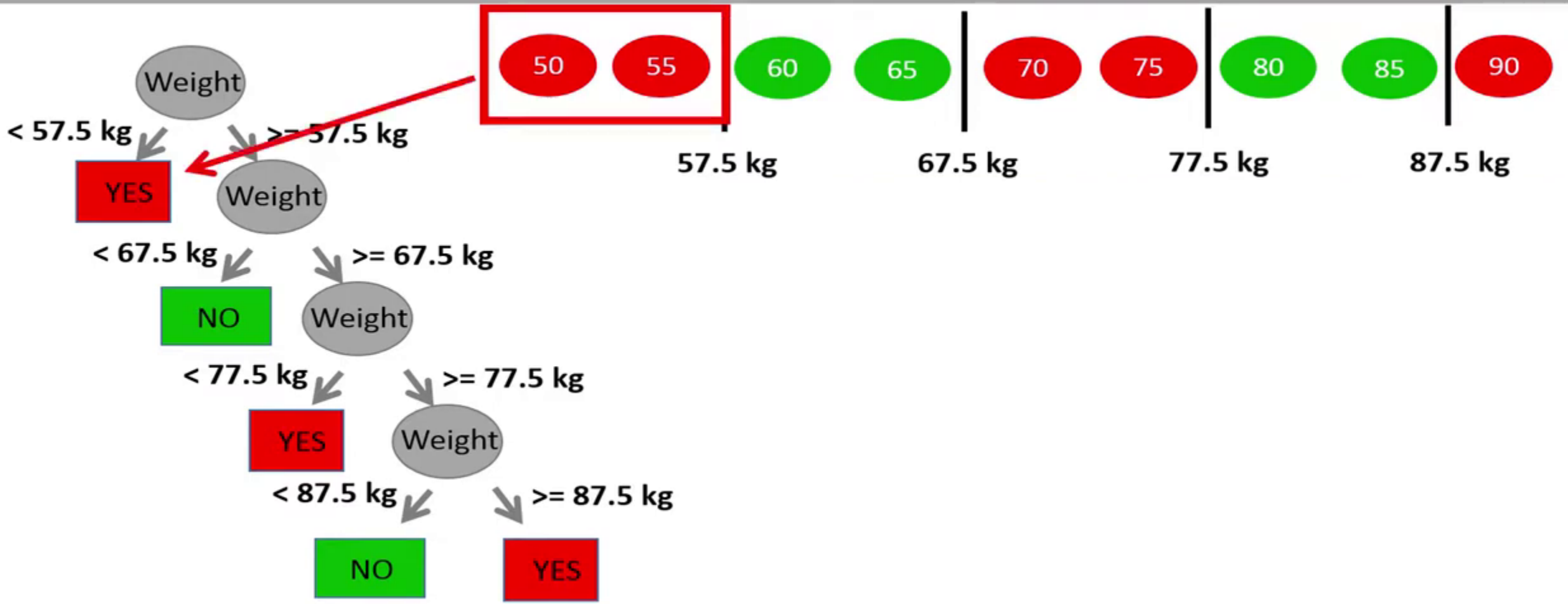
Overfitting





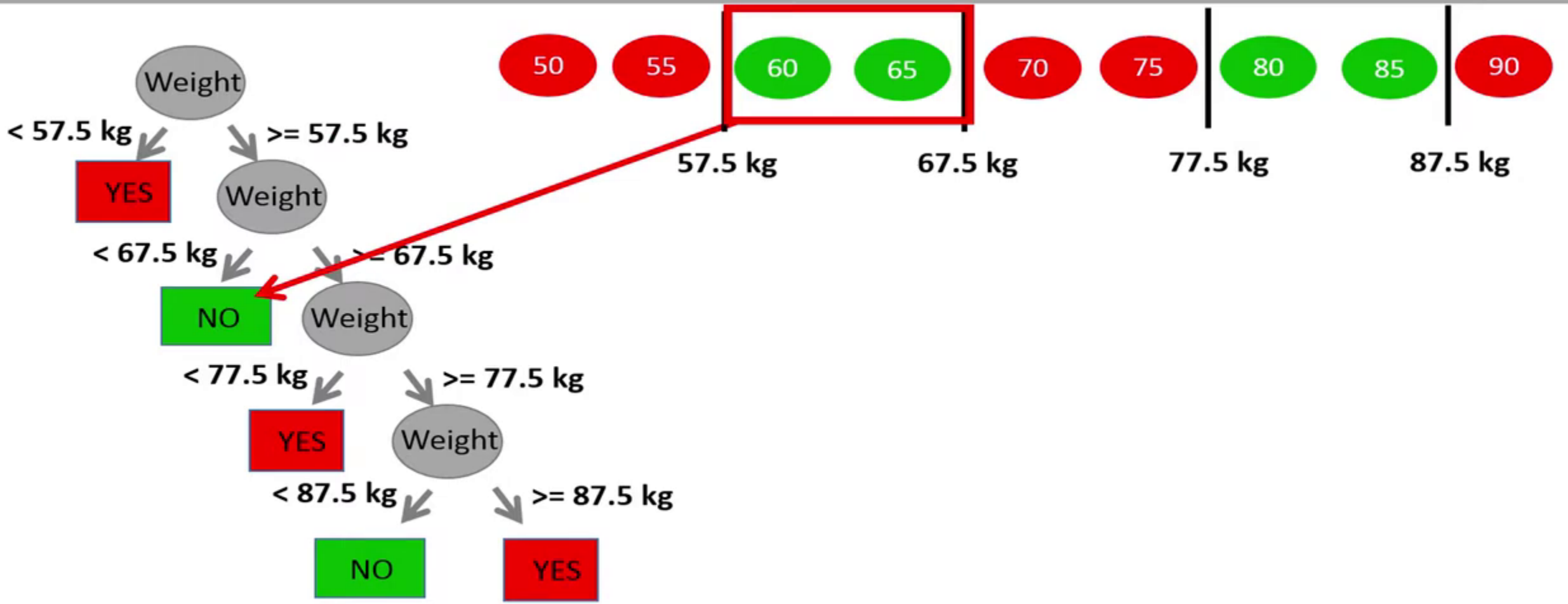
By using the following cutoff values, the decision tree would create 5 subgroups with 100% purity.

Overfitting



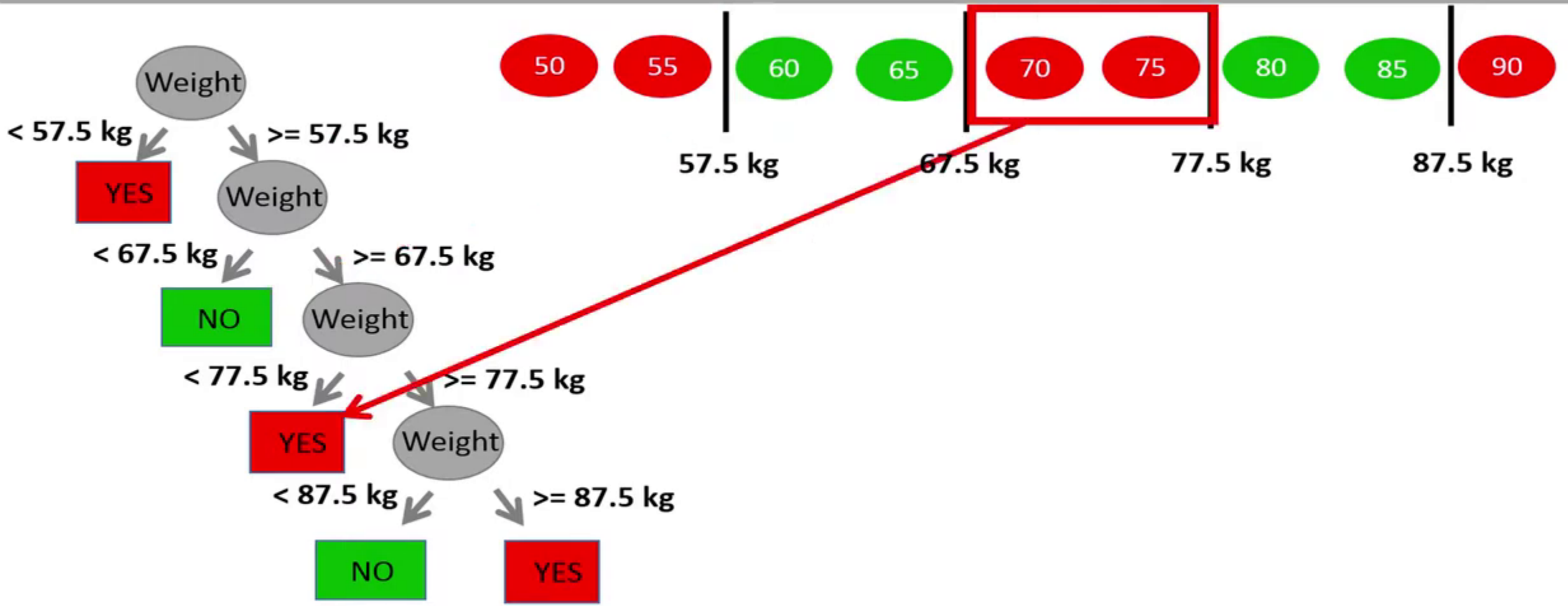
These two persons will be correctly predicted to have prostate cancer because their weights are less than 57.5 kg.

Overfitting



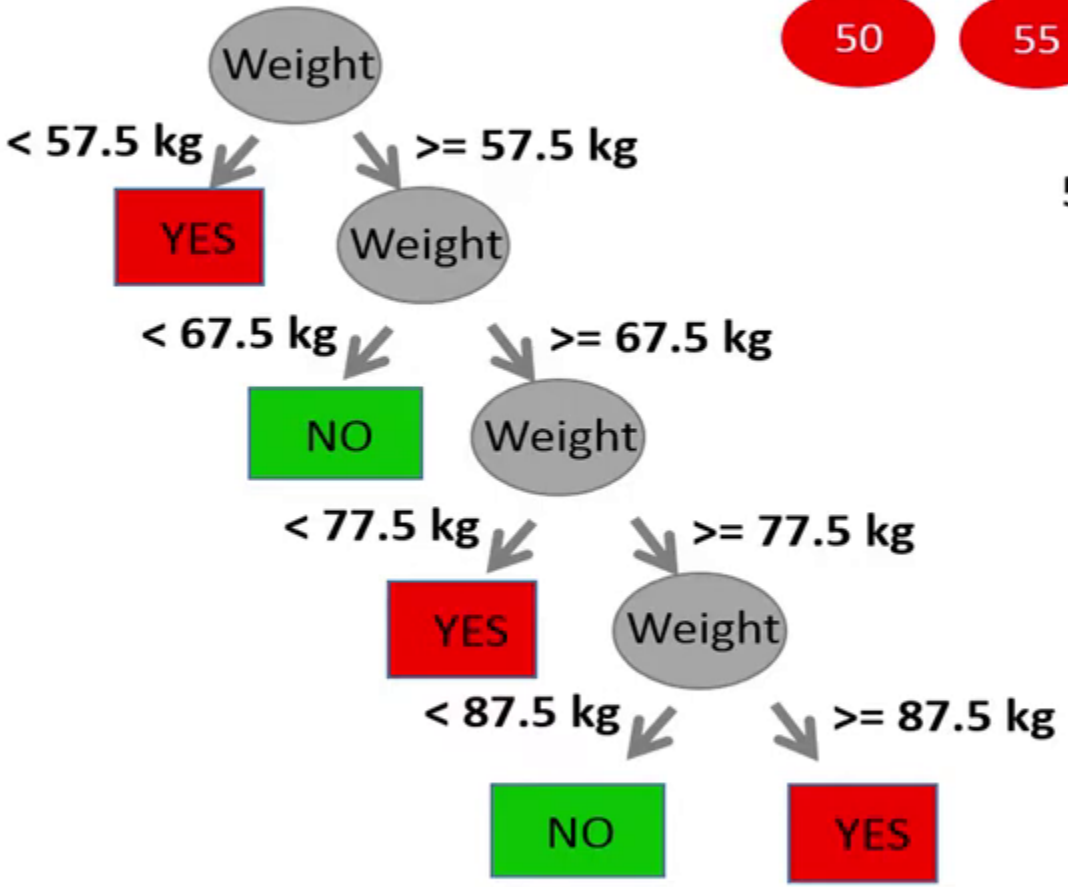
These two persons will be correctly classified as being healthy,

Overfitting



and these two will also be correctly predicted, and so forth.

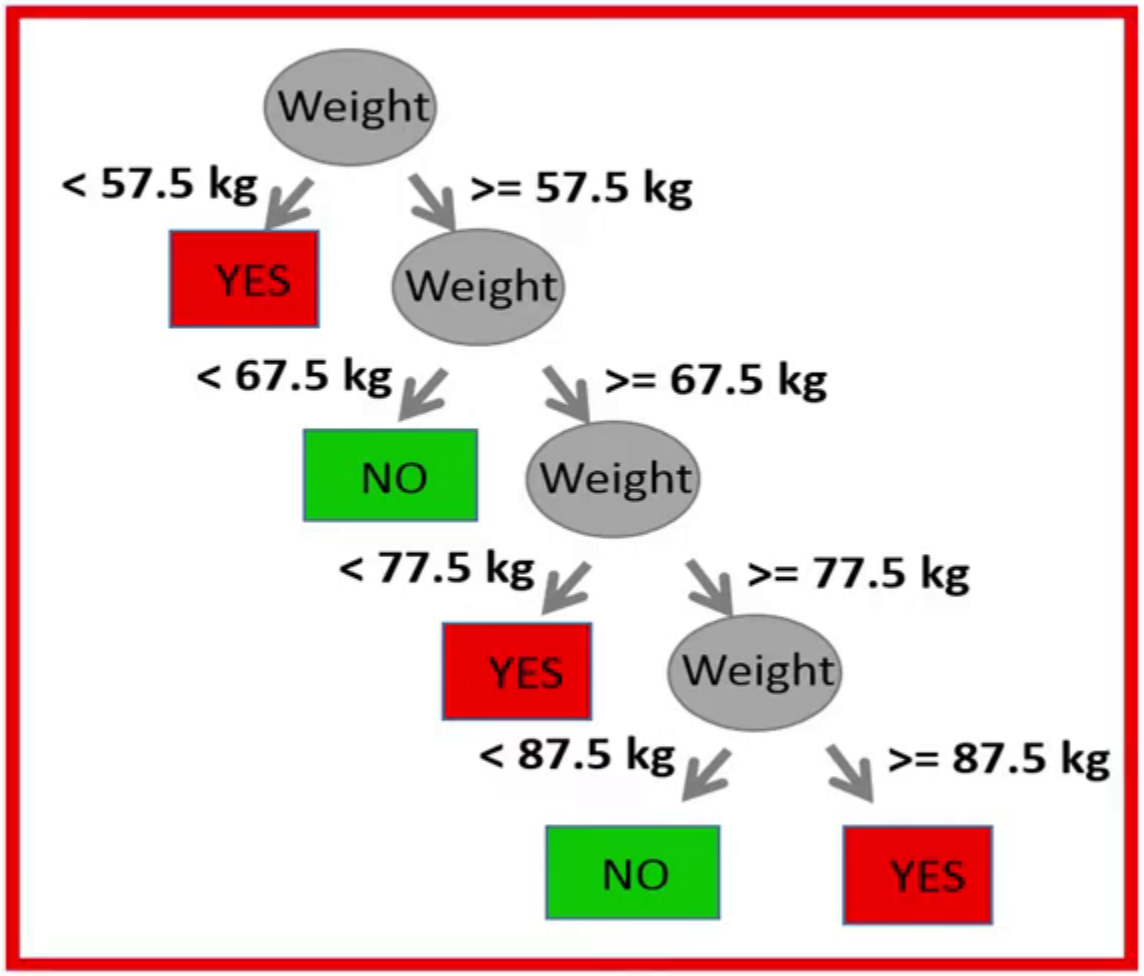
Overfitting





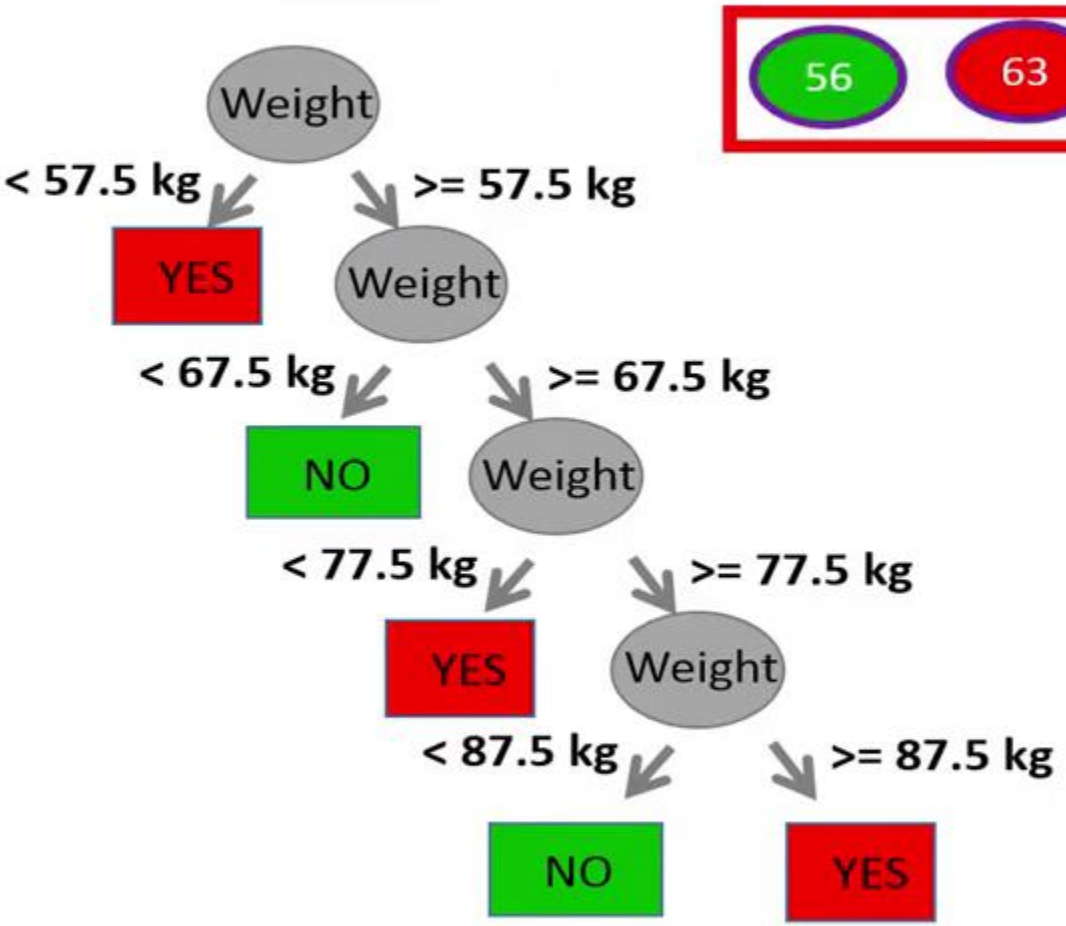
This seems too good to be true. Is this tree really useful?

Overfitting



What has happened is that we have encountered overfitting, which is highly problematic when we use decision trees.

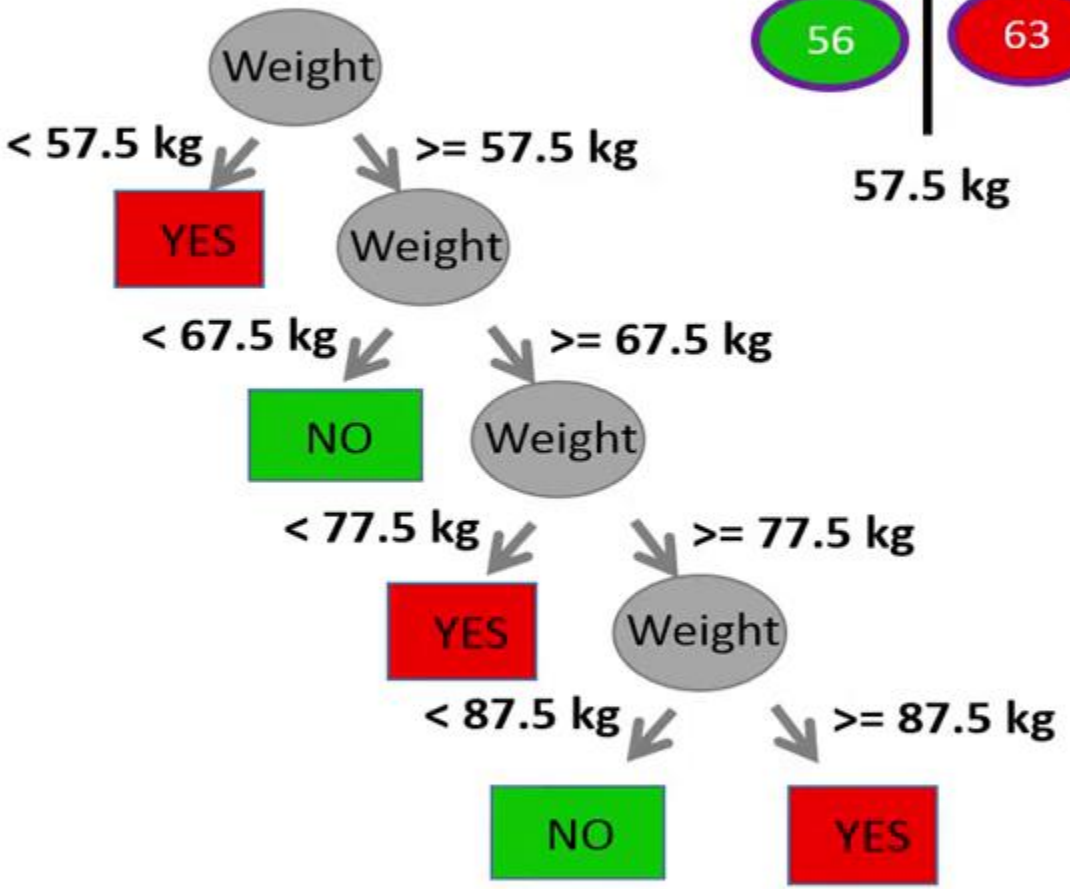
Overfitting

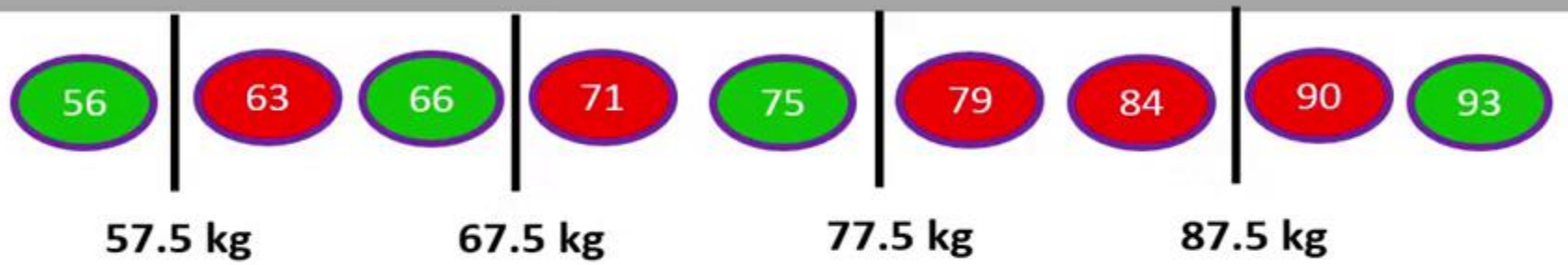




Let's say that we would collect a new data set, with five individuals who have prostate cancer and four healthy individuals. Since this is a new sample, we expect that the body weights are different from our previous sample.

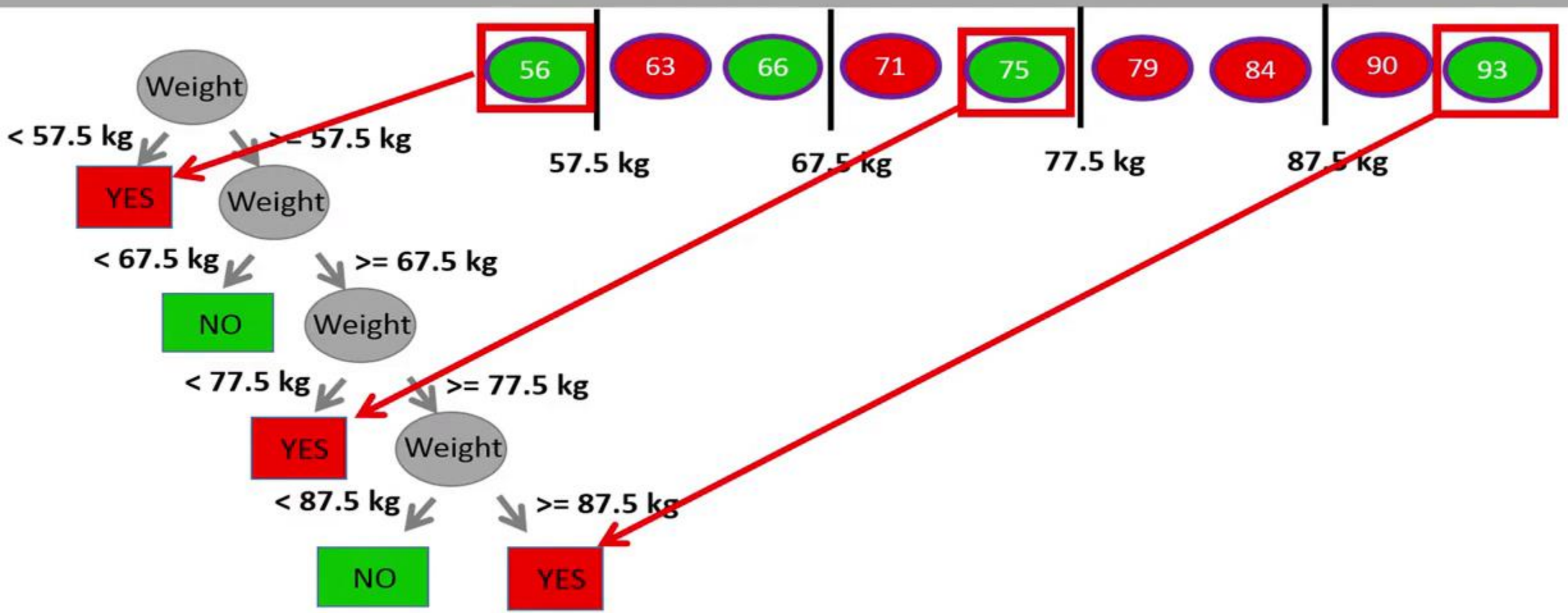
Overfitting





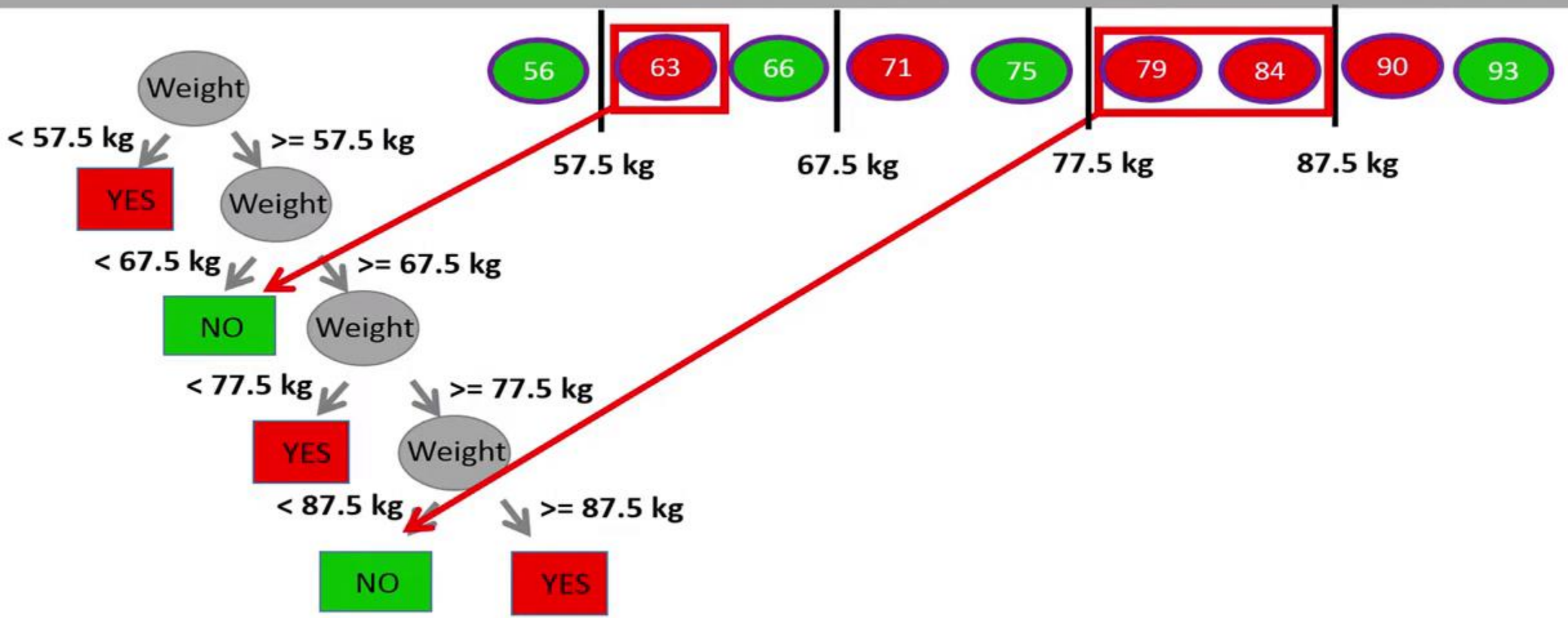
If we would use the decision tree that we created earlier to classify the new data set,

Overfitting



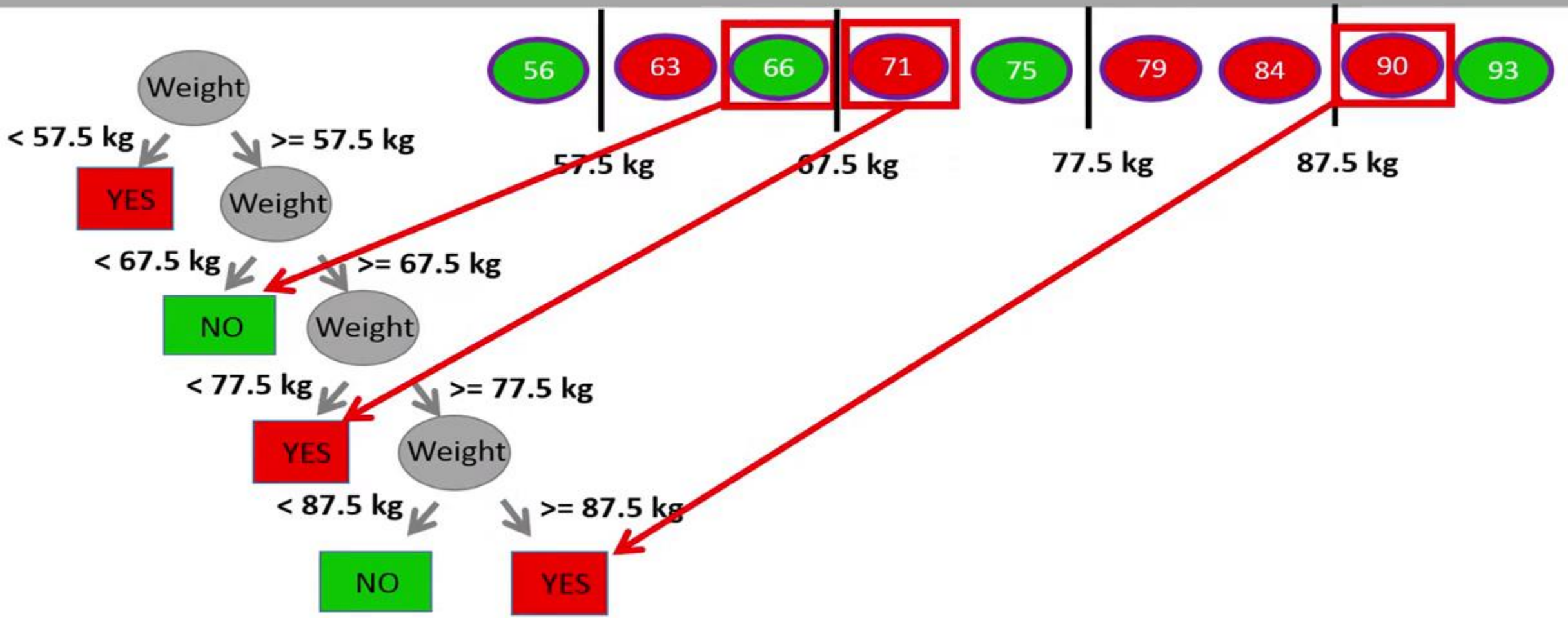
we see that the following healthy individuals would be incorrectly classified as having prostate cancer,

Overfitting



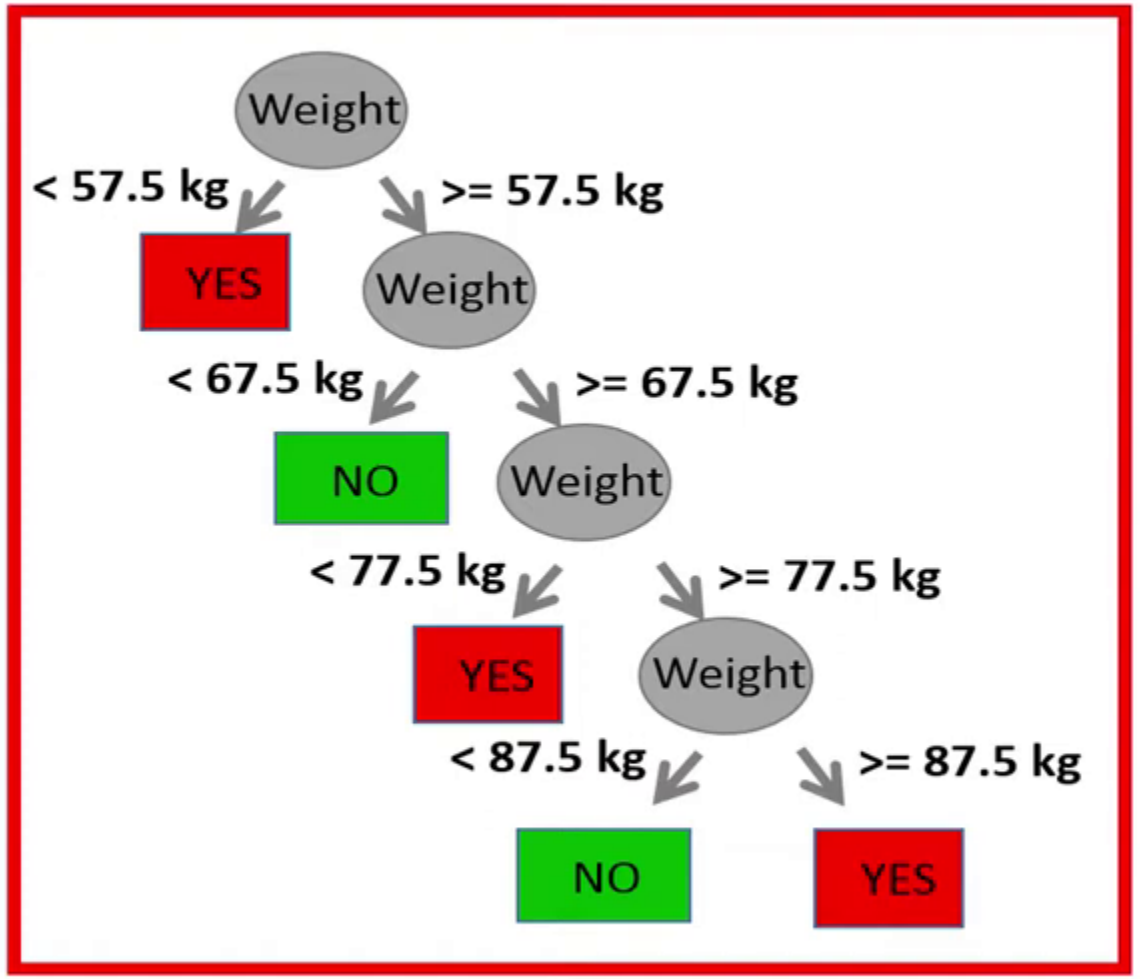
and that these patients with prostate cancer would be incorrectly classified as being healthy.

Overfitting



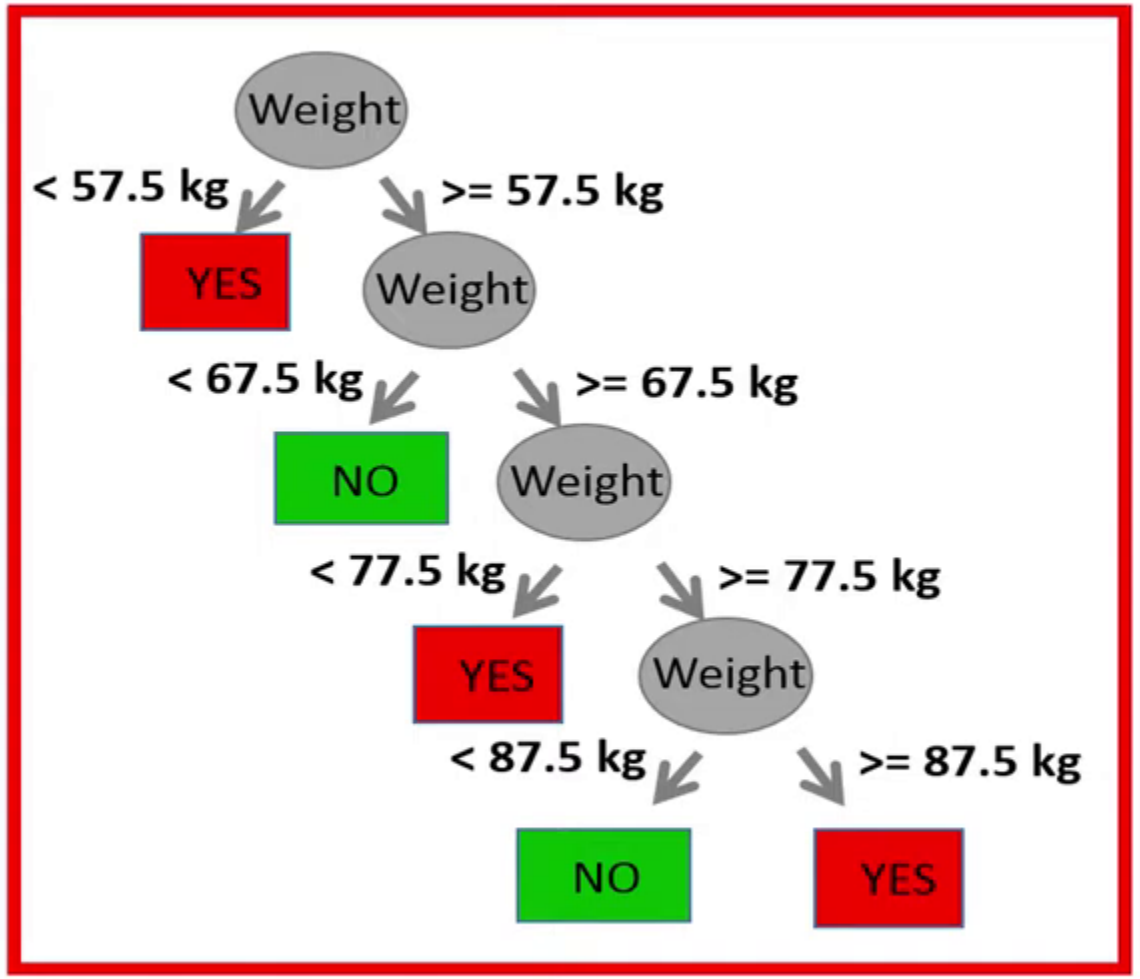
These individuals happened to be correctly predicted due to chance.

Overfitting



When the tree has many leaf nodes relative to the sample size, this indicates that the model has been overfitted, which means that we have built a perfect model on the training data.

Overfitting



The following decision tree would be completely useless if we apply it on new individuals. To know how well a decision tree would perform on new data, we have to use some sort of validation technique.

Pruning

Prune the tree

One way to overcome the problem with overfitting is to prune the tree. There are a number of different methods for pruning a tree.

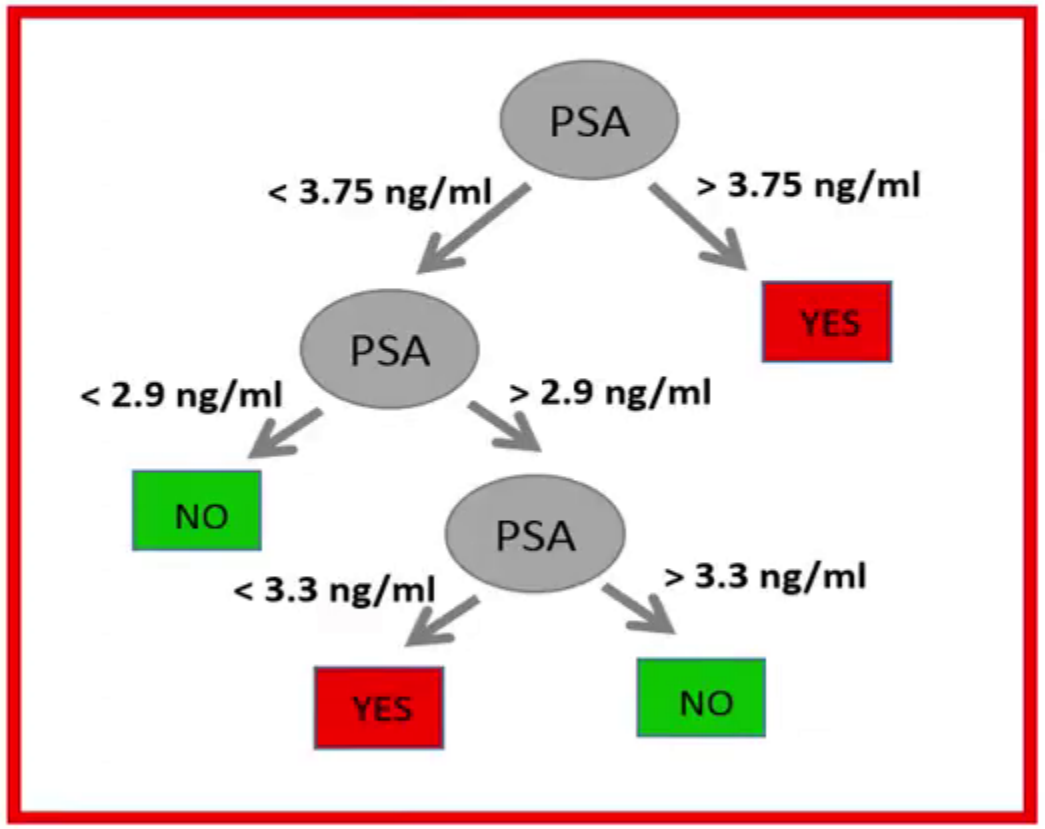
Pruning



To show how pruning works, we will use the same data on the PSA concentration of four healthy individuals and five patients with prostate cancer.

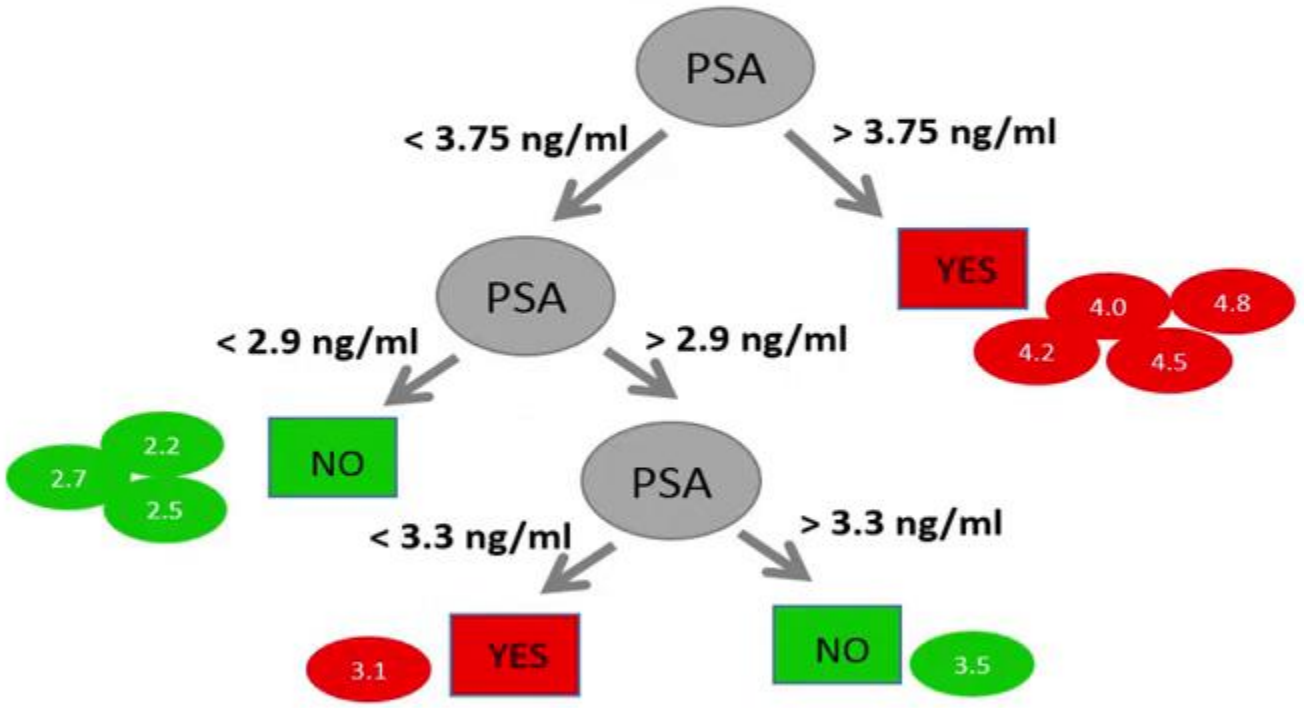
Pruning





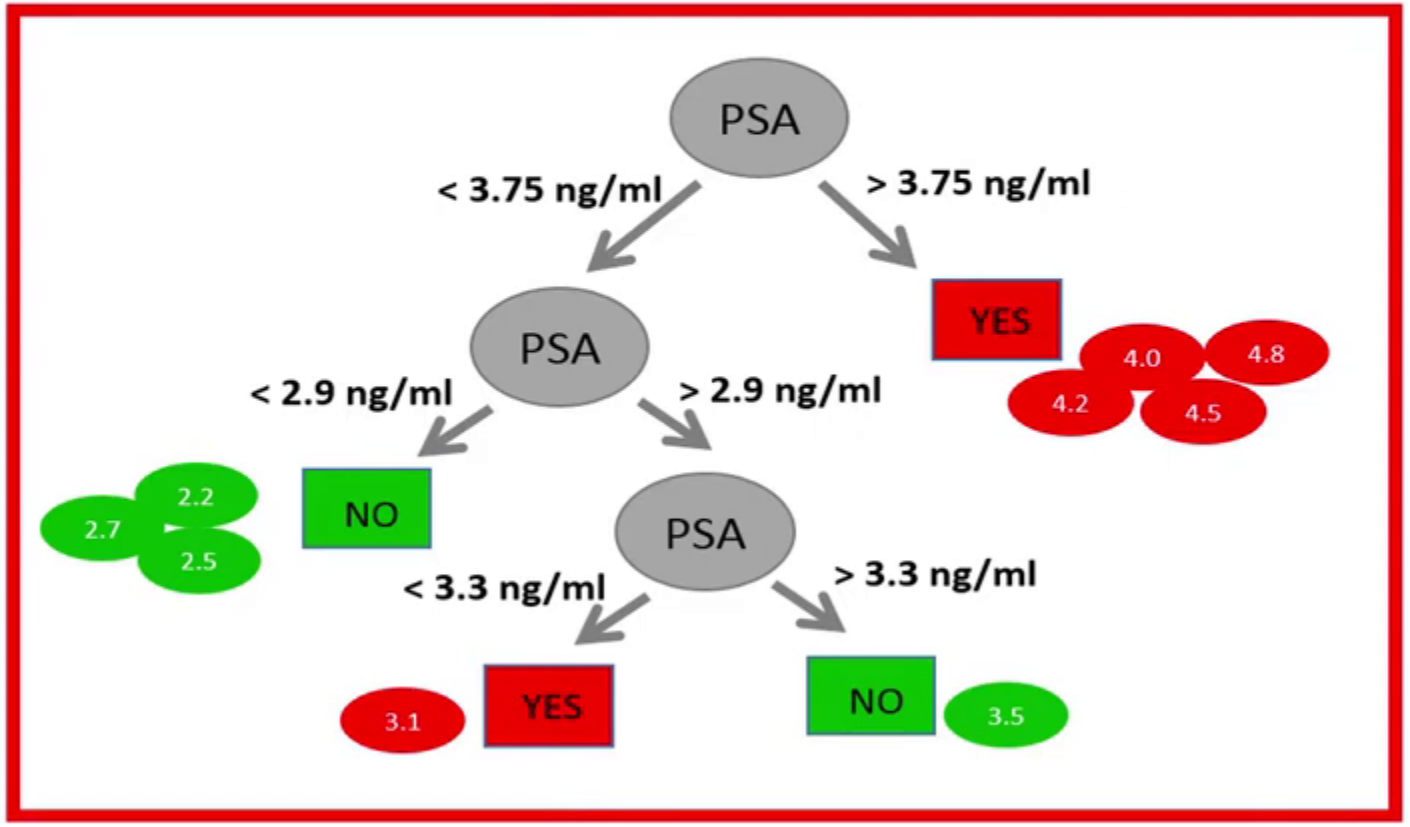
If we let the tree grow so that it perfectly separates the groups in the training data, the tree will look like this.

Pruning



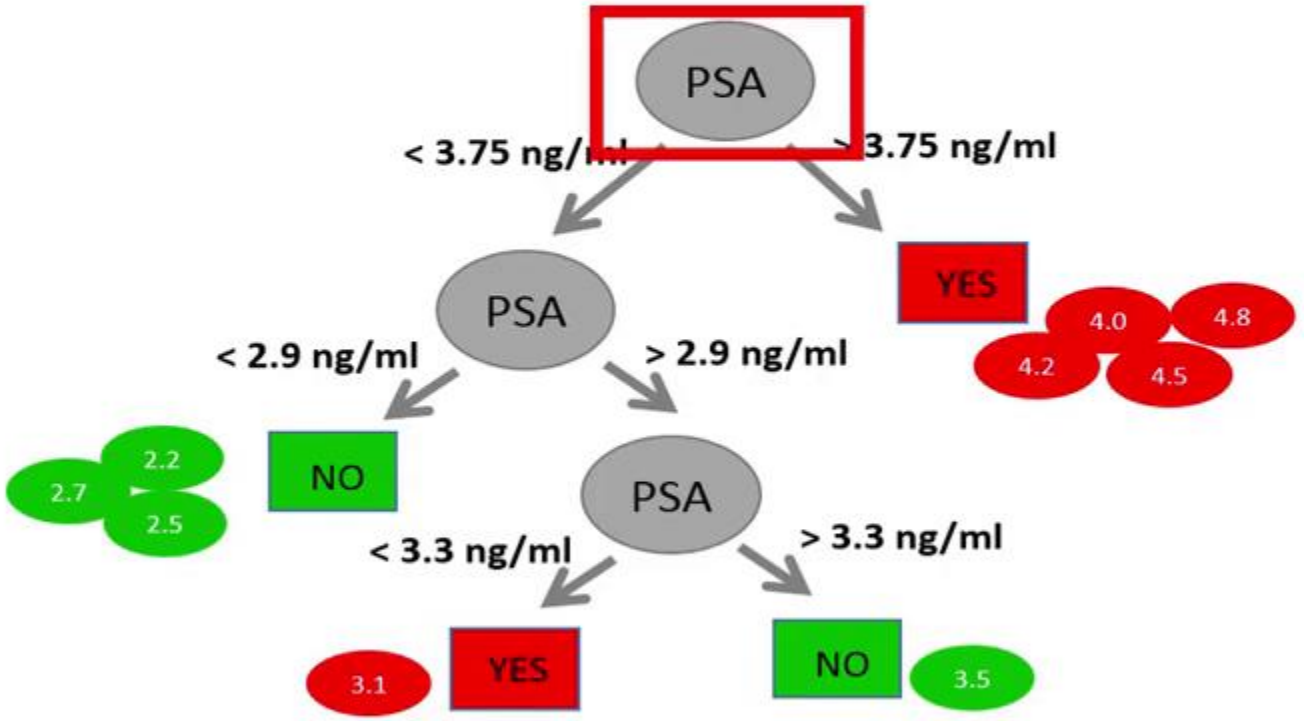
We see that all nine individuals are correctly predicted by this tree.

Minsplit = 6



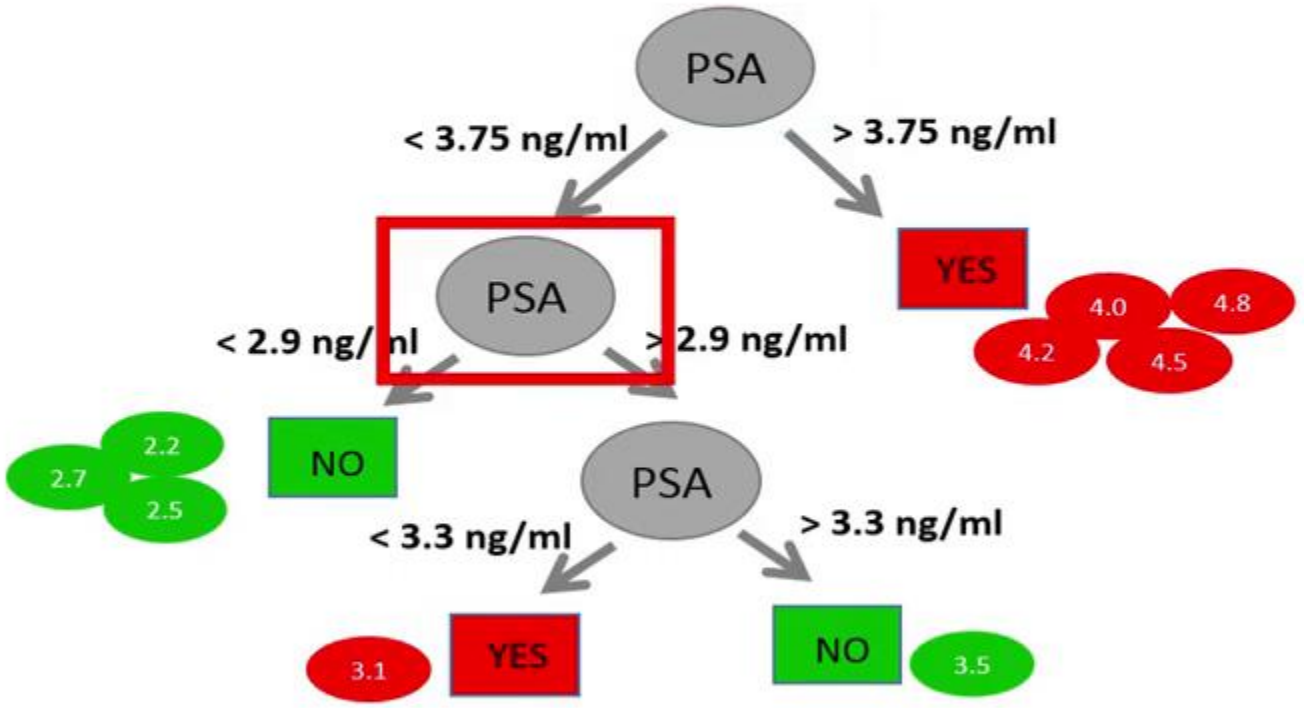
To prune this tree, we can, for example, specify that the minimum number of observations must be equal to six to allow a split.

Minsplit = 6



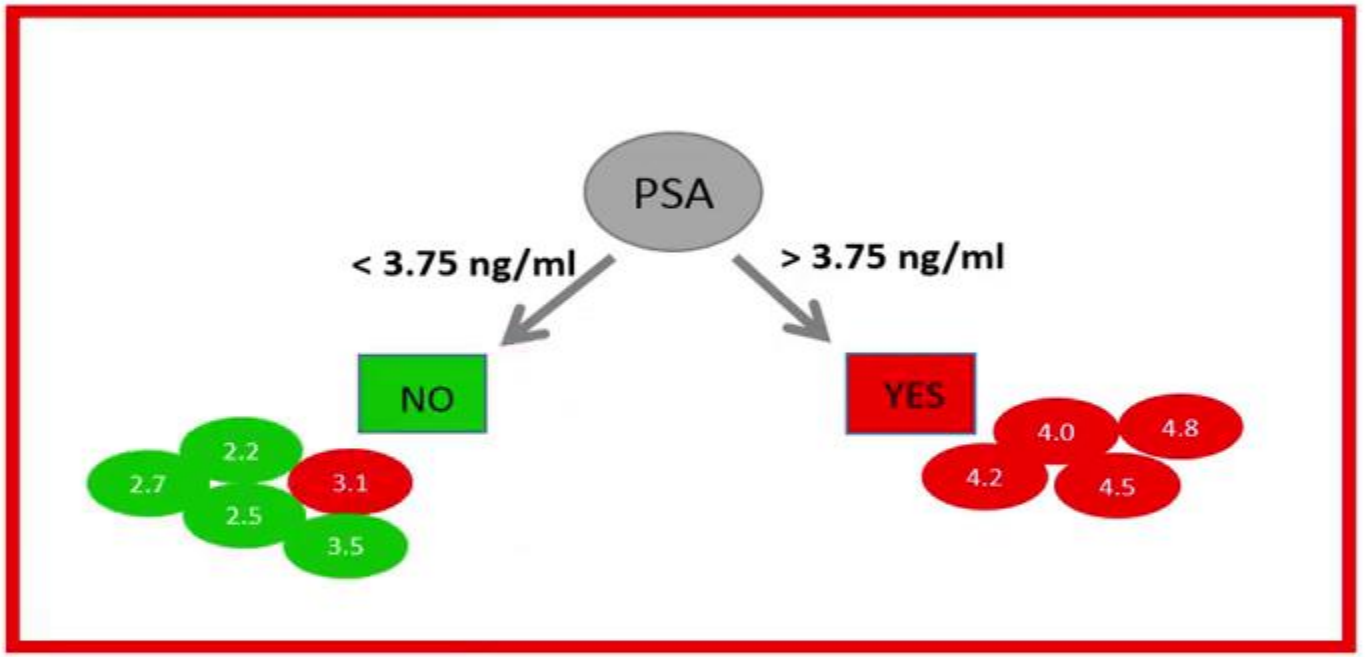
The first node was based on all nine data points,

Minsplit = 6



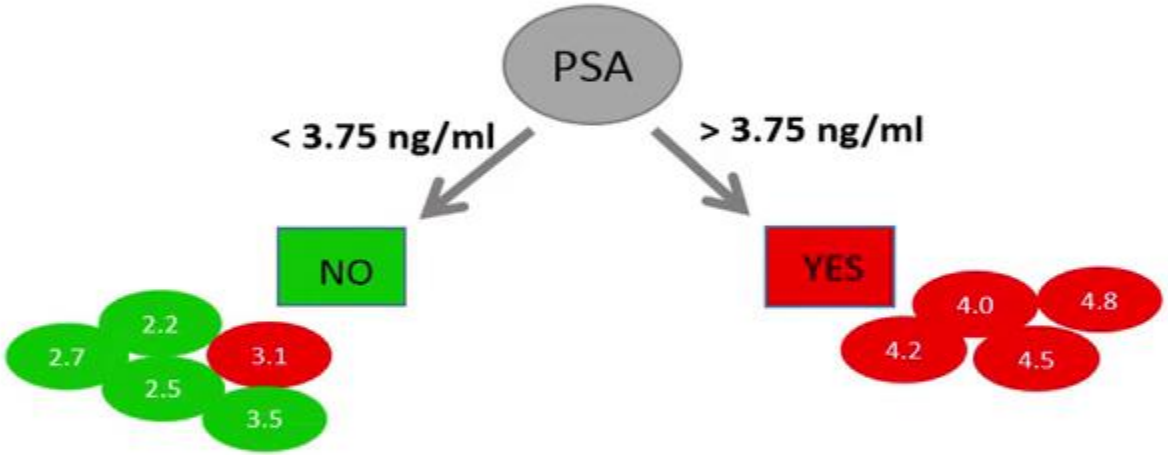
whereas the second node classified only five data points, which is less than our threshold value of six. The tree would therefore stop growing here,

Minsplit = 6



so that the tree only consists of one root node.

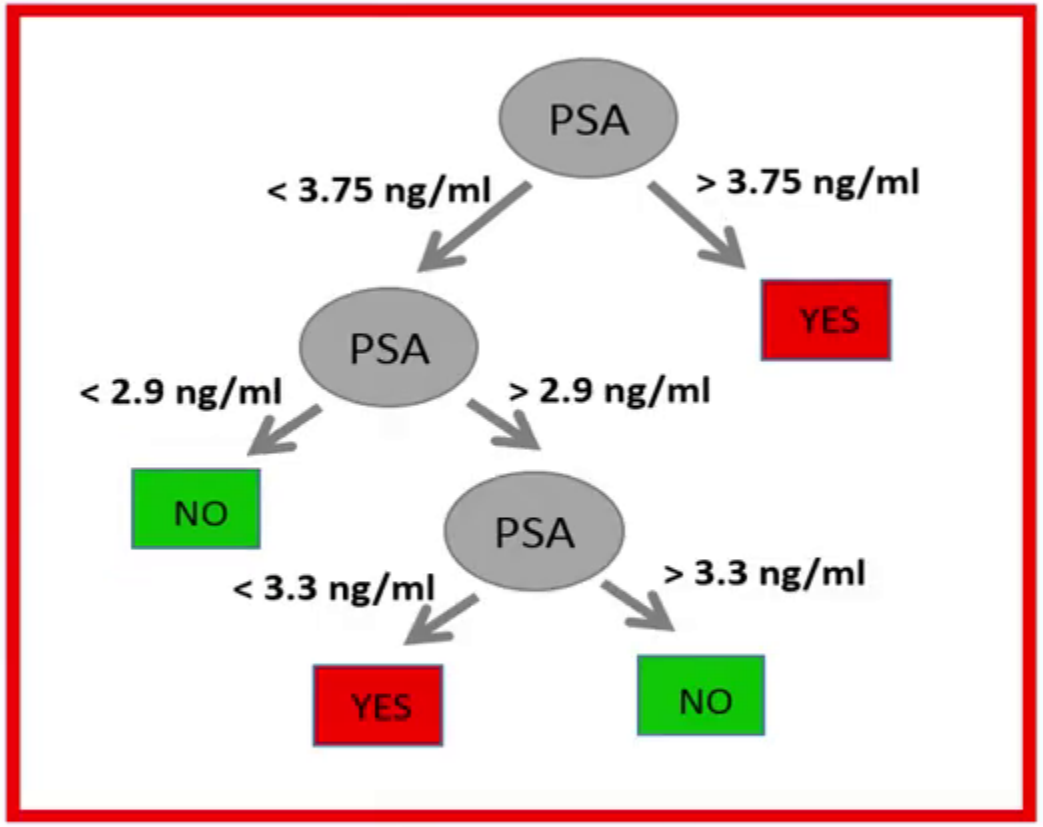
Minsplit = 6



$$Accuracy = \frac{8}{9} = 89\%$$

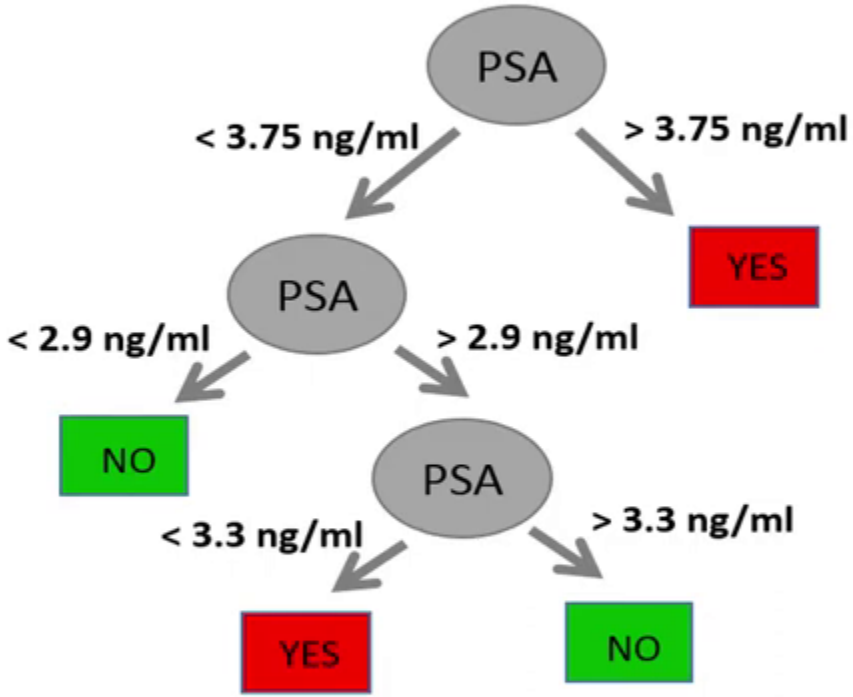
Although this tree consists of only one node, it still correctly predicts eight out of the nine observations.

Prune based on validation



Another strategy would be to prune the tree based on some sort of validation method.

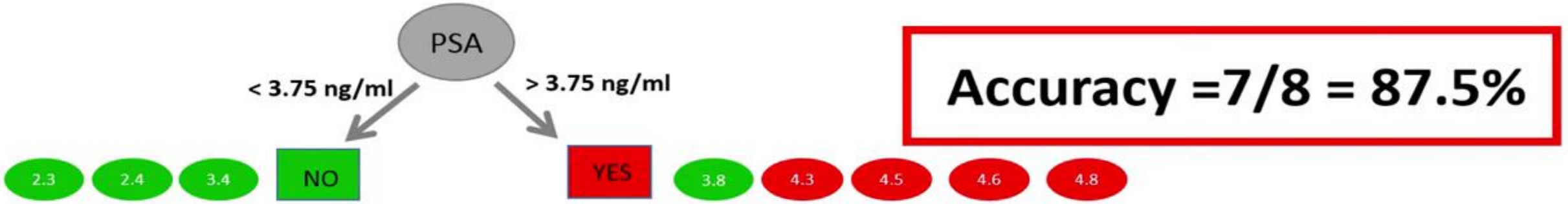
Prune based on validation





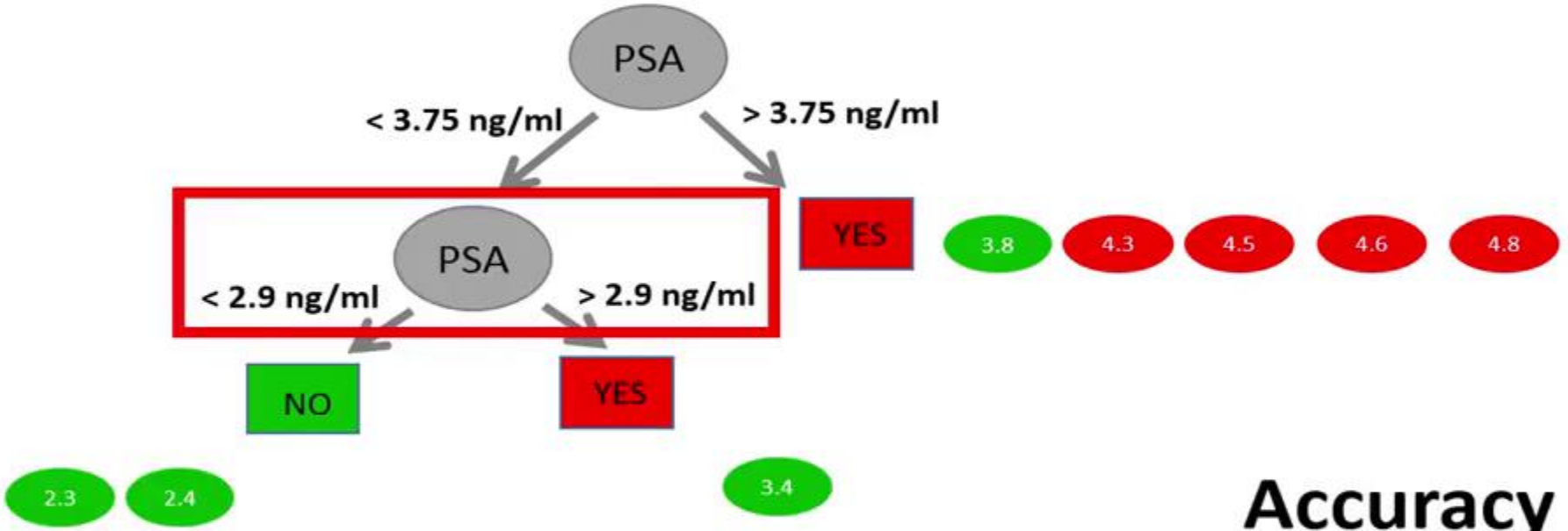
We therefore need a validation data set that has not been involved to create the tree. For example, let's say that we have collected four new healthy individuals and four new patients with prostate cancer.

Prune based on validation



We then analyze how well the tree would classify the validation data set based on only one node. In this example, a tree would make seven correct predictions out of the eight possible.

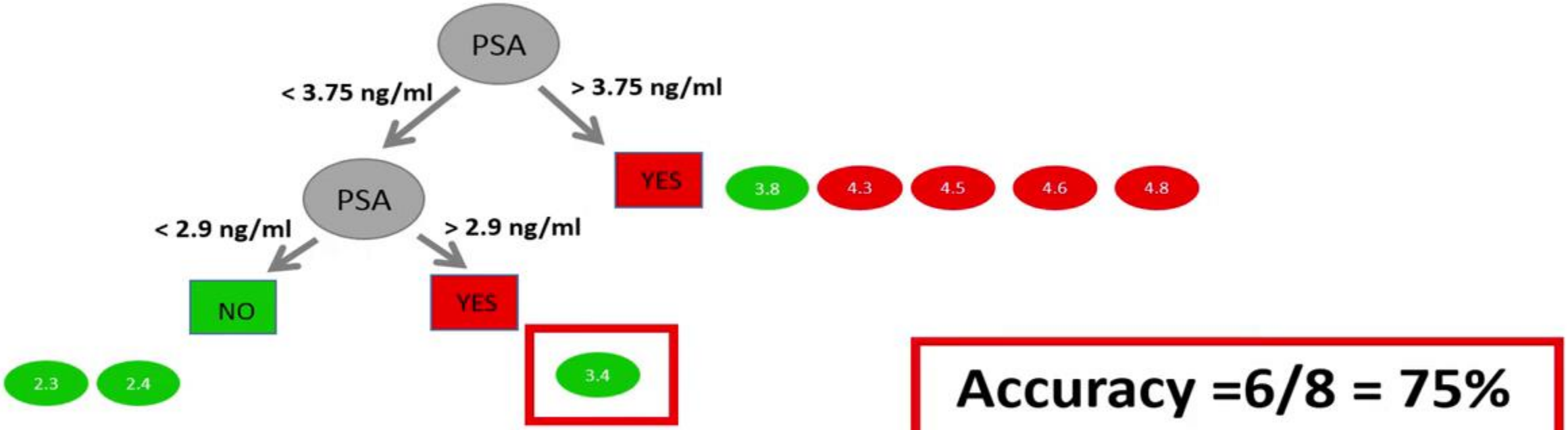
Prune based on validation



Accuracy =6/8 = 75%

If we would add the second node of the tree that was proposed based on the training data,

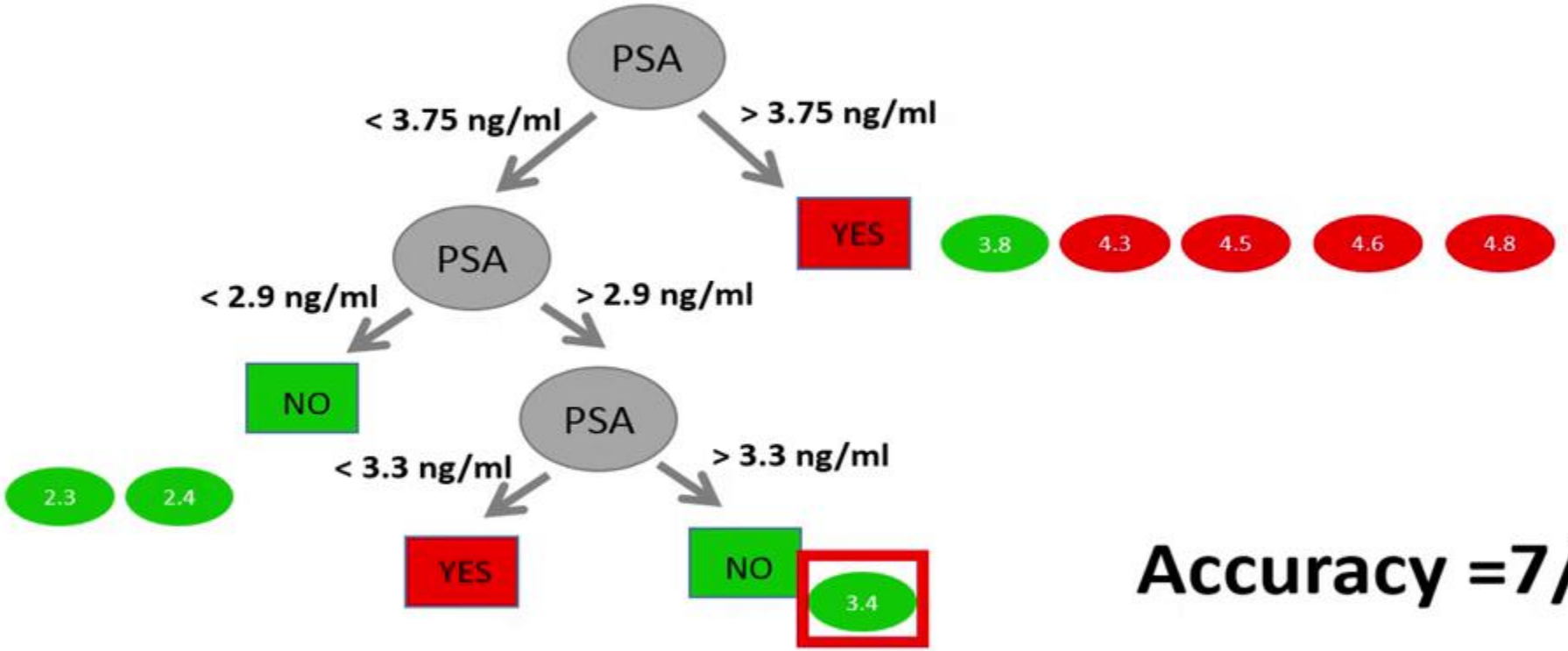
Prune based on validation



Accuracy =6/8 = 75%

and the accuracy would be reduced to 75%.

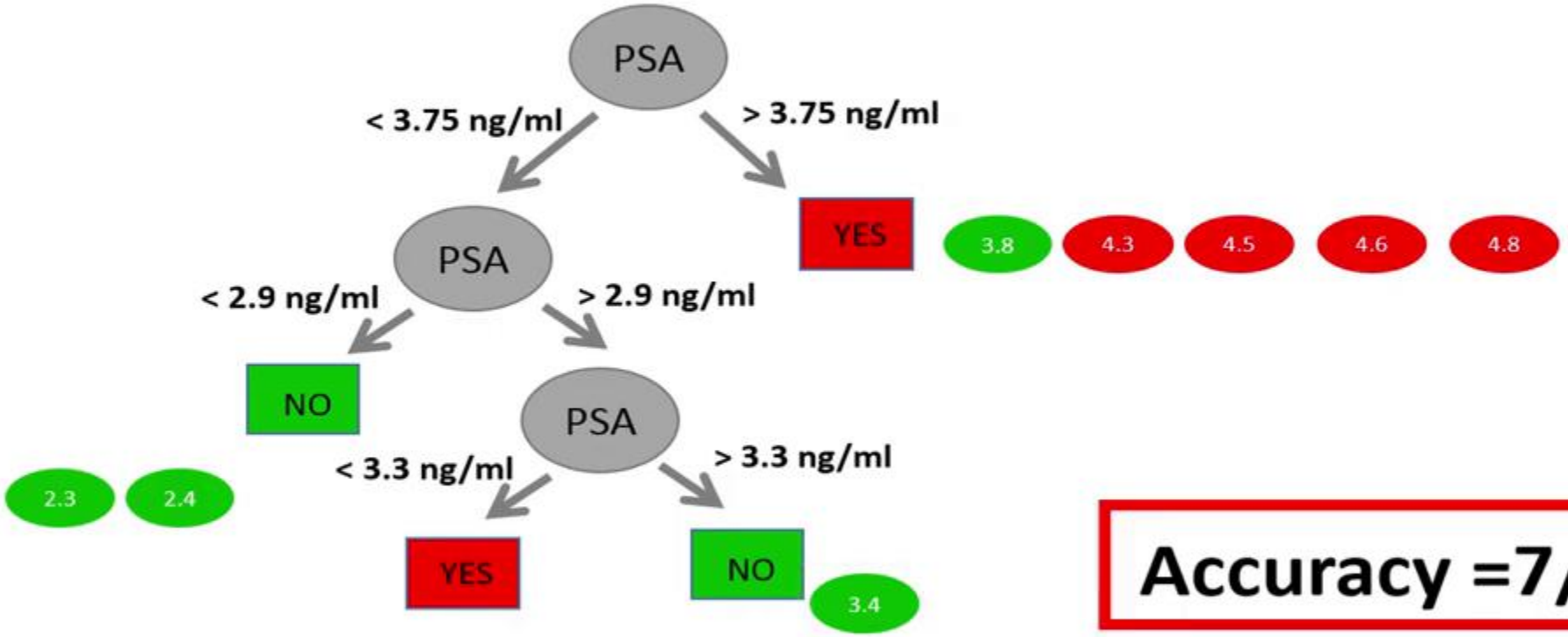
Prune based on validation



Accuracy =7/8 = 87.5%

Adding a third node will now correctly predict this observation. However, a tree with three nodes has the same accuracy as a tree with only one node.

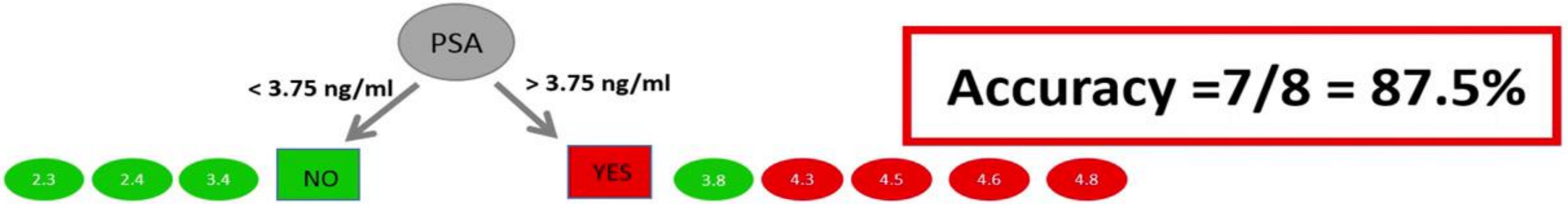
Prune based on validation



Accuracy =7/8 = 87.5%

If the accuracy does not increase, or only increases marginally, it might be a good idea to prune the tree from the node where the tree does not improve the prediction of the validation data set.

Prune based on validation



In this case, a tree with more than one node does improve the accuracy. In other words, a larger tree does not reduce the error, which means that we should select a tree with only one node in this case.